

# A Comprehensive Survey on World Models for Embodied AI

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**Abstract**—Embodied AI requires agents that perceive, act, and anticipate how actions reshape future world states. World models serve as internal simulators that capture environment dynamics, enabling forward and counterfactual rollouts to support perception, prediction, and decision making. This survey presents a unified framework for world models in embodied AI. Specifically, we formalize the problem setting and learning objectives, and propose a three-axis taxonomy encompassing: (1) *Functionality, Decision-Coupled vs. General-Purpose*; (2) *Temporal Modeling, Sequential Simulation and Inference vs. Global Difference Prediction*; (3) *Spatial Representation, Global Latent Vector, Token Feature Sequence, Spatial Latent Grid, and Decomposed Rendering Representation*. We systematize data resources and metrics across robotics, autonomous driving, and general video settings, covering pixel prediction quality, state-level understanding, and task performance. Furthermore, we offer a quantitative comparison of state-of-the-art models and distill key open challenges, including the scarcity of unified datasets and the need for evaluation metrics that assess physical consistency over pixel fidelity, the trade-off between model performance and the computational efficiency required for real-time control, and the core modeling difficulty of achieving long-horizon temporal consistency while mitigating error accumulation. Finally, we maintain a curated bibliography at <https://github.com/Li-Zn-H/AwesomeWorldModels>.

**Index Terms**—World Models, Embodied AI, Temporal Modeling, Spatial Representation.

## I. INTRODUCTION

EMBODIED AI aims to equip agents to perceive complex, multimodal environments, act within them, and anticipate how their actions will alter future world states [1], [2]. Central to this capability is the world model, an internal simulator that captures environment dynamics to support both forward and counterfactual rollouts for perception, prediction, and decision making [3], [4]. This survey focuses on world models that yield actionable predictions for embodied agents, distinguishing them from static scene descriptors or purely generative visual models that do not capture controllable dynamics.

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Cognitive science suggests humans construct internal models of the world by integrating sensory inputs. These models do not merely predict and simulate future events but also shape perception and guide action [5]–[7]. Motivated by this view, early AI research on world models was rooted in model-based reinforcement learning (RL), where latent state-transition models were used to improve sample efficiency and planning performance [8]. The seminal work of Ha and Schmidhuber [9] crystallized the term world model and inspired the Dreamer series [10]–[12], highlighting how learned dynamics can drive imagination-based policy optimization. More recently, advances in large-scale generative modeling and multimodal learning have expanded world models beyond their initial focus on policy learning into general-purpose environment simulators capable of high-fidelity future prediction, exemplified by models like Sora [13] and V-JEPA 2 [14]. This expansion has diversified functional roles, temporal modeling strategies, and spatial representations, while introducing inconsistencies in terminology and taxonomy across sub-communities.

Faithfully capturing environment dynamics requires addressing both the temporal evolution of states and the spatial encoding of scenes [3]. Long-horizon rollouts are susceptible to error accumulation, which establishes coherence as a central challenge in video prediction and policy imagination [15], [16]. Similarly, coarse or 2D-centric layouts provide insufficient geometric detail for handling challenges such as occlusion, object permanence, and geometry-aware planning. In contrast, volumetric or 3D occupancy representations such as neural fields [17] and structured voxel grids [18] provide geometric structure that better supports forecasting and control. Taken together, these points establish temporal modeling and spatial representation as core design dimensions that fundamentally influence the predictive horizon, physical fidelity, and downstream performance of embodied agents.

Several recent surveys have organized the rapidly growing literature on world models. Overall, these surveys follow two main approaches. The first is a function-oriented perspective. For example, Ding *et al.* [4] categorized relevant works based on the two core functions of understanding and prediction, while Zhu *et al.* [19] presented a framework based on the core capabilities of world models. The second approach is application-driven, focusing on specific domains such as autonomous driving. Notably, Guan *et al.* [20] and Feng *et al.* [21] provided overviews of world model techniques for autonomous driving.

To address the lack of a unified taxonomy in the context of embodied AI, this work introduces a framework centered on the three core axes of functionality, temporal modeling, and spatial representation. At the functional level, this framework distinguishes between decision-coupled and general-purpose

# 具身人工智能世界模型综述

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**摘要**——具身人工智能需要能够感知、行动并预测行为如何重塑未来世界状态的智能体。世界模型作为内部模拟器, 捕捉环境动态, 支持正向与反事实推演以辅助感知、预测和决策。本综述提出了具身人工智能中世界模型的统一框架, 具体包括: 我们形式化问题设定与学习目标, 并提出包含以下维度的三轴分类法: (1) 功能性, 决策耦合型 vs. 通用型; (2) 时序建模, 序列模拟与推理 vs. 全局差异预测; (3) 空间表征, 全局潜在向量、标记特征序列、空间潜在网络与分解渲染表征。我们系统化了机器人、自动驾驶及通用视频场景中的数据资源与评估指标, 涵盖像素预测质量、状态级理解能力及任务执行表现。此外, 我们对前沿模型进行了定量对比分析, 并提炼出关键开放性挑战: 包括统一数据集的稀缺性、需要建立既能评估像素保真度又能体现物理一致性的评价指标、模型性能与实时控制所需计算效率之间的权衡关系, 以及在实现长时域时间一致性的同时抑制误差累积这一核心建模难题。最后, 我们整理了精选参考文献, 详见<https://github.com/Li-Zn-H/AwesomeWorldModels>。

**索引术语**——世界模型、具身人工智能、时序建模、空间表示。

## 1. 引言

具身人工智能旨在使智能体能够感知复杂多模态环境、在其中执行动作, 并预测其行为将如何改变未来世界状态[1]。[2]这一能力的核心是世界模型——一个内部模拟器, 它能够捕捉环境动态, 从而支持感知、预测和决策方面的正向及反事实场景模拟[3]。[4]本研究聚焦于能够为具身智能体提供可操作预测结果的世界模型, 将其与静态场景描述符或仅能生成视觉模型(无法捕捉可控动态行为)区分开来。

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认知科学表明, 人类通过整合感官输入来构建对世界的内部模型。这些模型不仅能够预测和模拟未来事件, 还能塑造感知并指导行动[5]–[7]。基于这一观点, 早期人工智能对世界模型的研究根植于基于模型的强化学习(RL), 其中利用潜在状态转移模型来提升样本效率和规划性能[8]。Ha与Schmidhuber的开创性研究[9]将“世界模型”这一术语系统化, 并启发了Dreamer系列研究[10]–[12], 揭示了学习到的动力学机制如何驱动基于想象的策略优化。近年来, 大规模生成建模与多模态学习技术的进步, 使世界模型的研究突破了最初对策略学习的局限, 发展成为具备高保真未来预测能力的通用环境模拟器, Sora[13]和V-JEPA 2[14]等模型便是典型范例。此次扩展使功能角色、时间建模策略及空间表征实现了多样化, 但同时也导致不同子社区在术语体系和分类学方面存在不一致性。

要准确捕捉环境动态, 需要同时解决状态的时间演变和场景的空间编码问题[3]。长时域预测容易导致误差累积, 这使得连贯性成为视频预测和策略想象的核心挑战[15]。

[16]同样地, 粗略或以二维空间为中心的布局方式, 在处理遮挡效应、物体持久性以及几何感知规划等挑战时, 所提供的几何细节不足。相比之下, 神经场[17]和结构化体素网格[18]等体积化或三维空间占用表征方法, 能提供更优的几何结构, 从而有效支持预测与控制功能。综合来看, 这些观点确立了时间建模与空间表征作为核心设计维度, 它们从根本上影响着具身智能体的预测范围、物理仿真精度及下游系统性能表现。

近期多项调查系统梳理了关于世界模型的快速增长的文献。总体而言, 这些调查主要采用两种研究方法: 第一种是功能导向型视角。例如, Ding等人[4]根据理解与预测的两个核心功能对相关文献进行分类, 而Zhu等人[19]提出了一种基于世界模型核心能力的框架。第二种方法以应用为导向, 专注于自动驾驶等特定领域。值得注意的是, Guan等人[20]以及Feng等人[21]提供了自动驾驶领域全球模型技术的概述。

为解决具身人工智能领域缺乏统一分类体系的问题, 本研究提出了一套以功能、时间建模和空间表征三大核心维度为核心的框架。在功能层面, 该框架区分了决策耦合型与通用型两大类型。

models. At the temporal level, it differentiates between Sequential Simulation and Inference versus Global Difference Prediction. Finally, at the spatial level, it encompasses a range of representations from latent features to explicit geometry and neural fields. This framework offers a unified structure for organizing existing approaches and integrates standardized datasets and evaluation metrics. This structure facilitates quantitative comparisons and provides a panoramic, actionable knowledge map for future research.

Fig. 1 presents an overview of the structure and taxonomy of this paper. We begin in §II by outlining the core concepts and theoretical foundations of world models. §III introduces our three-axis taxonomy and maps representative methods onto this framework. §IV surveys the datasets and evaluation metrics used for training and assessment. §V offers a quantitative comparison of state-of-the-art models. §VI discusses open challenges and promising research directions, and §VII concludes the survey.

## II. BACKGROUND

### A. Core Concepts

As discussed in §I, a world model functions as an internal simulator of environmental dynamics. Its functionality rests on three aspects:

- **Simulation & Planning**, which uses learned dynamics to generate plausible future scenarios, allowing agents to assess potential actions through imagination without real-world interaction.
- **Temporal Evolution**, which learns how the encoded state evolves, enabling temporally consistent rollouts.
- **Spatial Representation**, which encodes scene geometry at an appropriate fidelity, using formats such as latent tokens or neural fields to provide context for control.

These three pillars provide the conceptual foundation for the taxonomy introduced in §III and are formalized in the mathematical framework that follows.

### B. Mathematical Formulation of World Models

We formalize the environment interaction as a POMDP [24]. For notational consistency, we define a null initial action  $a_0$  at  $t = 0$ , which allows the dynamics to be written uniformly. At each step  $t \geq 1$ , the agent receives an observation  $o_t$  and takes an action  $a_t$ , while the true state  $s_t$  remains unobserved. To handle this partial observability, world models infer a learned latent state  $z_t$  using a one-step filtering posterior, where the previous latent state  $z_{t-1}$  is assumed to summarize the relevant history. Finally,  $z_t$  is used to reconstruct  $o_t$ :

$$\begin{aligned} \text{Dynamics Prior:} & \quad p_\theta(z_t | z_{t-1}, a_{t-1}) \\ \text{Filtered Posterior:} & \quad q_\phi(z_t | z_{t-1}, a_{t-1}, o_t) \\ \text{Reconstruction:} & \quad p_\theta(o_t | z_t) \end{aligned} \quad (1)$$

Consistent with the Markovian structure, the joint distribution over observations and latent states factorizes as:

$$p_\theta(o_{1:T}, z_{0:T} | a_{0:T-1}) = p_\theta(z_0) \prod_{t=1}^T p_\theta(z_t | z_{t-1}, a_{t-1}) p_\theta(o_t | z_t). \quad (2)$$

To infer the latent states, we must approximate the intractable true posterior  $p_\theta(z_{0:T} | o_{1:T}, a_{0:T-1})$  with a time-factorized variational distribution:

$$q_\phi(z_{0:T} | o_{1:T}, a_{0:T-1}) = q_\phi(z_0 | o_1) \prod_{t=1}^T q_\phi(z_t | z_{t-1}, a_{t-1}, o_t), \quad (3)$$

which indeed reduces to the action-free case when ignoring  $a$  inputs. Directly maximizing the log-likelihood  $\log p_\theta(o_{1:T} | a_{0:T-1})$  is intractable. Instead, we optimize an ELBO using the approximate posterior  $q_\phi$ , which provides a tractable objective for learning the model parameters:

$$\begin{aligned} \log p_\theta(o_{1:T} | a_{0:T-1}) &= \log \int p_\theta(o_{1:T}, z_{0:T} | a_{0:T-1}) dz_{0:T} \\ &\geq \mathbb{E}_{q_\phi} \left[ \log \frac{p_\theta(o_{1:T}, z_{0:T} | a_{0:T-1})}{q_\phi(z_{0:T} | o_{1:T}, a_{0:T-1})} \right] =: \mathcal{L}(\theta, \phi). \end{aligned} \quad (4)$$

Under the assumption of Markov factorization for both  $p_\theta$  and  $q_\phi$ , this ELBO decomposes into a reconstruction objective and a KL regularization term:

$$\begin{aligned} \mathcal{L}(\theta, \phi) &= \sum_{t=1}^T \mathbb{E}_{q_\phi(z_t)} [\log p_\theta(o_t | z_t)] \\ &\quad - D_{\text{KL}}(q_\phi(z_{0:T} | o_{1:T}, a_{0:T-1}) \| p_\theta(z_{0:T} | a_{0:T-1})). \end{aligned} \quad (5)$$

Modern world models thus adopt a reconstruction-regularization training paradigm: the likelihood term  $\log p_\theta(o_t | z_t)$  encourages faithful observation prediction, and KL regularization terms align the filtered posterior  $q_\phi(z_t | z_{t-1}, a_{t-1}, o_t)$  with the dynamics prior  $p_\theta(z_t | z_{t-1}, a_{t-1})$ . Such world models can be instantiated with recurrent models [25]–[27], Transformer-based architectures [28]–[30], or diffusion-based decoders [31]–[35]. In all cases, the learned latent trajectory  $z_{1:T}$  serves as a compact, predictive memory to support downstream policy optimization, model-predictive control, and counterfactual reasoning in embodied AI.

## III. TAXONOMY

We categorize world models along three core dimensions, which provide the foundation for the subsequent analysis in this survey.

The first dimension, decision coupling, distinguishes between *Decision-Coupled* and *General-Purpose* world models. Decision-Coupled models are task-specific, learning dynamics optimized for a particular decision-making task. In contrast, General-Purpose models are task-agnostic simulators that focus on broad prediction, enabling generalization across various downstream applications.

The second dimension, temporal reasoning, delineates two distinct paradigms of prediction. *Sequential Simulation and Inference* models dynamics in an autoregressive manner, unfolding future states one step at a time. In contrast, *Global Difference Prediction* directly estimates entire future states in parallel, offering greater efficiency at the potential cost of reduced temporal coherence.



在时间维度上，该框架区分了顺序模拟与推理以及全局差异预测两种方法；在空间维度上，则涵盖了从潜在特征到显式几何结构及神经场等多种表征形式。该框架为现有研究方法提供了统一的组织结构，并整合了标准化数据集与评估指标。这种架构不仅便于进行定量比较，更为未来研究提供了全面且具有实践价值的知识图谱。

图1展示了本文的结构与分类体系概览。我们在§II中概述了世界模型的核心概念与理论基础；§III介绍了我们的三轴分类法，并将代表性方法映射至该框架；§IV综述了用于训练与评估的数据集及评估指标；§V对当前最先进的模型进行了定量比较；§VI讨论了开放性挑战与前景广阔的研究方向；§VII对全文进行了总结。

## II. 背景

### A. 核心概念

如第I节所述，世界模型作为环境动态的内部模拟器发挥作用。其功能基于三个方面：

- **模拟与规划**通过学习动态模型生成合理的未来场景，使智能体无需真实世界交互即可通过想象评估潜在行动方案。
- **时间演化**，该机制能够学习编码状态的演变规律，从而实现时间上一致的策略 rollout。
- **空间表征**通过潜在标记或神经场等格式以适当的保真度编码场景几何信息，从而为控制提供上下文信息。

这三大支柱为第III节中引入的分类法提供了概念基础，并在随后的数学框架中得到形式化表述。

**B. 世界模型的数学建模**我们将环境交互形式化为一个 POMDP[24]。

为保持符号一致性，我们在 $t=0$ 时刻定义一个初始空动作 $a_0$ ，这使得动力学过程能够统一表述。在每一步 $t \geq 1$ 中，智能体接收观测值 $o_t$ 并采取动作 $a_t$ ，而真实状态 $s_t$ 保持未被观测。为处理这种部分可观测性，世界模型通过一步滤波后验推断学习到的潜在状态 $z_t$ ，其中假设前一潜在状态 $z_{t-1}$ 能概括相关历史信息。最终， $z_t$ 用于重构 $o_t$ ：

$$\begin{aligned} \text{Dynamics Prior:} & \quad p_\theta(z_t | z_{t-1}, a_{t-1}) \\ \text{Filtered Posterior:} & \quad q_\phi(z_t | z_{t-1}, a_{t-1}, o_t) \\ \text{Reconstruction:} & \quad p_\theta(o_t | z_t) \end{aligned} \quad (1)$$

与马尔可夫结构一致，观测值与潜在状态的联合分布可分解为以下形式：

$$p_\theta(o_{1:T}, z_{0:T} | a_{0:T-1}) = p_\theta(z_0) \prod_{t=1}^T p_\theta(z_t | z_{t-1}, a_{t-1}) p_\theta(o_t | z_t). \quad (2)$$

为推断潜在状态，我们必须用时间因子化的变分分布来近似难以处理的真实后验 $p_\theta(z_{0:T} | o_{1:T}, a_{0:T-1})$ ：

$$q_\phi(z_{0:T} | o_{1:T}, a_{0:T-1}) = q_\phi(z_0 | o_1) \prod_{t=1}^T q_\phi(z_t | z_{t-1}, a_{t-1}, o_t), \quad (3)$$

这确实会在忽略一个输入时退化为无作用量的情况。直接最大化对数似然 $\log p_\theta(o_{1:T} | a_{0:T-1})$ 是不可行的。相反，我们通过近似后验 $q_\phi$ 来优化 ELBO，这为学习模型参数提供了可处理的目标函数：

$$\begin{aligned} \log p_\theta(o_{1:T} | a_{0:T-1}) &= \log \int p_\theta(o_{1:T}, z_{0:T} | a_{0:T-1}) dz_{0:T} \\ &\geq \mathbb{E}_{q_\phi} \left[ \log \frac{p_\theta(o_{1:T}, z_{0:T} | a_{0:T-1})}{q_\phi(z_{0:T} | o_{1:T}, a_{0:T-1})} \right] =: \mathcal{L}(\theta, \phi). \end{aligned} \quad (4)$$

在 $p_\theta$ 和 $q_\phi$ 均满足马尔可夫因子分解假设的情况下，该 ELBO 可分解为重构目标函数与KL正则化项：

$$\begin{aligned} \mathcal{L}(\theta, \phi) &= \sum_{t=1}^T \mathbb{E}_{q_\phi(z_t)} [\log p_\theta(o_t | z_t)] \\ &\quad - D_{\text{KL}}(q_\phi(z_{0:T} | o_{1:T}, a_{0:T-1}) \| p_\theta(z_{0:T} | a_{0:T-1})). \end{aligned} \quad (5)$$

因此，现代世界模型采用了重构-正则化训练范式：似然项 $\log p_\theta(o_t | \mathbf{z}_t)$ 鼓励对观测数据进行忠实预测，而KL正则化项则使滤波后的后验 $q_\phi(z_t | z_{t-1}, a_{t-1}, o_t)$ 与动力学先验 $p_\theta(z_t | z_{t-1}, a_{t-1})$ 保持一致。此类世界模型可通过循环模型[25]-[27]、基于Transformer的架构[28]-[30]或基于扩散的解编码器[31]-[35]进行实例化。

在所有情况下，学习到的潜在轨迹 $z_{1:T}$ 作为紧凑的预测性记忆，用于支持具身人工智能中的下游策略优化、模型预测控制和反事实推理。

## III. 分类学

我们将全球模型按三个核心维度进行分类，这些维度为本调查后续分析奠定了基础。

第一个维度是决策耦合，它区分了**决策耦合模型**与**通用型世界模型**。决策耦合模型具有任务特异性，其学习动态经过优化以适应特定决策任务；而通用型模型则是任务无关的模拟器，专注于广泛预测能力，能够实现跨多种下游应用场景的泛化。

第二个维度——时间推理，划分出两种截然不同的预测范式。**顺序模拟与推理**模型以自回归方式动态呈现未来状态，通过逐步展开实现预测。相比之下，**全局差异预测**模型能并行估算完整未来状态，虽然可能牺牲时间连贯性，但具有更高的计算效率。

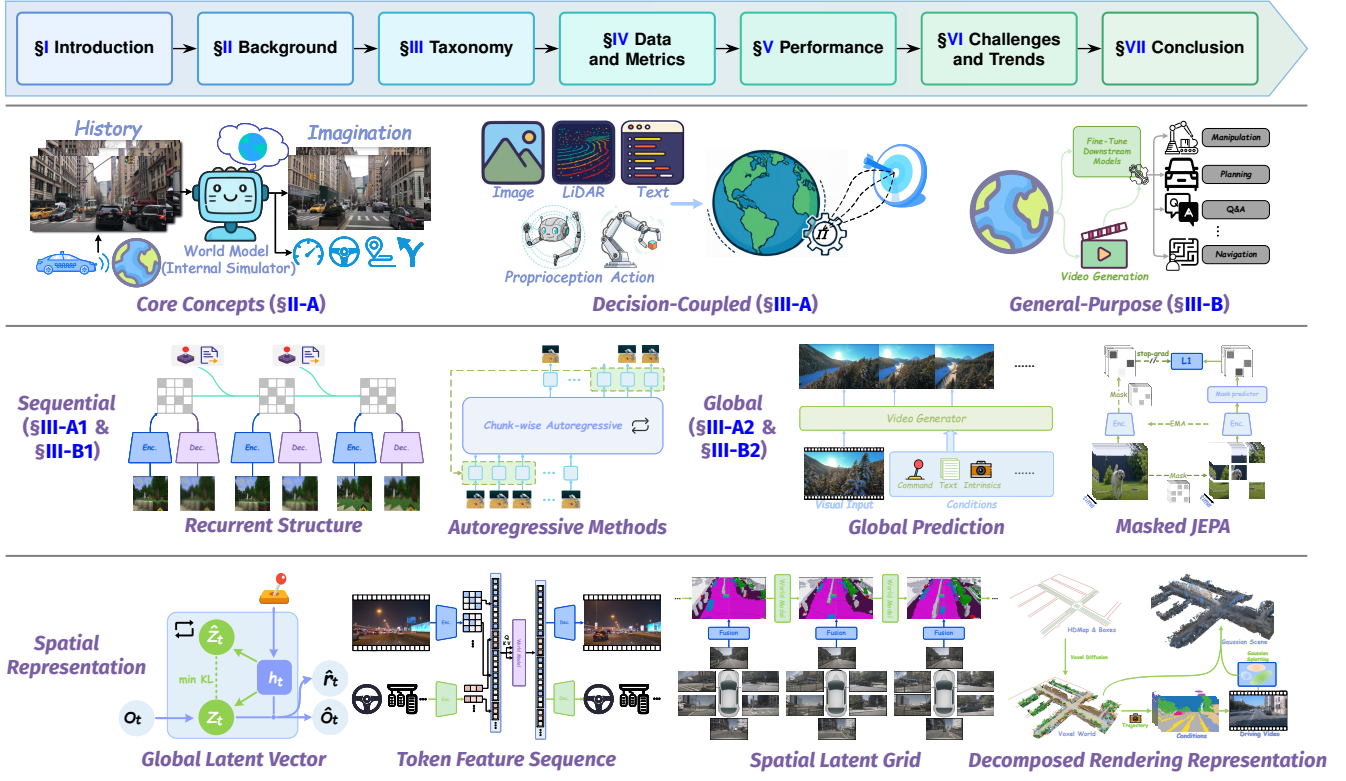


Fig. 1. Structure of this survey. The figure classifies world models along three axes and illustrates representative methods for each, providing a unified view of the field. Figure design inspired in part by [12], [14], [22], [23].

The third dimension, spatial representation, comprises four primary strategies used in current research to model spatial states:

- 1) **Global Latent Vector** representations encode complex world states into a compact vector, enabling efficient real-time computation on physical devices.
- 2) **Token Feature Sequence** representations model world states as sequences of tokens, focusing on capturing complex spatial, temporal, and cross-modal dependencies among tokens.
- 3) **Spatial Latent Grid** representations incorporate spatial inductive biases into world models by leveraging geometric priors such as Bird’s-Eye View (BEV) features or voxel grids.
- 4) **Decomposed Rendering Representation** involves decomposing 3D scenes into a set of learnable primitives, such as 3D Gaussian Splatting (3DGS) [36] or Neural Radiance Fields (NeRF) [37], and then using differentiable rendering to achieve high-fidelity novel view synthesis.

The following tables apply this taxonomy to classify representative works. Tab. I reviews approaches in robotics, while Tab. II focuses on autonomous driving. Together, they provide a roadmap for the detailed analyses in the subsequent sections.

#### A. Decision-Coupled World Models

1) *Sequential Simulation and Inference: Global Latent Vector.* Early decision-coupled world models combined sequential inference with global latent states. These approaches

primarily use Recurrent Neural Networks (RNNs) for efficient real-time and long-horizon prediction.

Ha and Schmidhuber [9] introduced an early world model that encodes observations into a latent space and employs an RNN to model dynamics for policy optimization. Building on this, PlaNet [38] introduced the Recurrent State-Space Model (RSSM), which fuses deterministic memory with stochastic components to enable robust long-horizon imagination. The successor models Dreamer, DreamerV2, and DreamerV3 [10]–[12] further advanced this formulation, inspiring a broad line of subsequent research.

Building on RSSM, several variants modify or eliminate the decoder to better capture dynamics. For example, Dreaming [110] uses contrastive learning and linear methods to mitigate state shifts, whereas DreamerPro [111] replaces the decoder with prototypes to suppress visual distractions. To further enhance robustness, HRSSM [25] was proposed, featuring a dual-branch architecture that aligns latent observations and shares information without reconstruction. Beyond architectural refinements, DisWM [112] disentangles semantic knowledge from video content, distilling it into a world model that enables cross-domain generalization.

A unifying theme across recent RSSM extensions is transferability, which reflects generalization across modalities, tasks, and embodiments for robust real-world robotics. At the representation level, PreLAR [52] learns implicit action abstractions to bridge video-pretrained representations and control fine-tuning. Similarly, Wang *et al.* [113] used optical flow as an embodiment-agnostic action representation to refine behavior-



图1. 本综述的结构。该图沿三个轴对全球模型进行分类，并展示了每种方法的代表性方案，从而为该领域提供了统一的视角。图的设计部分灵感来源于[12]、[14]、[22]、[23]。

第三个维度——空间表征，包含了当前研究中用于建模空间状态的四种主要策略：

- 1) **全局潜在向量**表示法将复杂世界状态编码为紧凑向量，从而实现在物理设备上的高效实时计算。
- 2) **令牌特征序列**表示模型将世界状态表示为令牌序列，重点捕捉令牌之间的复杂空间、时间及跨模态依赖关系。
- 3) **空间潜在网格**表示通过利用几何先验（如鸟瞰视角（BEV）特征或体素网格），将空间归纳偏置融入世界模型中。
- 4) **分解式渲染表示法**通过将三维场景分解为一系列可学习的基本元素（例如三维高斯泼溅技术 3DGS[36]或神经辐射场 NeRF[37]），随后运用可微分渲染技术实现高保真度的新视角合成。

以下表格运用该分类法对代表性作品进行归类：表I综述了机器人技术领域的研究方法，而表II则聚焦于自动驾驶领域。两者共同为后续章节的详细分析提供了路线图。

#### A. 决策耦合世界模型

1) **顺序模拟与推理：全局潜在向量**。早期的决策耦合世界模型将顺序推理与全局潜在状态相结合。这些方法主要采用循环神经网络（RNN）实现高效、实时且具有长时

程预测能力的模型。

Ha与Schmidhuber[9]提出了一个早期的全局模型，该模型将观测数据编码至潜在空间，并采用 RNN 来建模动态过程以实现策略优化。在此基础上，PlaNet[38]提出了循环状态空间模型（RSSM），该模型将确定性记忆与随机组件相结合，从而实现稳健的长时程想象能力。其后续模型Dreamer、DreamerV2和DreamerV3[10]–[12]进一步完善了这一框架，为后续大量研究奠定了基础。

基于 RSSM 模型，多种改进方案通过修改或移除解码器来更精准地捕捉动态特征。例如，Dreaming[110]采用对比学习与线性方法来缓解状态转换问题；而DreamerPro[111]则用原型表示替代解码器以抑制视觉干扰。为进一步提升鲁棒性，HRSSM[25]提出了双分支架构，该架构可对齐潜在观测结果并在无需重建的情况下共享信息。除架构优化外，DisWM[112]将视频内容的语义知识解耦，并将其提炼为世界模型，从而实现跨域泛化能力。

近期 RSSM 扩展研究的一个统一主题是可迁移性，这反映了跨模态、跨任务及跨实现形式的泛化能力，以实现稳健的现实世界机器人技术。在表征层面，PreLAR[52]通过学习隐式动作抽象来连接视频预训练表征与控制微调。类似地，Wang 等人 [113] 采用光流作为与具体实现无关的行为表征方式，以优化行为表现。



TABLE I  
A SUMMARY OF REPRESENTATIVE WORLD MODELS IN ROBOTICS AND GENERAL-PURPOSE DOMAINS.

| Paper                   | Publication | Taxonomy <sup>1</sup> | Characteristics <sup>2</sup> | Datasets Platform |       |         |      |     |            |        |      |        |          | Modality           |     |        |          |       |          | Reality <sup>4</sup> |          |
|-------------------------|-------------|-----------------------|------------------------------|-------------------|-------|---------|------|-----|------------|--------|------|--------|----------|--------------------|-----|--------|----------|-------|----------|----------------------|----------|
|                         |             |                       |                              | DMC               | Atari | RLBench | SSv2 | OXE | Meta-world | Franka | RT-1 | LIBERO | Other(s) | Total <sup>3</sup> | RGB | Action | Proprio. | Depth | Language |                      | Other(s) |
| PlaNet [38]             | ICML'19     | Dec/Seq/GLV           | RSSM                         | ✓                 |       |         |      |     |            |        |      |        |          | 1                  | ✓   | ✓      |          |       |          |                      |          |
| Dreamer [10]            | ICLR'20     | Dec/Seq/GLV           | RSSM                         | ✓                 | ✓     |         |      |     |            |        |      |        | ✓        | 3                  | ✓   | ✓      |          |       |          |                      |          |
| GLAMOR [39]             | ICLR'21     | Dec/Seq/GLV           | IDM                          | ✓                 |       |         |      |     |            |        |      |        |          | 2                  | ✓   | ✓      |          |       |          |                      |          |
| DreamerV2 [11]          | ICLR'21     | Dec/Seq/GLV           | RSSM                         | ✓                 | ✓     |         |      |     |            |        |      |        |          | 2                  | ✓   | ✓      |          |       |          |                      |          |
| TransDreamer [28]       | arXiv'22    | Dec/Seq/GLV           | TSSM                         | ✓                 | ✓     |         |      |     |            |        |      |        | ✓        | 4                  | ✓   | ✓      |          |       |          |                      |          |
| Iso-Dream [40]          | NeurIPS'22  | Dec/Seq/GLV           | IDM                          | ✓                 |       |         |      |     |            |        |      |        | ✓        | 4                  | ✓   | ✓      |          |       |          |                      |          |
| MWM [41]                | CoRL'22     | Dec/Seq/TFS           | RSSM                         | ✓                 |       | ✓       |      |     | ✓          |        |      |        |          | 3                  | ✓   | ✓      |          |       |          |                      |          |
| Inner Monologue [42]    | CoRL'22     | Dec/Seq/TFS           | CoT                          |                   |       |         |      |     |            |        |      |        | ✓        | 3                  | ✓   | ✓      |          |       | ✓        |                      | ✓        |
| DayDreamer [43]         | CoRL'22     | Dec/Seq/GLV           | RSSM                         |                   |       |         |      |     |            |        |      |        | ✓        | 4                  | ✓   | ✓      | ✓        | ✓     |          |                      | ✓        |
| TWM [29]                | ICLR'23     | Dec/Seq/TFS           | Transformer                  |                   | ✓     |         |      |     |            |        |      |        |          | 1                  | ✓   | ✓      |          |       |          |                      |          |
| IRIS [44]               | ICLR'23     | Dec/Seq/TFS           | Transformer                  |                   | ✓     |         |      |     |            |        |      |        |          | 1                  | ✓   | ✓      |          |       |          |                      |          |
| WorldDreamer [45]       | arXiv'24    | Gen/Glo/TFS           | Transformer                  |                   |       |         |      |     |            |        |      |        | ✓        | 4                  | ✓   | ✓      |          |       | ✓        |                      |          |
| Statler [46]            | ICRA'24     | Dec/Seq/TFS           | LLM                          |                   |       |         |      |     |            |        |      |        | ✓        | 2                  | ✓   | ✓      |          | ✓     | ✓        |                      | ✓        |
| Pandora [47]            | arXiv'24    | Gen/Seq/TFS           | Video Diffusion              |                   |       |         | ✓    |     |            |        |      |        | ✓        | 2                  | ✓   | ✓      |          |       | ✓        |                      |          |
| DWL [48]                | RSS'24      | Dec/Seq/GLV           | MLP                          |                   |       |         |      |     |            |        |      |        | ✓        | 4                  | ✓   | ✓      | ✓        |       |          | ✓                    | ✓        |
| RoboDreamer [49]        | ICML'24     | Dec/Glo/TFS           | IDM                          |                   |       | ✓       |      |     |            |        | ✓    |        |          | 2                  | ✓   | ✓      |          |       | ✓        | ✓                    |          |
| Genie [50]              | ICML'24     | Gen/Seq/TFS           | Transformer                  |                   |       |         |      |     |            |        | ✓    |        | ✓        | 3                  | ✓   | ✓      |          |       |          |                      |          |
| V-JEPA [51]             | TMLR'24     | Gen/Glo/TFS           | JEPA                         |                   |       |         | ✓    |     |            |        |      |        | ✓        | 6                  | ✓   | ✓      |          |       |          |                      |          |
| PreLAR [52]             | ECCV'24     | Dec/Seq/GLV           | RSSM                         |                   |       | ✓       | ✓    |     | ✓          |        |      |        |          | 3                  | ✓   | ✓      |          |       |          |                      |          |
| ManiGaussian [53]       | ECCV'24     | Dec/Seq/DRR           | 3DGS                         |                   |       | ✓       |      |     |            |        |      |        |          | 1                  | ✓   | ✓      | ✓        | ✓     | ✓        |                      |          |
| ECoT [54]               | CoRL'24     | Dec/Glo/TFS           | CoT                          |                   |       |         |      |     |            |        |      |        | ✓        | 3                  | ✓   | ✓      |          |       | ✓        |                      | ✓        |
| VidMan [55]             | NeurIPS'24  | Dec/Glo/TFS           | IDM                          |                   |       | ✓       |      |     | ✓          |        |      |        | ✓        | 4                  | ✓   | ✓      | ✓        | ✓     |          |                      |          |
| iVideoGPT [56]          | NeurIPS'24  | Gen/Seq/TFS           | Transformer                  |                   |       |         | ✓    | ✓   | ✓          | ✓      |      |        | ✓        | 6                  | ✓   | ✓      |          |       |          |                      |          |
| EnerVerse [34]          | arXiv'25    | Dec/Seq/SLG           | Video Diffusion              |                   |       |         |      |     |            |        | ✓    | ✓      | ✓        | 4                  | ✓   | ✓      |          |       | ✓        |                      | ✓        |
| GLAM [57]               | AAAI'25     | Dec/Seq/GLV           | Mamba                        |                   | ✓     |         |      |     |            |        |      |        |          | 1                  | ✓   | ✓      |          |       |          |                      |          |
| NavCoT [58]             | TPAMI'25    | Dec/Seq/TFS           | CoT                          |                   |       |         |      |     |            |        |      |        | ✓        | 4                  | ✓   | ✓      |          |       | ✓        |                      |          |
| DreamerV3 [12]          | Nature'25   | Dec/Seq/GLV           | RSSM                         | ✓                 | ✓     |         |      |     |            |        |      |        | ✓        | 8                  | ✓   | ✓      | ✓        |       |          |                      |          |
| MineWorld [59]          | arXiv'25    | Dec/Seq/TFS           | Transformer                  |                   |       |         |      |     |            |        |      |        | ✓        | 1                  | ✓   | ✓      |          |       |          |                      |          |
| DreMa [60]              | ICLR'25     | Dec/Seq/DRR           | 3DGS                         |                   |       | ✓       |      |     |            | ✓      |      |        |          | 2                  | ✓   | ✓      |          | ✓     | ✓        | ✓                    | ✓        |
| S2-SSM [61]             | arXiv'25    | Gen/Seq/TFS           | Mamba                        |                   |       |         |      |     |            |        |      |        | ✓        | 1                  | ✓   | ✓      |          |       |          | ✓                    |          |
| RLVR-World [62]         | arXiv'25    | Gen/Seq/TFS           | RLVR                         |                   |       |         |      |     |            | ✓      |      |        |          | 3                  | ✓   | ✓      |          |       | ✓        |                      |          |
| StateSpaceDiffuser [63] | arXiv'25    | Gen/Seq/TFS           | Mamba                        |                   |       |         |      |     |            |        |      |        | ✓        | 2                  | ✓   | ✓      |          |       |          |                      |          |
| DeepVerse [64]          | arXiv'25    | Gen/Seq/TFS           | DiT                          |                   |       |         |      |     |            |        |      |        | ✓        | 1                  | ✓   | ✓      |          | ✓     | ✓        | ✓                    |          |
| ORV [65]                | arXiv'25    | Gen/Glo/SLG           | DiT                          |                   |       |         |      |     |            |        | ✓    |        |          | 4                  | ✓   | ✓      |          | ✓     | ✓        | ✓                    |          |
| V-JEPA 2 [14]           | arXiv'25    | Gen/Glo/TFS           | JEPA                         |                   |       |         | ✓    |     |            | ✓      |      |        | ✓        | 15                 | ✓   | ✓      | ✓        |       | ✓        |                      | ✓        |
| NWM [66]                | CVPR'25     | Dec/Seq/TFS           | DiT                          |                   |       |         |      |     |            |        |      |        | ✓        | 6                  | ✓   | ✓      |          |       |          |                      |          |
| WorldVLA [67]           | arXiv'25    | Dec/Seq/TFS           | Transformer                  |                   |       |         |      |     |            |        | ✓    |        |          | 1                  | ✓   | ✓      |          |       | ✓        |                      |          |
| World4Omni [68]         | arXiv'25    | Gen/Seq/TFS           | VLM                          |                   |       | ✓       |      |     |            |        |      |        | ✓        | 2                  | ✓   | ✓      |          |       | ✓        | ✓                    | ✓        |
| Dyn-O [69]              | arXiv'25    | Dec/Seq/TFS           | Mamba                        |                   |       |         |      |     |            |        |      |        | ✓        | 1                  | ✓   | ✓      |          |       |          |                      |          |
| DINO-WM [70]            | ICML'25     | Dec/Seq/SLG           | Transformer                  | ✓                 |       |         |      |     |            |        |      |        | ✓        | 3                  | ✓   | ✓      | ✓        |       |          |                      |          |
| EVA [71]                | ICML'25     | Gen/Seq/TFS           | RoG                          |                   |       |         |      | ✓   |            |        | ✓    |        | ✓        | 4                  | ✓   | ✓      |          |       | ✓        |                      |          |
| AdaWorld [72]           | ICML'25     | Gen/Seq/TFS           | Video Diffusion              |                   |       |         | ✓    | ✓   |            |        |      | ✓      | ✓        | 6                  | ✓   | ✓      |          |       |          |                      |          |
| MindJourney [73]        | arXiv'25    | Gen/Seq/SLG           | VLM                          |                   |       |         |      |     |            |        |      |        | ✓        | 2                  | ✓   | ✓      |          |       | ✓        |                      |          |
| GAF [74]                | arXiv'25    | Dec/Seq/DRR           | 4DGS                         |                   |       | ✓       |      |     |            |        |      |        |          | 1                  | ✓   | ✓      | ✓        |       |          | ✓                    |          |
| Yume [75]               | arXiv'25    | Gen/Seq/TFS           | DiT                          |                   |       |         |      |     |            |        |      |        | ✓        | 1                  | ✓   | ✓      |          |       | ✓        |                      |          |
| villa-X [76]            | arXiv'25    | Dec/Glo/TFS           | IDM                          |                   |       |         | ✓    | ✓   |            |        | ✓    | ✓      |          | 5                  | ✓   | ✓      | ✓        |       | ✓        |                      | ✓        |
| AETHER [77]             | ICCV'25     | Gen/Glo/SLG           | DiT                          |                   |       |         |      |     |            |        |      |        | ✓        | 6                  | ✓   | ✓      |          | ✓     |          | ✓                    |          |
| Tesseract [78]          | ICCV'25     | Dec/Glo/SLG           | IDM                          |                   |       | ✓       | ✓    |     |            |        | ✓    |        | ✓        | 4                  | ✓   | ✓      |          | ✓     | ✓        | ✓                    |          |
| MineDreamer [79]        | IROS'25     | Dec/Seq/TFS           | CoI                          |                   |       |         |      |     |            |        |      |        | ✓        | 3                  | ✓   | ✓      |          |       | ✓        |                      |          |
| ManiGaussian++ [80]     | IROS'25     | Dec/Seq/DRR           | 3DGS                         |                   |       | ✓       |      |     |            |        |      |        | ✓        | 2                  | ✓   | ✓      | ✓        | ✓     | ✓        |                      | ✓        |

<sup>1</sup> **Taxonomy:** Abbreviations for the taxonomy categories defined in §III.

<sup>2</sup> **Characteristics:** Representative backbone or core technical approach.

<sup>3</sup> **Total:** Number of data platforms used. Underlined entries denote newly proposed or aggregated datasets.

<sup>4</sup> **Reality:** The check mark (✓) indicates validation on a physical robot.

cloned policies, facilitating transfer across diverse embodiments. SENSEI [114] distilled a Vision-Language Model (VLM) to derive semantic rewards and employed an RSSM that learns to predict and propagate these rewards internally. Under limited supervision, SR-AIF [115] exploits prior preference learning and self-revision to enable adaptive learning in sparse-reward, continuous-control settings. To mitigate the Sim-to-Real (S2R) gap, ReDRAW [116] is pretrained in simulation and adapted to the real environment using a limited amount of reward-free data, applying residual corrections to the latent dynamics. To handle mismatches, AdaWM [117] identifies discrepancies between the learned dynamics and the planner and selectively fine-tunes critical components. Other methods like WMP [118] address S2R transfer for challenging tasks, and DayDreamer [43]

demonstrated sample-efficient deployment on physical robots. To broaden transfer, FOUNDER [119] grounds representations from foundation models in the world-model state space, using temporal-distance prediction to handle flexible goals, and LUMOS [120] introduced a language-conditioned imitation framework that operates on-policy in latent space with intrinsic rewards, enabling zero-shot transfer to real-world robotics.

RSSM-based models have also been developed for autonomous driving. MILE [81] leverages offline expert data to enable imagined future states for planning. SEM2 [83] integrates semantic filtering with multi-source sampling to extract driving-relevant features and balance data distribution. Popov *et al.* [121] addressed covariate shift through a latent generative world model that realigns policies with expert states.

表1  
机器人技术与通用领域代表性世界模型概要

| 纸张                      | 出版物         | 分类学 <sup>1</sup> | 特性 <sup>2</sup> | 数据集平台 |     |         |      |     |     |     |      |        |        | 模态              |     |    |      |   | 现实 <sup>4</sup> |              |
|-------------------------|-------------|------------------|-----------------|-------|-----|---------|------|-----|-----|-----|------|--------|--------|-----------------|-----|----|------|---|-----------------|--------------|
|                         |             |                  |                 | DMC   | 阿基里 | RLBench | SSv2 | OXE | 元世界 | 泰坦卡 | RT-1 | LIBERO | 其他 (个) | 总计 <sup>3</sup> | RGB | 行动 | 感知交互 | 深 |                 | 语言<br>其他 (个) |
| PlaNet [38]             | ICML '19    | 12月/序列/ GLV      | RSSM            | ✓     |     |         |      |     |     |     |      |        |        | 1               | ✓   | ✓  |      |   |                 |              |
| 梦想家[10]                 | ICLR '20    | 12月/序列/ GLV      | RSSM            | ✓     | ✓   |         |      |     |     |     |      |        | ✓      | 3               | ✓   | ✓  |      |   |                 |              |
| GLAMOR [39]             | ICLR '21    | 12月/序列/ GLV      | IDM             | ✓     | ✓   |         |      |     |     |     |      |        |        | 2               | ✓   | ✓  |      |   |                 |              |
| DreamerV2 [11]          | ICLR '21    | 12月/序列/ GLV      | RSSM            | ✓     | ✓   |         |      |     |     |     |      |        |        | 2               | ✓   | ✓  |      |   |                 |              |
| TransDreamer [28]       | arXiv '22   | 12月/序列/ GLV      | TSSM            | ✓     | ✓   |         |      |     |     |     |      |        | ✓      | 4               | ✓   | ✓  |      |   |                 |              |
| 异梦[40]                  | NeurIPS '22 | 12月/序列/ GLV      | IDM             | ✓     |     |         |      |     |     |     |      |        | ✓      | 4               | ✓   | ✓  |      |   |                 |              |
| MWM [41]                | CoRL '22    | 12月/序列/时间序列分析    | RSSM            | ✓     |     | ✓       |      |     | ✓   |     |      |        |        | 3               | ✓   | ✓  |      |   |                 |              |
| 内心独白[42]                | CoRL '22    | 12月/序列/时间序列分析    | CoT             |       |     |         |      |     |     |     |      |        | ✓      | 3               | ✓   | ✓  |      | ✓ |                 | ✓            |
| 白日梦者[43]                | CoRL '22    | 12月/序列/ GLV      | RSSM            |       |     |         |      |     |     |     |      |        | ✓      | 4               | ✓   | ✓  | ✓    | ✓ |                 | ✓            |
| TWM [29]                | ICLR '23    | 12月/序列/时间序列分析    | Transformer     |       | ✓   |         |      |     |     |     |      |        |        | 1               | ✓   | ✓  |      |   |                 |              |
| IRIS [44]               | ICLR '23    | 12月/序列/时间序列分析    | Transformer     |       | ✓   |         |      |     |     |     |      |        |        | 1               | ✓   | ✓  |      |   |                 |              |
| WorldDreamer [45]       | arXiv '24   | 基因组/表型/表型组合      | Transformer     |       |     |         |      |     |     |     |      |        | ✓      | 4               | ✓   | ✓  |      |   | ✓               |              |
| 斯塔特勒[46]                | ICRA '24    | Dec/Seq/TFS      | LLM             |       |     |         |      |     |     |     |      |        | ✓      | 2               | ✓   | ✓  |      | ✓ |                 | ✓            |
| 潘多拉[47]                 | arXiv '24   | 基因组/测序/转录组测序     | 视频传播            |       |     |         | ✓    |     |     |     |      |        | ✓      | 2               | ✓   |    |      | ✓ |                 |              |
| DWL [48]                | RSS '24     | 12月/序列/ GLV      | MLP             |       |     |         |      |     |     |     |      |        | ✓      | 4               |     | ✓  | ✓    |   | ✓               | ✓            |
| RoboDreamer [49]        | ICML '24    | 12月/全球/过渡期       | IDM             |       |     | ✓       |      |     |     |     | ✓    |        |        | 2               | ✓   | ✓  |      |   | ✓               |              |
| 精灵[50]                  | ICML '24    | 基因组/测序/转录组测序     | Transformer     |       |     |         |      |     |     |     | ✓    |        | ✓      | 3               | ✓   | ✓  |      |   | ✓               | ✓            |
| V-JEPA [51]             | TMLR '24    | 基因组/表型/表型组合      | JEPA            |       |     |         | ✓    |     |     |     |      |        | ✓      | 6               | ✓   |    |      |   |                 |              |
| PreLAR [52]             | ECCV '24    | 12月/序列/ GLV      | RSSM            |       |     | ✓       | ✓    |     | ✓   |     |      |        |        | 3               | ✓   | ✓  |      |   |                 |              |
| ManiGaussian [53]       | ECCV '24    | 12月/序列/ DRR      | 3DGS            |       |     | ✓       |      |     |     |     |      |        |        | 1               | ✓   | ✓  | ✓    | ✓ | ✓               |              |
| ECoT [54]               | CoRL '24    | 12月/全球/过渡期       | CoT             |       |     |         |      |     |     |     |      |        | ✓      | 3               | ✓   | ✓  |      |   | ✓               | ✓            |
| VidMan [55]             | NeurIPS '24 | 12月/全球/过渡期       | IDM             |       |     | ✓       |      |     | ✓   |     |      |        | ✓      | 4               | ✓   | ✓  | ✓    |   | ✓               |              |
| iVideoGPT [56]          | NeurIPS '24 | 基因组/测序/转录组测序     | Transformer     |       |     |         | ✓    |     | ✓   | ✓   |      |        |        | 6               | ✓   | ✓  |      |   |                 |              |
| EnerVerse [34]          | arXiv '25   | 12月/序列/ SLG      | 视频扩散            |       |     |         |      |     |     |     | ✓    | ✓      | ✓      | 4               | ✓   | ✓  |      |   | ✓               | ✓            |
| GLAM [57]               | AAAI '25    | 12月/序列/ GLV      | 曼巴蛇             |       | ✓   |         |      |     |     |     |      |        |        | 1               | ✓   | ✓  |      |   |                 |              |
| NavCoT [58]             | TPAMI '25   | 12月/序列/时间序列分析    | CoT             |       |     |         |      |     |     |     |      |        | ✓      | 4               | ✓   | ✓  |      |   | ✓               |              |
| DreamerV3 [12]          | 《自然》2025年   | 12月/序列/ GLV      | RSSM            | ✓     | ✓   |         |      |     |     |     |      |        | ✓      | 8               | ✓   | ✓  | ✓    |   |                 |              |
| MineWorld [59]          | arXiv '25   | 12月/序列/时间序列分析    | Transformer     |       |     |         |      |     |     |     |      |        | ✓      | 1               | ✓   | ✓  |      |   |                 |              |
| DreMa [60]              | ICLR '25    | 12月/序列/ DRR      | 3DGS            |       |     | ✓       |      |     |     | ✓   |      |        |        | 2               | ✓   | ✓  |      | ✓ | ✓               | ✓            |
| S2-SSM [61]             | arXiv '25   | 基因组/测序/转录组测序     | 曼巴蛇             |       |     |         |      |     |     |     |      |        | ✓      | 1               | ✓   |    |      |   | ✓               |              |
| RLVR-World [62]         | arXiv '25   | 基因组/测序/转录组测序     | RLVR            |       |     |         |      |     |     | ✓   |      |        | ✓      | 3               | ✓   | ✓  |      |   | ✓               |              |
| StateSpaceDiffuser [63] | arXiv '25   | 基因组/测序/转录组测序     | 曼巴蛇             |       |     |         |      |     |     |     |      |        | ✓      | 2               | ✓   | ✓  |      |   |                 |              |
| DeepVerse [64]          | arXiv '25   | 基因组/测序/转录组测序     | DiT             |       |     |         |      |     |     |     |      |        | ✓      | 1               | ✓   |    |      | ✓ | ✓               | ✓            |
| ORV [65]                | arXiv '25   | 基因型/表型/ SLG      | DiT             |       |     |         |      |     |     | ✓   |      |        | ✓      | 4               | ✓   | ✓  |      | ✓ | ✓               | ✓            |
| V-JEPA 2 [14]           | arXiv '25   | 基因型/表型/表型组合      | JEPA            |       |     |         | ✓    |     |     | ✓   |      |        | ✓      | 15              | ✓   | ✓  | ✓    |   | ✓               | ✓            |
| NWM [66]                | CVPR '25    | 12月/序列/时间序列分析    | DiT             |       |     |         |      |     |     |     |      |        | ✓      | 6               | ✓   | ✓  |      |   |                 |              |
| WorldVLA [67]           | arXiv '25   | 12月/序列/时间序列分析    | Transformer     |       |     |         |      |     |     |     |      | ✓      |        | 1               | ✓   | ✓  |      |   | ✓               |              |
| World4Omni [68]         | arXiv '25   | 基因组/测序/转录组测序     | VLM             |       |     | ✓       |      |     |     |     |      |        | ✓      | 2               | ✓   | ✓  |      |   | ✓               | ✓            |
| Dyn-O [69]              | arXiv '25   | 12月/序列/时间序列分析    | 曼巴蛇             |       |     |         |      |     |     |     |      |        | ✓      | 1               | ✓   | ✓  |      |   |                 |              |
| DINO-WM [70]            | ICML '25    | 12月/序列/ SLG      | Transformer     | ✓     |     |         |      |     |     |     |      |        | ✓      | 3               | ✓   | ✓  | ✓    |   |                 |              |
| EVA [71]                | ICML '25    | 基因组/测序/转录组测序     | RoG             |       |     |         |      | ✓   |     |     | ✓    |        | ✓      | 4               | ✓   | ✓  |      |   | ✓               |              |
| AdaWorld [72]           | ICML '25    | 基因组/测序/转录组测序     | 视频扩散            |       |     |         | ✓    | ✓   |     |     |      | ✓      | ✓      | 6               | ✓   | ✓  |      |   |                 |              |
| MindJourney [73]        | arXiv '25   | 基因组/测序/ SLG      | VLM             |       |     |         |      |     |     |     |      |        | ✓      | 2               | ✓   | ✓  |      |   | ✓               |              |
| GAF [74]                | arXiv '25   | 12月/序列/ DRR      | 4DGS            |       | ✓   |         |      |     |     |     |      |        |        | 1               | ✓   | ✓  | ✓    |   |                 | ✓            |
| 梦[75]                   | arXiv '25   | 基因组/测序/转录组测序     | DiT             |       |     |         |      |     |     |     |      |        | ✓      | 1               | ✓   | ✓  |      |   | ✓               |              |
| Villa-X[76]             | arXiv '25   | 12月/全球/过渡期       | IDM             |       |     |         | ✓    | ✓   |     |     | ✓    | ✓      |        | 5               | ✓   | ✓  | ✓    |   | ✓               | ✓            |
| AETHER [77]             | ICCV '25    | 基因型/表型/ SLG      | DiT             |       |     |         |      |     |     |     |      |        | ✓      | 6               | ✓   | ✓  |      | ✓ | ✓               |              |
| Tesseract [78]          | ICCV '25    | Dec/Glo/SLG      | IDM             |       | ✓   | ✓       |      |     |     |     | ✓    |        | ✓      | 4               | ✓   |    |      | ✓ | ✓               | ✓            |
| MineDreamer [79]        | IROS '25    | 12月/序列/时间序列分析    | CoI             |       |     |         |      |     |     |     |      |        | ✓      | 3               | ✓   | ✓  |      |   | ✓               | ✓            |
| ManiGaussian++ [80]     | IROS '25    | 12月/序列/ DRR      | 3DGS            |       | ✓   |         |      |     |     |     |      |        | ✓      | 2               | ✓   | ✓  | ✓    | ✓ | ✓               | ✓            |

<sup>1</sup> 分类学: 第三节中定义的分类学类别的缩写。  
<sup>2</sup> 特性: 具有代表性的核心技术方案。  
<sup>3</sup> 总计: 所使用的数据集平台数量。带下划线的条目表示新提出的或聚合的数据集。  
<sup>4</sup> 实际情况: 勾选标记 (✓) 表示在物理机器人上完成验证。

克隆策略可实现跨不同实施形态的迁移。SENSEI[114]提出了一种视觉-语言模型（VLM）来推导语义奖励，并采用了一种能够内部预测和传播这些奖励的 RSSM。在有限监督下，SR-AIF[115]通过利用先验偏好学习和自我修正机制，在稀疏奖励、连续控制场景中实现自适应学习。为缓解仿真到真实（S2R）迁移差距，REDRAW[116]在仿真环境中进行预训练，并通过少量无奖励数据对真实环境进行适配，对潜在动力学进行残差校正。针对参数失配问题，AdaWM[117]通过识别学习动力学与规划器之间的差异，对关键组件进行选择性的微调。其他方法如 WMP[118]针对复杂任务解决S2R迁移问题，DayDreamer[43]则在物理机器人上实现了样本高效部署。为拓展迁移能力，创始人

[119]将表征从基础模型中提取到世界模型状态空间，通过时间距离预测处理灵活目标；LUMOS[120]提出了一种基于语言条件的模仿框架，该框架在潜在空间中采用策略内奖励进行操作策略，实现了对真实机器人领域的零样本迁移。

基于 RSSM 的模型也已被开发用于自动驾驶领域。MILE[81]利用离线专家数据来实现规划所需的未来状态想象。SEM2[83]将语义过滤与多源采样相结合，以提取驾驶相关特征并平衡数据分布。Popov 等人 [121] 通过一个潜在生成世界模型来解决协变量偏移问题，该模型可使策略与专家状态重新对齐。



TABLE II  
A SUMMARY OF REPRESENTATIVE WORLD MODELS FOR THE AUTONOMOUS DRIVING DOMAIN.

| Paper                  | Publication | Taxonomy <sup>1</sup> | Characteristics <sup>2</sup> | Datasets Platform |          |        |       |       |        |          |                    | Input Modality |        |     |       |           |          |           |          |
|------------------------|-------------|-----------------------|------------------------------|-------------------|----------|--------|-------|-------|--------|----------|--------------------|----------------|--------|-----|-------|-----------|----------|-----------|----------|
|                        |             |                       |                              | CARLA             | nuScenes | nuPlan | Waymo | Occ3D | OpenDV | Other(s) | Total <sup>3</sup> | RGB            | Motion | Map | LiDAR | Bound box | Language | Occupancy | Other(s) |
| MILE [81]              | NeurIPS'22  | Dec/Seq/GLV           | RSSM                         | ✓                 |          |        |       |       |        |          | 1                  | ✓              | ✓      | ✓   |       |           |          |           |          |
| Copilot4D [82]         | ICLR'24     | Gen/Seq/SLG           | Video Diffusion              |                   | ✓        |        |       |       |        | ✓        | 3                  |                | ✓      |     | ✓     |           |          |           |          |
| SEM2 [83]              | TITS'24     | Dec/Seq/GLV           | RSSM                         | ✓                 |          |        |       |       |        |          | 1                  | ✓              | ✓      | ✓   | ✓     |           |          |           |          |
| MagicDrive3D [84]      | arXiv'24    | Gen/Glo/DRR           | 3DGS                         |                   | ✓        |        |       |       |        |          | 1                  | ✓              | ✓      | ✓   |       | ✓         | ✓        |           | ✓        |
| OccSora [85]           | arXiv'24    | Gen/Glo/SLG           | Diffusion                    |                   | ✓        |        |       | ✓     |        |          | 2                  |                | ✓      |     |       |           |          | ✓         |          |
| Delphi [86]            | arXiv'24    | Gen/Seq/SLG           | Video Diffusion              |                   | ✓        |        |       |       |        |          | 1                  | ✓              |        | ✓   |       | ✓         | ✓        |           |          |
| DriveWorld [87]        | CVPR'24     | Dec/Seq/SLG           | RSSM                         |                   | ✓        |        |       |       |        | ✓        | 2                  | ✓              | ✓      |     |       |           | ✓        | ✓         |          |
| Drive-WM [88]          | CVPR'24     | Dec/Glo/SLG           | Video Diffusion              |                   | ✓        |        |       |       |        |          | 1                  | ✓              | ✓      | ✓   |       | ✓         | ✓        |           | ✓        |
| ViDAR [89]             | CVPR'24     | Gen/Seq/SLG           | Transformer                  |                   | ✓        |        |       |       |        |          | 1                  | ✓              | ✓      |     | ✓     |           |          |           |          |
| GenAD [90]             | CVPR'24     | Gen/Seq/TFS           | Video Diffusion              |                   | ✓        | ✓      | ✓     |       | ✓      | ✓        | 4                  | ✓              | ✓      |     |       |           | ✓        |           |          |
| OccLLaMA [18]          | arXiv'24    | Dec/Seq/SLG           | Transformer                  |                   | ✓        |        |       | ✓     |        | ✓        | 3                  | ✓              | ✓      |     |       |           | ✓        | ✓         |          |
| DriveDreamer [91]      | ECCV'24     | Dec/Seq/SLG           | GRU                          |                   | ✓        |        |       |       |        |          | 1                  | ✓              | ✓      | ✓   |       | ✓         | ✓        |           |          |
| GenAD [92]             | ECCV'24     | Dec/Seq/SLG           | GRU                          |                   | ✓        |        |       |       |        |          | 1                  | ✓              |        |     |       |           |          |           |          |
| OccWorld [93]          | ECCV'24     | Dec/Seq/SLG           | Transformer                  |                   | ✓        |        |       | ✓     |        |          | 2                  | ✓              | ✓      |     | ✓     |           |          | ✓         |          |
| DOVE [94]              | arXiv'24    | Gen/Seq/SLG           | DiT                          |                   | ✓        |        |       | ✓     |        |          | 2                  | ✓              | ✓      |     |       |           |          | ✓         |          |
| TOKEN [95]             | CoRL'24     | Dec/Glo/TFS           | Transformer                  |                   | ✓        |        |       |       |        | ✓        | 2                  | ✓              | ✓      | ✓   |       |           | ✓        |           |          |
| Vista [96]             | NeurIPS'24  | Gen/Seq/SLG           | Video Diffusion              |                   | ✓        |        | ✓     |       | ✓      | ✓        | 4                  | ✓              | ✓      |     |       |           | ✓        |           |          |
| DriveDreamer-2 [97]    | AAAI'25     | Gen/Glo/SLG           | Video Diffusion              |                   | ✓        |        |       |       |        |          | 1                  | ✓              | ✓      | ✓   |       | ✓         | ✓        |           |          |
| DTT [98]               | arXiv'25    | Dec/Seq/DRR           | Transformer                  |                   | ✓        |        |       | ✓     |        |          | 2                  | ✓              | ✓      |     |       |           | ✓        | ✓         |          |
| DynamicCity [99]       | ICLR'25     | Gen/Glo/SLG           | DiT                          | ✓                 | ✓        |        | ✓     | ✓     |        |          | 4                  | ✓              | ✓      |     |       |           | ✓        | ✓         |          |
| LidarDM [100]          | ICRA'25     | Gen/Seq/SLG           | Diffusion                    |                   |          |        | ✓     |       |        | ✓        | 3                  |                | ✓      | ✓   |       |           |          |           |          |
| FutureSightDrive [101] | arXiv'25    | Dec/Seq/TFS           | CoT(VLM)                     |                   | ✓        |        |       |       |        | ✓        | 3                  | ✓              | ✓      |     |       |           |          |           |          |
| GEM [102]              | CVPR'25     | Gen/Seq/SLG           | Video Diffusion              |                   | ✓        |        |       |       | ✓      |          | 1                  | ✓              | ✓      |     |       |           |          |           | ✓        |
| GaussianWorld [103]    | CVPR'25     | Gen/Seq/DRR           | Transformer                  |                   | ✓        |        |       |       |        | ✓        | 2                  | ✓              | ✓      |     |       |           |          |           |          |
| MaskGWM [104]          | CVPR'25     | Gen/Glo/TFS           | DiT                          |                   | ✓        |        | ✓     |       | ✓      |          | 3                  | ✓              | ✓      |     |       |           |          | ✓         |          |
| DriveDreamer4D [105]   | CVPR'25     | Gen/Glo/DRR           | 4DGS                         |                   | ✓        | ✓      |       |       | ✓      | 3        | ✓                  |                | ✓      | ✓   | ✓     | ✓         | ✓        |           | ✓        |
| ReconDreamer [106]     | CVPR'25     | Gen/Glo/DRR           | 3DGS                         |                   | ✓        |        | ✓     |       |        | ✓        | 3                  | ✓              | ✓      | ✓   | ✓     | ✓         | ✓        |           |          |
| WoTE [107]             | ICCV'25     | Dec/Seq/SLG           | Transformer                  | ✓                 |          | ✓      |       |       |        |          | 2                  | ✓              | ✓      |     | ✓     |           |          |           |          |
| HERMES [108]           | ICCV'25     | Gen/Glo/SLG           | LLM                          |                   | ✓        |        |       |       |        | ✓        | 4                  | ✓              | ✓      |     |       |           |          | ✓         |          |
| InfiniCube [22]        | ICCV'25     | Gen/Seq/DRR           | 3DGS                         |                   |          |        | ✓     |       |        |          | 1                  | ✓              | ✓      | ✓   |       | ✓         | ✓        |           | ✓        |
| DriveVerse [109]       | ACMMM'25    | Gen/Seq/TFS           | DiT                          |                   | ✓        |        | ✓     |       |        |          | 2                  | ✓              | ✓      |     |       |           | ✓        |           | ✓        |

<sup>1</sup> **Taxonomy:** Abbreviations for the taxonomy categories defined in §III.

<sup>2</sup> **Characteristics:** Representative backbone or core technical approach.

<sup>3</sup> **Total:** Number of data platforms used. Underlined entries denote newly proposed or aggregated datasets.

For safety, VL-SAFE [122] supervises world models using safety scores derived from a VLM to generate safe trajectories. Finally, CALL [123] extended the RSSM framework to Multi-Agent RL by introducing ego-centric information sharing to enhance planning capabilities.

In contrast to RSSM, TransDreamer [28] introduced a Transformer State-Space Model (TSSM) that replaces the recurrent core in Dreamer, thereby substantially enhancing its capacity to capture long-horizon dependencies. The complementary OSVI-WM [124] employs a causal Transformer for one-shot Imitation Learning (IL), autoregressively predicting the future latent trajectory and decoding it into physical waypoints for robot control.

Some approaches continue to employ RNNs to capture temporal dependencies. On the modeling side, RWM [125] introduced a dual-autoregressive, domain-agnostic neural simulator for long-horizon prediction. X-MOBILITY [126], in contrast, disentangles modeling from policy learning, using multi-head decoders for large-scale pretraining followed by supervised fine-tuning to derive strong policies. For humanoid locomotion, DWL [48] and WMR [127] adopted end-to-end (E2E) frameworks. These frameworks reconstruct states from partial observations using either denoising or a gradient-blocked state estimator, which enables zero-shot transfer across complex real-world terrains.

Recently, State Space Models (SSMs), exemplified by

Mamba, have emerged as alternatives to RNNs and Transformers, combining linear-time complexity with long-horizon modeling capacity. Building on this, GLAM [57] improves both fidelity and efficiency via a Mamba-based parallel framework that integrates global and local modules to capture contextual and fine-grained dynamics.

Beyond forward temporal modeling, Inverse Dynamics Modeling (IDM) is a key paradigm in world model construction. An IDM infers the actions required to transition between an initial and a target state. Agrawal *et al.* [128] integrated a forward model with an IDM for multi-step prediction, establishing the basis for subsequent research. More recent work includes GLAMOR [39], which trains an object-conditioned IDM to predict the actions necessary to reach a specified target. In Dreamer-style agents, Iso-Dream [40] leverages IDM to decompose the world model into controllable and uncontrollable components, using the rollout of uncontrollable states to guide policy learning.

**Token Feature Sequence.** The Token Feature Sequence paradigm centers on modeling dependencies among discrete tokens. This representation supports causal inference, multi-modal integration, and the reuse of Large Language Model (LLM).

Recent RSSM-centric studies have begun to exploit token-level dependencies to strengthen representation learning and

表 II  
A 自动驾驶领域代表性世界模型概览

| 纸张                     | 出版物         | 分类学 <sup>1</sup> | 特性 <sup>2</sup> | 数据集平台 |     |     |       |       |      |        | 输入模式            |     |    |    |      |     |    |    |
|------------------------|-------------|------------------|-----------------|-------|-----|-----|-------|-------|------|--------|-----------------|-----|----|----|------|-----|----|----|
|                        |             |                  |                 | CARLA | 无场景 | 核计划 | Waymo | Occ3D | 开放DV | 其他 (个) | 总计 <sup>3</sup> | RGB | 运动 | 地图 | 激光雷达 | 限定框 | 语言 | 占有 |
| MILE [81]              | NeurIPS '22 | 12月/序列/ GLV      | RSSM            | ✓     |     |     |       |       |      |        | 1               | ✓   | ✓  | ✓  |      |     |    |    |
| Copilot4D [82]         | ICLR '24    | 基因组/测序/ SLG      | 视频扩散            |       | ✓   |     |       |       |      | ✓      | 3               |     | ✓  |    | ✓    |     |    |    |
| SEM2 [83]              | TITS '24    | 12月/序列/ GLV      | RSSM            | ✓     |     |     |       |       |      |        | 1               | ✓   | ✓  | ✓  | ✓    |     |    |    |
| MagicDrive3D [84]      | arXiv '24   | 基因型/表型/ DRR      | 3DGS            |       | ✓   |     |       |       |      |        | 1               | ✓   | ✓  | ✓  |      | ✓   | ✓  | ✓  |
| OccSora [85]           | arXiv '24   | 基因型/表型/ SLG      | 扩散              |       | ✓   |     |       | ✓     |      |        | 2               |     | ✓  |    |      |     | ✓  |    |
| Delphi [86]            | arXiv '24   | 基因组/测序/ SLG      | 视频扩散            |       | ✓   |     |       |       |      |        | 1               | ✓   |    | ✓  |      | ✓   | ✓  |    |
| DriveWorld [87]        | CVPR '24    | 12月/序列/ SLG      | RSSM            |       | ✓   |     |       |       |      | ✓      | 2               | ✓   | ✓  |    |      |     | ✓  | ✓  |
| Drive-WM [88]          | CVPR '24    | Dec/Glo/SLG      | 视频扩散            |       | ✓   |     |       |       |      |        | 1               | ✓   | ✓  | ✓  |      | ✓   | ✓  | ✓  |
| ViDAR [89]             | CVPR '24    | 基因组/测序/ SLG      | Transformer     |       | ✓   |     |       |       |      |        | 1               | ✓   | ✓  |    | ✓    |     |    |    |
| GenAD [90]             | CVPR '24    | 基因组/测序/转录组测序     | 视频扩散            |       | ✓   | ✓   | ✓     |       | ✓    | ✓      | 4               | ✓   | ✓  |    |      |     | ✓  |    |
| OccLLaMa [18]          | arXiv '24   | 12月/序列/ SLG      | Transformer     |       | ✓   |     |       | ✓     |      | ✓      | 3               | ✓   | ✓  |    |      |     | ✓  | ✓  |
| DriveDreamer [91]      | ECCV '24    | 12月/序列/ SLG      | GRU             |       | ✓   |     |       |       |      |        | 1               | ✓   | ✓  | ✓  |      | ✓   | ✓  |    |
| GenAD [92]             | ECCV '24    | 12月/序列/ SLG      | GRU             |       | ✓   |     |       |       |      |        | 1               | ✓   |    |    |      |     |    |    |
| OccWorld [93]          | ECCV '24    | 12月/序列/ SLG      | Transformer     |       | ✓   |     |       | ✓     |      |        | 2               | ✓   | ✓  |    | ✓    |     |    | ✓  |
| DOME [94]              | arXiv '24   | 基因组/测序/ SLG      | DiT             |       | ✓   |     |       | ✓     |      |        | 2               | ✓   | ✓  |    |      |     |    | ✓  |
| 令牌[95]                 | CoRL '24    | 12月/全球/过渡期       | Transformer     |       | ✓   |     |       |       |      | ✓      | 2               | ✓   | ✓  | ✓  |      |     | ✓  |    |
| Vista [96]             | NeurIPS '24 | 基因组/测序/ SLG      | 视频扩散            |       | ✓   |     | ✓     |       | ✓    | ✓      | 4               | ✓   | ✓  |    |      |     | ✓  |    |
| DriveDreamer-2 [97]    | AAAI '25    | 基因型/表型/ SLG      | 视频扩散            |       | ✓   |     |       |       |      |        | 1               | ✓   | ✓  | ✓  |      | ✓   | ✓  |    |
| DTT [98]               | arXiv '25   | 12月/序列/ DRR      | Transformer     |       | ✓   |     |       | ✓     |      |        | 2               | ✓   | ✓  |    |      |     | ✓  | ✓  |
| DynamicCity [99]       | ICLR '25    | 基因型/表型/ SLG      | DiT             | ✓     | ✓   |     | ✓     | ✓     |      |        | 4               | ✓   | ✓  |    |      |     | ✓  | ✓  |
| LidarDM [100]          | ICRA '25    | 基因组/测序/ SLG      | 扩散              |       |     |     | ✓     |       |      | ✓      | 3               |     | ✓  | ✓  |      |     |    |    |
| FutureSightDrive [101] | arXiv '25   | 12月/序列/时间序列分析    | CoT(VLM)        |       | ✓   |     |       |       |      | ✓      | 3               | ✓   | ✓  |    |      |     | ✓  |    |
| GEM [102]              | CVPR '25    | 基因组/测序/ SLG      | 视频扩散            |       | ✓   |     |       |       | ✓    |        | 1               | ✓   | ✓  |    |      |     |    | ✓  |
| GaussianWorld [103]    | CVPR '25    | 基因组/测序/ DRR      | Transformer     |       | ✓   |     |       |       |      | ✓      | 2               | ✓   | ✓  |    |      |     |    |    |
| MaskGWM [104]          | CVPR '25    | 基因型/表型/表型组合      | DiT             |       | ✓   |     | ✓     |       | ✓    |        | 3               | ✓   | ✓  |    |      |     | ✓  |    |
| DriveDreamer4D [105]   | CVPR '25    | 基因型/表型/ DRR      | 4DGS            |       | ✓   | ✓   |       |       | ✓    | ✓      | 3               | ✓   | ✓  | ✓  | ✓    | ✓   | ✓  | ✓  |
| ReconDreamer [106]     | CVPR '25    | 基因型/表型/ DRR      | 3DGS            |       | ✓   |     | ✓     |       |      | ✓      | 3               | ✓   | ✓  | ✓  | ✓    | ✓   |    |    |
| WoTE [107]             | ICCV '25    | 12月/序列/ SLG      | Transformer     | ✓     |     | ✓   |       |       |      |        | 2               | ✓   | ✓  |    | ✓    |     |    |    |
| 赫尔墨斯[108]              | ICCV '25    | 基因型/表型/ SLG      | LLM             |       | ✓   |     |       |       |      | ✓      | 4               | ✓   | ✓  |    |      |     | ✓  |    |
| InfiniCube [22]        | ICCV '25    | 基因组/测序/ DRR      | 3DGS            |       |     |     | ✓     |       |      |        | 1               | ✓   | ✓  | ✓  |      | ✓   |    | ✓  |
| Driverse [109]         | ACMMM '25   | 基因组/测序/转录组测序     | DiT             |       | ✓   |     | ✓     |       |      |        | 2               | ✓   | ✓  |    |      |     | ✓  |    |

<sup>1</sup> 分类学: 第三节中定义的分类学类别的缩写。  
<sup>2</sup> 特征: 具有代表性的核心技术方案。  
<sup>3</sup> 总计: 所使用的数据集平台数量。带下划线的条目表示新提出的或聚合的数据集。

出于安全考虑，VL-SAFE[122]通过基于 VLM 的安全评分对世界模型进行监督，以生成安全轨迹。最后，CALL [123]通过引入以自我为中心的信息共享机制，将 RSSM 框架扩展至多智能体强化学习领域，从而提升规划能力。

与 RSSM 不同，TransDreamer[28]引入了Transformer状态空间模型（TSSM），取代了Dreamer中的循环核心结构，从而显著提升了其捕捉长期依赖关系的能力。其互补模型 OSVI -WM[124]采用因果Transformer实现单次模仿学习（IL），通过自回归方式预测未来的潜在轨迹，并将其解码为用于机器人控制的物理路径点。

部分研究方法仍采用循环神经网络（RNN）来捕捉时间依赖性。在建模层面，RWM[125]提出了一种双自回归、领域无关的神经模拟器用于长时域预测。相比之下，X-mobility[126]通过采用多头解码器进行大规模预训练，再结合监督微调来解耦建模与策略学习，从而推导出强策略。针对人形机器人运动学问题，DWL[48]与 WMR[127]采用了端到端（E2E）框架。这些框架通过去噪或梯度块状态估计器从部分观测数据中重构状态，从而实现在复杂真实地形环境中的零样本迁移能力。

近年来，以Mamba为代表的状态空间模型（SSMs）逐渐成

为RNN和Transformer的替代方案，其优势在于兼具线性时间复杂度与长时程建模能力。基于此，GLAM[57]通过采用Mamba架构的并行框架，将全局模块与局部模块有机结合以捕捉上下文信息和细粒度动态特征，从而显著提升了模型的保真度与计算效率。

除了前向时间建模之外，逆向动力学建模（IDM）是世界模型构建中的关键方法。IDM 能够推断出在初始状态与目标状态之间转换所需的动作。Agrawal 等人 [128] 该研究将前向模型与多步预测 IDM 相结合，为后续研究奠定了基础。近期的研究包括glamor[39]，该方法通过训练对象条件 IDM 来预测达成指定目标所需的动作。在Dreamer风格智能体中，Iso-Dream[40]利用 IDM 将世界模型分解为可控与不可控两部分，并通过不可控状态的策略展开来指导策略学习。

**词元特征序列。**词元特征序列范式的核心在于对离散词元之间依赖关系的建模。该表征方法能够支持因果推断、多模态整合以及大型语言模型（LLM）的复用。

近期以 RSSM 为中心的研究开始利用标记层级依赖关系来增强表示学习效果。

temporal reasoning. For example, MWM [41] decouples visual tokens from RSSM-based dynamics via a masked autoencoder, improving both performance and data efficiency. NavMorph [129] introduced a self-evolving RSSM with a contextual evolution memory for online adaptation. For temporal abstraction, WISTER [130] employed action-conditioned contrastive predictive coding to train a TSSM that captures high-level temporal features. Similarly, TWM [29] used a Transformer to align multimodal tokens with historical states during training, while relying on a lightweight policy at inference. To handle long-horizon tasks, some approaches integrate LLMs with RSSMs to decompose objectives into subtasks. EvoAgent [131], for example, uses an LLM to guide low-level actions and regularizes RSSM updates. RoboHorizon [132], in contrast, enhances task recognition with dense rewards and leverages key task segments via a masked autoencoder.

In autonomous driving, token-based sequence representations are increasingly adopted to model cross-modal interactions and spatiotemporal structures. DrivingWorld [133] pairs next-state prediction for temporal dynamics with next-token prediction for spatial structure. For multimodal control, Doe-1 [134] formulates closed-loop driving as autoregressive prediction over perception-description-action tokens, which unifies perception, prediction, and planning, and DrivingGPT [23] interleaves vision and action tokens and casts world modeling and trajectory planning as next-token prediction. To enhance diversity and safety, LatentDriver [135] models future actions as a mixture distribution and actuates the world model with planner-sampled intermediate actions. At the same time, Vasudevan *et al.* [136] proposed an adaptive model that predicts surrounding agents for safe navigation.

The token-based paradigm also extends to broader robotics. Within RL, IRIS [44] and TWM [137] leverage discrete tokens to enable data-efficient policy learning via imagined or hybrid rollouts. DyWA [138] improves action learning by conditioning on trajectory dynamics and jointly predicting future states with single-view point-cloud and proprioceptive modalities. EgoAgent [139] interleaves state-action sequence modeling within a Transformer, enabling unified perception, prediction, and action inference. Tokenized representations unify multimodal inputs, including vision, language, and action (VLA), enabling generalist agents with cross-domain adaptability, as shown in WorldVLA [67]. Recent studies encode environmental states as discrete symbolic tokens and condition next-token prediction on action, as demonstrated by DCWM [140] and TrajWorld [141].

Recent studies have strengthened the link between tokenized representations and planning, particularly through object-centric approaches. These models, such as CarFormer [142], the work of Jeong *et al.* [143], and Dyn-O [69], represent scenes as a collection of slots. CarFormer autoregressively models the relationships between these slots in BEV. Jeong *et al.* added language-guided manipulation, while Dyn-O used a Mamba with dropout scheduling for temporal modeling and to disentangle static from dynamic elements.  $\Delta$ -IRIS [144] introduced a hybrid Transformer that integrates tokens with stochastic  $\Delta$ -tokens to capture dynamics. D<sup>2</sup>PO [145] employed preference learning to jointly optimize state prediction and action selection,

enhancing the model’s understanding of underlying dynamics. For efficiency, MineWorld [59] accelerated token generation by predicting sequences in parallel and introduced an IDM as a controllability metric. Meanwhile, PIVOT-R [146] and ReOI [147] incorporated VLMs into control. PIVOT-R parses instructions to produce waypoint-based plans that an action module decodes into low-level controls, whereas ReOI detects implausible prediction elements, reimagines the distractors, and reintegrates the corrected content.

Based on tokenization, some studies employ autoregressive diffusion to achieve stable generation and long-horizon planning. Epona [148] decouples spatiotemporal modeling, handled by a Transformer, from long-horizon multimodal generation, realized through trajectory and vision Diffusion Transformers (DiTs). Goff *et al.* [149] used a DiT to instantiate the state transition, which enables on-policy training and multi-second closed-loop rollouts. SceneDiffuser++ [150] pushes this further to city-scale traffic simulation, applying multi-tensor diffusion over agents and traffic lights to produce stable closed-loop rollouts. For navigation, NWM [66] introduced an efficient conditional DiT to simulate visual trajectories for zero-shot planning.

Another emerging direction is to inject explicit reasoning into the world model using LLMs and Chain-of-Thought (CoT). NavCoT [58] decomposes navigation into imagination, filtering, and prediction, enabling parameter-efficient in-domain training, and ECoT [54] leverages a pipeline of foundation models to generate reasoning labels for training a VLA policy. Variants like MineDreamer [79] introduced Chain-of-Imagination (CoI), where a multimodal LLM imagines future observations to steer diffusion and guide actions, and FSDrive [101] generates physics-constrained future scenes and treats them as CoT supervision, enabling VLMs to function as IDMs for planning.

Other approaches directly couple LLMs with world models to operationalize planning and data generation. Dyna-Think [151] fuses reasoning and acting via a distilled LLM, and RIG [152] unifies reasoning and imagination end-to-end generalist policy. In terms of explicit dynamics and long-horizon, Gkountouras *et al.* [153] trained a causal world-model simulator that grounds an LLM environment causal reasoning and planning skills, Statler [46] enables LLMs to keep a structured world-state, using a reader for planning and a writer for updating, and Inner Monologue [42] incorporates a closed-loop feedback into LLMs, enabling agents to reason and deliberate more akin to human thinking. Finally, WoMAP [154] synthesized 3DGS scenes and trained a world model that refines VLM instructions for precise execution.

**Spatial Latent Grid.** By encoding features on geometry-aligned grids or incorporating explicit spatial priors, this paradigm preserves locality, enables efficient convolutional or attention-based updates and streaming rollouts.

In autonomous driving, many studies couple RNN-based dynamics with spatial grids to guide planning. For instance, DriveDreamer [91] and GenAD [92] adopt GRU-based dynamics over grid or instance-centric tokens to predict motion and decode trajectories. In contrast, DriveWorld [87] and Raw2Drive [155] instantiate RSSM dynamics on BEV



时间推理方面,例如 MWM[41]通过掩码自编码器将视觉标记与基于 RSSM 的动力学过程解耦,从而同时提升了性能和数据效率。NavMorph[129]引入了一种带有上下文演化记忆的自进化 RSSM,以实现在线适应。在时间抽象方面, wister[130]采用动作条件对比预测编码来训练能够捕捉高层次时间特征的 TSSM 模型;类似地, TWM[29]在训练阶段使用Transformer将多模态标记与历史状态对齐,而在推理阶段则依赖轻量级策略。为处理长时域任务,部分方法将大语言模型(LLM)与随机状态马尔可夫模型(RSSM)结合,将目标分解为子任务:例如EvoAgent[131]利用LLM指导低层级动作并规范RSSM更新;而RoboHorizon[132]则通过密集奖励机制增强任务识别能力,并借助掩码自编码器提取关键任务片段。

在自动驾驶领域,基于标记的序列表征方法正日益被采用以建模跨模态交互与时空结构。DrivingWorld[133]将时间动态的下一状态预测与空间结构的下一标记预测相结合。对于多模态控制,Doe-I[134]将闭环驾驶建模为感知-描述-动作标记上的自回归预测,从而统一感知、预测与规划过程;DrivingGPT[23]则通过视觉与动作标记的交织,将世界建模与轨迹规划转化为下一标记预测。为增强多样性和安全性,LatentDriver[135]将未来动作建模为混合分布,并通过规划器采样的中间动作驱动世界模型。与此同时,Vasudevan等人[136]提出了一种自适应模型,用于预测周围智能体以实现安全导航。

基于令牌的范式也延伸至更广泛的机器人领域。在强化学习(RL)中,IRIS[44]和TWM[137]通过离散令牌实现数据高效的策略学习,采用想象式或混合式策略展开策略。DyWA[138]通过结合轨迹动力学条件,并利用单视点云与本体感觉模态联合预测未来状态,从而提升动作学习效果。EgoAgent[139]在Transformer模型中嵌入状态-动作序列建模机制,实现了统一的感知、预测与动作推理能力。分词表示法将视觉、语言和动作等多模态输入(VLA)统一整合,如WorldVLA[67]所示,可构建具备跨域适应能力的通用智能体。近期研究将环境状态编码为离散符号令牌,并通过DCWM[140]和TrajWorld[141]验证了基于动作条件的下令牌预测方法。

最新研究进一步证实了标记化表征与规划之间的关联性,尤其通过以物体为中心的研究方法。这类模型包括CarFormer[142]以及Jeong等的研究成果。[143],以及Dyn-O[69],将场景表示为一系列槽位集合。CarFormer通过自回归模型建模BEV中这些槽位之间的关系。Jeong等人引入了语言引导式操作机制,而Dyn-O采用带有丢弃调度机制的Mamba模型进行时序建模,并区分静态与动态元素。<sup>u</sup> IRIS[144]提出了一种混合Transformer架构,通过将标记与随机<sup>u</sup>标记相结合来捕捉动态特征。D<sup>2</sup>PO[145]采用偏好学习方法联合优化状态预测与动作选择,提升了模型对

底层动态的理解能力。为提升效率,MineWorld[59]通过并行预测序列加速标记生成,并引入IDM作为可控性指标。与此同时,pivot-R[146]与ReOI[147]将VLMs融入控制机制:pivot-R解析指令生成基于航点的规划方案,由动作模块解码为低级控制指令;ReOI则检测不可信预测元素、重构干扰项并重新整合校正内容。

基于分词技术,一些研究采用自回归扩散机制以实现稳定的生成能力和长期规划能力。Epona[148]将时空建模(由Transformer处理)与长期多模态生成(通过轨迹和视觉扩散Transformer(DiTs)实现)解耦。Goff等人[149]采用DiT模型实例化状态转移机制,该机制支持策略内训练及多秒级闭环部署。SceneDiffuser++[150]将该技术进一步拓展至城市规模交通仿真领域,通过在智能体与交通信号灯上应用多张量扩散算法,实现稳定的闭环部署。在导航领域,NWM[66]提出高效条件DiT模型,用于零样本规划场景下的视觉轨迹模拟。

另一个新兴方向是利用大语言模型(LLMs)和思维链(CoT)将显式推理注入世界模型。NavCoT[58]将导航任务分解为想象、过滤和预测三个阶段,实现了参数高效的领域内训练;而ECoT[54]则通过基础模型流水线生成推理标签来训练视觉语言模型(VLA)策略。MineDreamer[79]等变体引入了想象链(CoI)机制,多模态大语言模型通过想象未来观测场景来引导扩散过程并指导行动决策;FSDrive[101]则生成物理约束下的未来场景并将其视为思维链监督数据,使视觉语言模型能够作为智能决策模型(IDMs)用于规划决策。

其他方法直接将大语言模型(LLMs)与世界模型相结合,以实现规划和数据生成的可操作性。Dyna-Think[151]通过蒸馏式LLM融合推理与行动能力,而RIG[152]则将推理与想象力端到端统一为通用策略。在显式动态机制与长时域考量方面,Gkountouras等人的研究具有重要参考价值。[153]我们训练了一个因果世界模型模拟器,该模型为大语言模型(LLM)环境中的因果推理与规划能力提供了基础支撑。Statler[46]使LLM能够通过读取器进行规划、写入器进行更新,从而保持结构化的世界状态;而Inner Monologue[42]则将闭环反馈机制融入LLM系统,使智能体的推理与决策过程更接近人类思维模式。最后,WoMAP[154]实现了三维地理场景的合成,并训练了一个世界模型,该模型通过优化VLM指令实现精准执行。

**空间潜在网格。**通过在几何对齐网格上编码特征或引入显式空间先验,该范式能够保持局部性,支持高效的卷积或注意力机制更新,并实现流式部署。

在自动驾驶领域,许多研究将基于RNN的动力学模型与空间网格相结合以指导规划。例如,DriveDreamer[91]和GenAD[92]采用基于GRU的动力学模型,通过网格或实例中心化标记来预测运动并解码轨迹。相比之下,DriveWorld[87]和Raw2Drive[155]则在BEV上实例化RSSM动力学模型。

tokens. DriveWorld conditions on tokens and actions for joint prediction, while Raw2Drive adopts a dual-stream design for spatiotemporal learning.

Numerous studies focus on autoregressively predicting future 3D occupancy representations to enable motion planning for autonomous driving. One strand discretizes scenes into occupancy tokens for sequential prediction, exemplified by OccWorld [93] and RenderWorld [156]. Another strand directly forecasts volumetric features or embeddings, as in Drive-OccWorld [157] and PreWorld [158]. Self-supervised variants predict future representations from current cues. For example, LAW [159] conditions on current representations and trajectories, SSR [160] compresses scenes into sparse BEV tokens for future BEV features, and NeMo [161] voxelizes multi-frame images and predicts occupancy to support imitation-based planning. Building on these representations, FASTopoWM [162] employs a unified decoder to align fast and slow systems from vehicle poses, enabling lane-topology reasoning, and WoTE [107] simulates candidate trajectories in BEV and selects among them with a reward model. Extending the paradigm, OccLLaMA [18] unifies occupancy, actions, and text within a single token vocabulary and employs a LLaMA for next token forecasting, planning, and question answering.

Beyond autonomous driving, similar formulations have been extended to broader domains of robotics. WMNav [163] leverages a VLM to maintain a curiosity-driven value map and adopts phased decision making to enable zero-shot, object-driven navigation. RoboOccWorld [164] targets indoor robotics by predicting fine-grained 3D occupancy with a pose-conditioned autoregressive Transformer, thereby supporting exploration and decision making. To achieve high-fidelity dynamics, EnerVerse [34] applies chunk-wise autoregressive video diffusion with a sparse memory mechanism to produce 4D latent dynamics and integrates 4DGS to mitigate the S2R gap in robotic execution. For manipulation, ParticleFormer [165] forecasts future point clouds with a Transformer-based particle-tized dynamics model, enabling robust handling of multi-object and multi-material interactions. At the representation level, DINO-WM [70] learns dynamics in the DINOv2 feature space and predicts future states to support zero-shot planning.

**Decomposed Rendering Representation.** This paradigm represents scenes with explicit renderable primitives such as NeRFs and 3DGS, updating them to simulate dynamics and render future observations. It provides view-consistent forecasts, object-level compositionality, and seamless integration with physics priors and digital twins, thereby supporting long-horizon rollouts.

Building on 3DGS, GAF [74] augments splats with learnable motion attributes to forecast future states and refines initial actions with diffusion. ManiGaussian [53] predicts per-point variations to generate future Gaussian scenes for manipulation under current states and actions, and ManiGaussian++ [80] adds a hierarchical leader-follower design with task-oriented splats to model primitive deformations for multi-body and bimanual skills. Within simulation and digital twin coupling, DreMa [60] integrates GS with a physics simulator to build twins for data synthesis in imitation learning, Abou-Chakra *et al.* [166]

introduced a dual Gaussian-Particle representation where Gaussian points are attached to particles driven by visual loss forces, DexSim2Real<sup>2</sup> [167] builds twins of articulated objects with generative models and uses sampling based planning for precise manipulation, PIN-WM [168] combines 3DGS with differentiable physics to estimate physical parameters from limited observations and generates digital cousins for zero-shot S2R policy learning, and PWTF [169] constructs an interactive twin that simulates candidate action outcomes and uses VLMs for evaluation and selection. At the representation level, DTT [98] adopts a triplane representation with multiscale Transformers to autoregressively capture incremental changes, forming a 4D world model for prediction and planning.

**2) Global Difference Prediction: Token Feature Sequence.** The compact Global Latent Vector representation discards fine-grained spatiotemporal detail and is therefore rarely used for global prediction. In contrast, Token Feature Sequences predict the future sequence in parallel, reducing error accumulation while enabling multimodal diversity.

On the representation side, TOKEN [95] tokenizes scenes into object-level tokens, aligning world representations with reasoning and leveraging LLMs to predict full future trajectories for long-tail scenarios. GeoDrive [170] extracts 3D representations, renders trajectory-conditioned views, and edits the position of vehicles to guide DiT in producing editable generations. For control, FLARE [171] aligns diffusion policies with latent future representations, avoiding pixel-space video generation and learning effectively from action-free videos. In a related vein, LaDi-WM [172] predicts future states through interactive diffusion in a latent space aligned with visual foundation models, integrating geometric and semantic features while iteratively refining the diffusion policy to improve performance and generalization. villa-X [76] and VidMan [55] both couple diffusion-based models with IDM for control. villa-X infers latent actions, aligns them with ego-centric forward dynamics, and maps them via joint diffusion, while VidMan adapts a pretrained video-diffusion model into an IDM using a self-attention adapter for accurate action prediction.

**Spatial Latent Grid.** Spatial grid models parallel-forecast BEV or voxel maps from ego-stabilized views, preserving locality and uncertainty while producing planner-ready maps for fast control.

Diffusion-based world models are commonly used for parallel generation. EmbodiedDreamer [173] couples differentiable physics with video diffusion to render photorealistic and physically consistent futures. Tesseract [78] reconstructs 4D spatiotemporal consistent scenes by jointly generating RGB, depth, and normal videos for IDM-based action learning. DFIT-OccWorld [174] reformulates prediction as decoupled voxel warping and adopts image-assisted single-stage training for reliable and efficient dynamic scene modeling. For instruction-conditioned control, RoboDreamer [49] decomposes instructions into low-level primitives that steer video diffusion, synthesizing novel compositional scenes beyond the training distribution while grounding execution via an IDM, and ManipDreamer [175] extends this design with an action-tree prior together with depth and semantic guidance to improve



tokens。DriveWorld采用基于tokens和动作的联合预测机制，而Raw2Drive则采用双流设计以实现时空学习。

大量研究聚焦于通过自回归预测未来三维占用表示来实现自动驾驶的运动规划。一种方法将场景离散化为占用标记进行序列预测，例如OccWorld[93]和RenderWorld[156]。另一种方法直接预测体积特征或嵌入向量，如Drive-OccWorld[157]和PreWorld[158]。自监督变体通过当前线索预测未来表示，例如LAW[159]基于当前表示和轨迹进行条件化，SSR[160]将场景压缩为稀疏BEV标记以生成未来BEV特征，NeMo[161]对多帧图像进行体素化并预测占用状态以支持模仿式规划。基于这些表示，FASTopoWM[162]采用统一解码器从车辆姿态对快速与慢速系统进行对齐以实现车道拓扑推理，而WoTE[107]则在BEV中模拟候选轨迹并通过奖励模型进行选择。扩展这一范式，OccLLaMA[18]将占用状态、动作和文本统一在一个单一的标记词汇表中，并采用LLaMA技术进行下一个标记的预测、规划及问答处理。

除了自动驾驶领域，类似的算法框架已扩展到更广泛的机器人技术领域。WMNav[163]利用VLM技术构建基于好奇心的价值地图，并采用分阶段决策机制实现零样本、基于物体的导航。RoboOccWorld[164]针对室内机器人应用，通过姿态条件自回归Transformer预测精细的三维空间占用信息，从而支持探索与决策过程。为实现高保真动态效果，EnerVerse[34]采用基于稀疏记忆机制的块状自回归视频扩散技术生成4D潜在动力学，并整合4DGS模型以缩小机器人执行中的S2R差距。在物体操控方面，ParticleFormer[165]通过基于Transformer的粒子化动力学模型预测未来点云分布，实现对多物体及多材料交互的稳健处理。在表征层面，DINOWM[70]在DINOv2特征空间中学习动力学规律，并预测未来状态以支持零样本规划。

**分解式渲染表征。**该范式通过显式可渲染基元（如NeRFs和3DGS）来表征场景，通过动态更新这些基元以模拟场景变化并生成未来观测结果。该方法可提供视图一致性预测、对象级组合性，并与物理先验及数字孪生系统无缝集成，从而支持长期部署场景的持续演进。

基于3DGS，GAF[74]通过添加可学习的运动属性来增强轨迹迹图，以预测未来状态，并利用扩散技术优化初始动作。ManiGaussian[53]预测每个点的变异值，生成可在当前状态和动作下进行操作的未来高斯场景；而ManiGaussian++[80]则采用分层的领导者-跟随者设计，并结合任务导向型轨迹迹图来建模多体及双臂操作技能的基本形变。在仿真与数字孪生耦合领域，DreMa[60]将GS与物理模拟器集成，用于构建用于模仿学习中数据合成的孪生模型；Abou-Chakra等人[166]引入了一种双高斯-粒子表示

法，其中高斯点附着在由视觉损失力驱动的粒子上；De-xSim2Real<sup>2</sup>[167]通过生成模型构建关节对象的孪生体，并采用基于采样的规划实现精确操作；PIN-WM[168]将3DGS与可微分物理方法结合，从有限观测中估计物理参数并生成数字孪生体以实现零样本S2R策略学习；PWTF[169]构建了可模拟候选动作结果的交互式孪生体，并利用VLM进行评估与选择。在表示层面，DTT[98]采用三平面表示法结合多尺度Transformer自回归捕捉增量变化，形成用于预测与规划的4D世界模型。

2) **全局差异预测：令牌特征序列。**紧凑的全局潜在向量表示会舍弃精细的时空细节，因此很少用于全局预测。相比之下，令牌特征序列能够并行预测未来序列，既能减少误差累积，又能实现多模态多样性。

在表征层面，Token[95]将场景分解为对象级标记，使世界表征与推理过程相匹配，并利用大语言模型（LLMs）为长尾场景预测完整的未来轨迹。GeoDrive[170]提取三维表征、渲染轨迹相关的视图，并调整车辆位置以指导DiT生成可编辑的图像序列。在控制方面，Flare[171]将扩散策略与潜在的未来表征对齐，避免像素空间视频生成，并能从无动作视频中有效学习。类似地，LaDi-WM[172]通过与视觉基础模型对齐的潜在空间中的交互式扩散来预测未来状态，整合几何与语义特征，并迭代优化扩散策略以提升性能和泛化能力。Villa-X[76]和VidMan[55]均将基于扩散的模型与IDM相结合用于控制。villaX推断潜在动作，将其与以自我为中心的前向动态对齐，并通过联合扩散进行映射；而VidMan则将预训练视频扩散模型转化为IDM，通过自注意力适配器实现精准的动作预测。

**空间潜在网格。**空间网格模型可并行预测来自自我稳定视图的BEV或体素图，既保持局部性与不确定性，又能生成适用于快速控制的规划器就绪地图。

基于扩散的世界模型常被用于并行生成。EmbodiedDriver[173]将可微分物理引擎与视频扩散技术相结合，能够生成逼真且符合物理规律的未来场景。TesserAct[78]通过联合生成RGB图像、深度图和法线图视频，实现基于IDM的动作学习，从而重建具有时空一致性的4D场景。DFIT-OccWorld[174]将预测问题重新表述为解耦的体素变形任务，并采用图像辅助的单阶段训练方法，实现可靠高效的动态场景建模。在指令条件控制方面，RoboDreamer[49]将指令分解为低级基本单元以引导视频扩散，能够在训练分布之外合成新颖的组合场景，同时通过IDM确保执行一致性；而ManipDreamer[175]则在此基础上引入动作树先验以及深度与语义引导机制，进一步提升了生成效果。



instruction following and temporal consistency.

On the planning side, 3DFlowAction [176] employs a pretrained 3D optical flow world model to treat future motion as a unified action cue, enabling label-free and cross-robot manipulation through closed-loop optimization. Imagine-2-Drive [177] integrates video diffusion with a multimodal diffusion policy to accelerate policy learning. Drive-WM [88] uses multi-view diffusion with image-based rewards to select safer trajectories, while World4Drive [178] leverages vision-based priors to construct intent-aware world models that support self-supervised multi-intent imagination. COMBO [179] composes multi-agent actions with diffusion, leverages VLMs to infer purposes, and integrates tree search for online cooperative planning.

## B. General-Purpose World Models

1) *Sequential Simulation and Inference: Token Feature Sequence.* General-purpose models pretrain task-agnostic dynamics to capture environmental physics and generate future scenes, prioritizing transferability over specific tasks.

Some general world models increasingly pretrain on unlabeled video and use tokenized latent space for robust forecasting and generation. iVideoGPT [56] was pretrained on large-scale interaction videos for action-free forecasting and later adapted to downstream control. Genie [50] learned discrete latent actions and spatiotemporal tokens, enabling user-controllable interactive environments via autoregressive dynamics. RoboScape [180] jointly learned video generation with temporal depth and keypoint dynamics to improve physical realism. PACT [181] tokenized multimodal perception and action and trained a causal Transformer to obtain a unified representation for diverse tasks, and DINO-world [182] learns generalizable dynamics by predicting the temporal evolution of DINOv2 features from large-scale unlabeled video corpora. Building on language priors, EVA [71] introduced a Reflection-of-Generation (RoG) policy that used a VLM for iterative self-correction, strengthening long-horizon anticipation. In the same vein, Owl-1 [183] employs a VLM to forecast world dynamics conditioned on current states and generated fragments, explicitly guiding the subsequent fragments and enabling coherent long-horizon video synthesis, while World4Omni [68] employs a reflective world model, where a VLM refines subgoal images from an image generator, and integrates them with pretrained modules for zero-shot robotic manipulation.

Recent work adapts video diffusion models into controllable world models that autoregressively imagine future scenes. AdaWorld [72] introduced an action-aware pretraining scheme that extracted self-supervised latent actions between adjacent frames to condition diffusion, enabling efficient transfer with minimal interaction. Vid2World [184] adapts pretrained video diffusion models into autoregressive interactive world models via causalization and a causal action-guidance mechanism. GenAD [90] employs a two-stage strategy to adapt diffusion into a general video-prediction model conditioned on text and actions, enabling large-scale driving simulation and planning. Pandora [47] uses an instruction-tuned LLM to autoregressively steer a separate video-diffusion generator for explicit, goal-directed control, while Yume [75] quantized camera motions

into text tokens to guide a masked Video DiT, enabling autoregressive synthesis of dynamic 3D exploratory worlds.

To maintain geometric fidelity and long-horizon stability, recent methods couple explicit 3D priors with temporal-consistency modules in diffusion-based world models. At the geometric level, Geometry Forcing [185] aligns latent features with a geometric foundation model to inject explicit 3D priors, improving geometric consistency, while DeepVerse [64] integrates visual and geometric prediction targets and introduces a geometry-aware memory to sustain consistent long-horizon generation. For temporal stability, VRAG [186] proposed a Video Retrieval-Augmented Generation (RAG) framework that retrieves historical frames conditioned on a global state to stabilize autoregressive rollouts, StateSpaceDiffuser [63] combines Mamba with diffusion to alleviate long-term memory loss and content drift under short context windows, and Infinity-Drive [187] injects memory and adopts an adaptive loss within DiT, producing minute-scale driving videos with high fidelity, temporal consistency, and diverse content. Complementing these designs, LongDWM [188] mitigates error accumulation in long-horizon video generation through distillation—where a fine-grained DiT learns continuous motion to guide a coarse model, whereas MiLA [189] adopts a coarse-to-fine strategy that predicts sparse anchor frames and refines them during interpolation to improve temporal consistency and long-term fidelity. Finally, for dynamics and conditioning, Orbis [190] employs a continuous-space flow-matching formulation that demonstrates greater robustness than discrete-token schemes for long-horizon rollouts, and DriVerse [109] leverages multimodal trajectory prompts with latent motion alignment to synthesize long-horizon driving videos from a single image and navigation trajectories.

Sequential world models increasingly act as learned simulators, providing action-conditioned rollouts for policy evaluation and training. WorldGym [191] and WorldEval [192] generate action-conditioned rollouts and use VLM-based critics for evaluation, while WorldEval further leverages latent action representations to drive a DiT-based synthesizer. RLVR-World [62] fine-tunes world models with RL with Verifiable Rewards (RLVR), using explicit metrics to close the pretraining–task objective gap. For safety risk prediction, Guan *et al.* [193] presented a framework for autonomous driving accident prediction that augments data with a domain-informed world model and enhances spatiotemporal reasoning using graph and temporal convolutions.

Beyond diffusion, sequence models broadened the capacity for long-range consistency. Po *et al.* [194] integrated block-wise state space models for long-term memory with local attention for short-term coherence, enabling video generation with sustained memory and consistent dynamics. S2-SSM [61] employs a Mamba layer to model the independent evolution of object slots and a sparsity-regularized cross-attention mechanism to capture their causal interactions, enabling causal reasoning over the environment.

**Spatial Latent Grid.** Pretraining geometry-aligned spatial maps with self-supervised spatiotemporal objectives, the spatial latent grid paradigm preserves locality and enables efficient

指令跟随性与时间一致性

在规划层面, 3DFlowAction[176]采用预训练的三维光流世界模型, 将未来运动视为统一的动作线索, 通过闭环优化实现无标签标注和跨机器人协同操控。Imagine-2-Drive[177]将视频扩散与多模态扩散策略相结合以加速策略学习。Drive-WM[88]运用多视角扩散结合基于图像的奖励机制来选择更安全的轨迹, 而World4Drive[178]则借助基于视觉的先验知识构建意图感知世界模型, 支持自监督多意图想象。combo[179]通过扩散技术组合多智能体动作, 利用视觉语言模型推断目标意图, 并整合树搜索算法实现在线协同规划。

## B. 通用世界模型

1) **顺序模拟与推理: 词元特征序列。**通用模型通过预训练任务无关的动力学机制来捕捉环境物理特性并生成未来场景, 其核心目标是提升迁移能力而非特定任务表现。

一些通用世界模型越来越多地采用无标注视频进行预训练, 并利用分词化潜在空间实现稳健的预测与生成。iVideoGPT[56]最初在大规模交互视频上进行无动作预测预训练, 随后被适配用于下游控制任务。Genie[50]通过自回归动力学学习离散潜在动作与时空标记, 实现了用户可控的交互环境。RoboScape[180]联合学习视频生成与时序深度及关键点动态, 以提升物理真实感。PACT[181]对多模态感知与动作进行分词化处理, 并训练因果Transformer获得跨任务统一表征; DINO-world[182]则通过从大规模无标注视频语料库预测DINOv2特征的时序演变来学习泛化动力学。基于语言先验知识, EVA[71]提出反射生成 (RoG) 策略, 该策略采用迭代自校正 VLM 以增强长时程预测能力。同理, Owl-1[183]采用 VLM 来预测基于当前状态和生成片段的世界动态, 明确引导后续片段并实现连贯的长时域视频合成; 而World4Omni[68]则采用反射式世界模型, 其中 VLM 从图像生成器中优化子目标图像, 并将其与预训练模块整合以实现零样本机器人操控。

近期研究将视频扩散模型转化为可控制的世界模型, 实现对未来场景的自回归想象。AdaWorld[72]提出了一种动作感知预训练方案, 通过提取相邻帧间的自监督潜在动作来调控扩散过程, 从而实现最小交互下的高效迁移。Vid2World[184]通过因果化与因果动作引导机制, 将预训练视频扩散模型适配为自回归交互式世界模型。GenAD[90]采用两阶段策略, 将扩散模型转化为基于文本和动作的通用视频预测模型, 支持大规模驾驶模拟与规划。Pandora[47]利用指令调优的大型语言模型, 通过自回归方式引导独立视频扩散生成器实现显式目标导向控制; 而Yume[75]将摄像机运动量化为文本标记以引导掩码视频扩散生成器, 实

现动态3D探索世界的自回归合成。

为保持几何保真度和长时域稳定性, 最新方法将显式三维先验与基于扩散的世界模型中的时序一致性模块相结合。在几何层面, Geometry Forcing[185]通过将潜在特征与几何基础模型对齐来注入显式三维先验, 从而提升几何一致性; DeepVerse[64]则整合视觉与几何预测目标, 并引入几何感知记忆机制以维持长时域生成的一致性。为增强时序稳定性, VRAG[186]提出视频检索增强生成 (RAG) 框架, 通过全局状态条件检索历史帧来稳定自回归输出; StateSpaceDiffuser[63]将Mamba与扩散机制结合以缓解短上下文窗口下的长期记忆缺失与内容漂移问题; InfinityDrive[187]则在DiT框架中注入记忆机制并采用自适应损失函数, 生成具有高保真度、时序一致性及内容多样性的微米级驾驶视频。与这些设计相辅相成的是, LongDWM[188]通过蒸馏技术缓解了长时域视频生成中的误差累积问题——其中细粒度DiT学习连续运动以引导粗粒度模型, 而MiLA[189]采用从粗到细的策略, 通过预测稀疏锚帧并在插值过程中进行优化来提升时间一致性与长期保真度。最后, 在动态建模与条件处理方面, Orbis[190]采用连续空间流匹配公式, 相较于离散标记方案展现出更强的鲁棒性, 适用于长时域生成任务; DriVerse[109]则通过多模态轨迹提示结合潜在运动对齐技术, 实现了从单张图像和导航轨迹合成长时域驾驶视频。

序贯世界模型日益演变为学习型模拟器, 为策略评估与训练提供动作条件化输出。WorldGym[191]和WorldEval[192]生成动作条件化输出, 并采用基于 VLM 的评估器进行评价, 其中WorldEval进一步利用潜在动作表征驱动基于DiT的合成器。RLVR-World[62]通过可验证奖励强化学习 (RLVR) 对世界模型进行微调, 采用显式指标来缩小预训练与任务目标之间的差距。在安全风险预测领域, Guan等人[193]本文提出了一种自动驾驶事故预测框架, 该框架通过引入领域知识驱动的世界模型来增强数据, 并利用图结构与时域卷积技术提升时空推理能力。

除扩散效应外, 序列模型还增强了长程一致性能力。Po等人[194]针对具有局部注意力机制的长时记忆模型, 我们构建了集成块状状态空间模型以实现短期信息一致性, 从而生成具有持久记忆特征和动态一致性特征的视频。S2-SSM[61]采用Mamba层来模拟对象槽位的独立演化过程, 并通过稀疏正则化交叉注意力机制捕捉其因果交互关系, 从而支持对环境进行因果推理。

**空间潜在网络。**该范式通过自监督时空目标对几何对齐的空间地图进行预训练, 保留了局部性特征并实现了高效计算。

rollouts, multimodal fusion, and transferable planner-ready maps.

Building on this paradigm, structured-grid and physics-informed methods encode geometry and dynamics for controllable rollouts. PhyDNet [195] disentangles physical priors expressed as partial differential equations from visual factors, improving prediction. ViDAR [89] unifies semantics, geometry, and dynamics through a pre-training task of point cloud forecasting and a latent rendering operator, enabling a scalable foundation for downstream autonomous driving tasks. FOLIAGE [196] models dynamics with an Accretive Graph Network and a Transformer-based predictor, executing rollouts on simulated data. Complementing these grid and physics lines, MindJourney [73] couples VLMs with a controllable world model to render egocentric rollouts along planned camera trajectories, enabling multi-view reasoning.

Building on grid-based representations, diffusion-based forecasting has become dominant for stable long-horizon generation. Within grid-centric predictors, DOME [94] encodes observations into a continuous latent space and applies a spatiotemporal DiT for scene forecasting, Copilot4D [82] tokenizes point clouds and couples a spatiotemporal Transformer with discrete diffusion to improve fidelity and coherence, and LidarDM [100] generates layout-conditioned static scenes, composes them with dynamic objects, integrating LiDAR simulation to produce controllable videos. For long-video generation, Vista [96] adopts a two-stage large-scale training regime to produce controllable, high-fidelity, driving videos, and Delphi [86] enforces long-horizon multiview consistency through shared noise and feature alignment and a failure-driven framework to synthesize targeted data for improved planning. To strengthen long-horizon stability, GEM [102] achieves controllable egovision generation through large-scale training and fine-grained control over motion, dynamics, and posture, Zhou *et al.* [197] maintains a persistent RGB-D 3D memory map to guide subsequent frames, and STAGE [198] introduced hierarchical temporal feature transfer with multi-stage training.

**Decomposed Rendering Representation.** Scenes are decomposed into explicit primitives to synthesize view-consistent, simulatable trajectories over long horizons. Within this paradigm, GaussianWorld [103] models scene evolution as ego-motion, object dynamics, and newly observed regions, iteratively updating 3D Gaussian primitives to enable accurate and efficient dynamic perception. InfiniCube [22] introduced a hybrid pipeline that combines voxel-based generation, video synthesis, and dynamic Gaussian reconstruction, enabling large-scale dynamic 3D driving scenes conditioned on HDmaps, bounding boxes, and text. Complementarily, Wu *et al.* [199] augmented a video world model with long-term spatial memory grounded in reconstructed geometry and an episodic memory, which together condition sequential generation for long-range consistency.

2) *Global Difference Prediction: Token Feature Sequence.* For general-purpose world models, tokenized feature sequences support global prediction via masked and generative modeling, enabling parallel long-horizon rollouts with global constraints and multimodal conditioning.

Within Joint-Embedding Predictive Architecture (JEPA) [200], V-JEPA [51] extends this architecture to video by predicting latent features of occluded spatiotemporal regions, learning generalizable representations for both appearance and motion without pixel reconstruction or contrastive learning. Building on this, V-JEPA 2 [14] scales pretraining to large-scale Internet videos with larger models and incorporates limited robot interaction data for post-training, transferring to robotic planning. AD-L-JEPA [201] adapts JEPA to BEV LiDAR, predicting masked embeddings in a self-supervised manner. Beyond JEPA-style prediction, WorldDreamer [45] frames world modeling as masked visual sequence prediction to learn physics and motion for diverse video generation and editing, while MaskGWM [104] combines diffusion with masked feature reconstruction and a dual-branch masking strategy to improve long-horizon consistency and generalization.

In parallel, diffusion-based methods have become central to global-difference modeling. Sora [13] represents video as unified spacetime patches and uses a DiT to generate long, coherent sequences at scale. ForeDiff [202] decouples conditioning from denoising by adding a deterministic predictive stream and employing a pretrained predictor to guide generation, improving accuracy and consistency. For domain-specific synthesis, AirScape [203] introduces an aerial video-intention dataset, applies supervised fine-tuning for controllability, and leverages VLMs to impose spatiotemporal constraints; MarsGen [204] builds a multimodal Mars dataset from NASA’s sparse rover stereo imagery using geometric foundation models, then trains a controllable generator to produce visually realistic, geometry-consistent Martian videos. In clinical guidance, EchoWorld [205] proposes a motion-aware world model for echocardiography probe control, pretraining on region- and motion-outcome prediction and fine-tuning attention to fuse visual and motion cues for precise guidance.

**Spatial Latent Grid.** Spatial-grid models forecast voxel grids in parallel and fuse multi-view visual features into a unified map, learning a general-purpose world model.

Recent work converges on unified scene understanding and future prediction. UniFuture [206] couples Dual Latent Sharing with multi-scale latent interaction to jointly model appearance and depth in future driving scenes, and HERMES [108] integrates multiview BEV features into an LLM with world queries that link scene understanding to future prediction within a single framework. BEVWorld [207] maps images and LiDAR into a compact BEV latent space through a unified tokenizer and applies a latent BEV diffusion model for synchronized multimodal forecasting. Progress in grid and occupancy forecasting includes differentiable raycasting with a proxy reformulation for sensor agnostic motion learning by Khurana *et al.* [208], [209] and LiDAR to range images 3D spatiotemporal convolutions by Mersch *et al.* [210]. Cam4DOcc [211] established the first vision-only benchmark with an E2E 3D CNN baseline, and Liu *et al.* [212] enhanced cross-task transfer through high-ratio compression and latent flow matching.

On the generative front, tokenized 4D representations enable controllable scene synthesis. OccSora [85] uses a 4D tokenizer



系统部署、多模态融合及可转移的规划器就绪地图。

基于这一范式，结构化网格与物理信息方法通过编码几何结构和动力学特性实现可控展开。PhyDNet[195]将以偏微分方程形式表达的物理先验与视觉因素解耦，显著提升了预测精度。ViDAR[89]通过点云预测预训练任务与潜在渲染算子相结合，实现了语义、几何与动力学的统一，为下游自动驾驶任务构建了可扩展基础框架。Foliage[196]采用累积图网络与基于Transformer的预测器对动力学进行建模，并在模拟数据上执行展开操作。与网格和物理信息方法相辅相成的是MindJourney[73]，该方法将视觉语言模型与可控世界模型相结合，沿规划的相机轨迹渲染以自我为中心的展开场景，从而实现多视角推理能力。

基于网格表示方法，扩散式预测技术已成为稳定长时域生成领域的主流方案。在网格中心预测器中，DOME[94]将观测数据编码至连续潜在空间，并采用时空DiT模型进行场景预测；Copilot4D[82]对点云进行分词处理，结合时空Transformer与离散扩散模型以提升保真度与连贯性；LidarDM[100]生成布局条件静态场景，通过动态物体合成与LiDAR仿真整合生成可控视频。针对长视频生成，Vista[96]采用两阶段大规模训练方案生成可控、高保真驾驶视频；Delphi[86]通过共享噪声与特征对齐机制及故障驱动框架强制实现长时域多视角一致性，并合成目标数据以优化规划。为增强长时域稳定性，GEM[102]通过大规模训练与对运动、动力学及姿态的精细控制实现可控自动驾驶视觉生成，Zhou等人[197]该模型维护一个持久的RGB-D三维内存映射，用于指导后续帧的处理；而[198]提出的模型则采用了多阶段训练机制，实现了分层时间特征迁移。

**分解式渲染表征。**通过将场景分解为显式基本元素，可合成具有视图一致性且可长期模拟的轨迹。基于该范式，GaussianWorld[103]将场景演化建模为自我运动、物体动力学及新观测区域，通过迭代更新三维高斯基本元素实现精准高效的动态感知。InfiniCube[22]提出了一种混合流程，整合基于体素的生成、视频合成与动态高斯重构技术，能够根据高清地图、边界框及文本条件生成大规模动态三维驾驶场景。此外，Wu等人[199]通过基于重构几何结构和情景记忆构建的长期空间记忆，对视频世界模型进行了增强，二者共同为序列生成过程提供条件以确保长距离一致性。

2) **全局差异预测：标记特征序列。**对于通用世界模型而言，标记化特征序列通过掩码模型与生成模型支持全局预测，从而实现具有全局约束条件和多模态条件的并行长时域部署。

在联合嵌入预测架构（JEPA）[200]中，V-JEPA[51]将该架构扩展至视频领域，通过预测被遮挡时空区域的潜在特征，无需像素重建或对比学习即可学习适用于外观和运动的通用表示。基于此，V-JEPA 2[14]采用更大模型将预训练扩展至大规模互联网视频，并在后训练阶段整合有限的机器人交互数据，从而应用于机器人规划任务。AD-L-JEPA[201]将JEPA适配至BEV激光雷达系统，以自监督方式预测掩码嵌入特征。除JEPA风格的预测方法外，WorldDreamer[45]将场景建模视为掩码视觉序列预测任务，以学习适用于多样化视频生成与编辑的物理规律和运动规律；而MaskGWM[104]则结合扩散机制与掩码特征重建技术，并采用双分支掩码策略，以提升长期一致性与泛化能力。

与此同时，基于扩散的方法已成为全局差异建模的核心。Sora[13]将视频表示为统一的时空片段，并利用DiT生成大规模下长而连贯的序列。ForeDiff[202]通过添加确定性预测流并采用预训练预测器引导生成过程，将条件化与去噪过程解耦，从而提升准确性和一致性。在领域特定合成方面，AirScape[203]引入了航拍视频-意图数据集，采用监督微调实现可控性，并利用VLMs施加时空约束；MarsGen[204]基于NASA的稀疏火星车立体影像，利用几何基础模型构建多模态火星数据集，随后训练可控生成器生成视觉逼真且几何一致的火星视频。在临床引导领域，EchoWorld[205]提出了一种用于超声心动图探头控制的运动感知世界模型，通过区域与运动结果预测的预训练及注意力机制的微调，融合视觉与运动线索以实现精准引导。

**空间潜在网格。**空间网格模型通过并行预测体素网格，并将多视角视觉特征融合为统一地图，从而学习通用世界模型。

近期研究聚焦于统一场景理解与未来预测领域。UniFuture[206]通过将双重潜在共享与多尺度潜在交互相结合，实现了对未来驾驶场景中外观与深度的联合建模；hermes[108]则将多视角BEV特征与世界查询整合至大语言模型中，构建了将场景理解与未来预测统一于单一框架的解决方案。BEVWorld[207]通过统一分词器将图像与激光雷达数据映射至紧凑的BEV潜在空间，并采用潜在BEV扩散模型实现同步多模态预测。在网格与占用预测方面，Khurana等人提出了基于代理重构的可微射线投射方法，实现了传感器无关的运动学习。[208]，[209]以及通过Mersch等人提出的激光雷达对测距图像进行三维时空卷积[210]Cam4DOcc[211]建立了首个仅基于视觉的基准测试，并采用了端到端3D CNN基线；Liu等人[212]通过高比率压缩与潜在流匹配技术实现增强的跨任务迁移能力。

在生成方面，分词化的4D表示能够实现可控的场景合成。OccSora[85]采用了一种4D分词器。

to derive compact representations for trajectory conditioned diffusion, and DynamicCity [99] encodes 4D occupancy as HexPlanes representations with a VAE and employs a conditional DiT for high fidelity controllable dynamics. Fidelity and consistency improve through decoupling ego-motion with scene evolution in COME [213], physics-informed constraints in DrivePhysica [214], cross-view point map alignment in Liu *et al.* [215], and photometric warping-based supervision in PosePilot [216]. For controllable conditioning, DriveDreamer 2 [97] translates prompts into agent trajectories and HDMaps for customizable video generation, EOT-WM [217] encodes ego and surrounding trajectories as trajectory videos for trajectory consistent synthesis, and ORV [65] uses 4D semantic occupancy sequences to guide action conditioned video with S2R transfer. AETHER [77] unifies dynamic 4D reconstruction, action-conditioned video prediction, and vision-based planning under training on synthetic 4D data and achieves zero-shot generalization to real-world scenarios.

**Decomposed Rendering Representation.** This paradigm performs global prediction by combining explicit 3D structure with video generative priors.

A trend is combining video generation with Gaussian Splatting. DriveDreamer4D [105] exploits complex driving trajectories such as lane changes to guide video synthesis and optimize a 4DGS model, which enhances reconstruction fidelity and spatiotemporal coherence from novel viewpoints. ReconDreamer [106] introduces an online restoration module together with progressive data reuse to correct artifacts in Gaussian-rendered views and enables reliable reconstruction of large-scale trajectories. MagicDrive3D [84] generates multi-view street scenes conditioned on BEV maps, 3D boxes, and text, and further converts the outputs into full 3D environments through fault-tolerant GS. In contrast, implicit-field methods replace GS with continuous neural representations. UnO [17] leverages future point clouds to learn a NeRF-style 4D occupancy field, which allows annotation-free prediction and achieves strong transfer performance beyond supervised baselines in point-cloud forecasting.

#### IV. DATA RESOURCES & METRICS

World models in embodied AI are required to address diverse tasks spanning manipulation, navigation, and autonomous driving, requiring heterogeneous resources and rigorous evaluation. Accordingly, we present Data Resources in §IV-A and Metrics in §IV-B, focusing on the most widely adopted platforms and evaluation measures as a unified foundation for cross-domain assessment.

##### A. Data Resources

To meet the diverse demands of embodied AI, we categorize data resources into four categories: Simulation Platforms, Interactive Benchmarks, Offline Datasets, and Real-world Robot Platforms, as detailed in the following subsections. Tab. III provides a comprehensive overview of these resources.

1) *Simulation Platforms:* Simulation platforms provide controllable and scalable virtual environments for training and evaluating world models.

- **MuJoCo** [218] is a customizable physics engine widely adopted for its efficient robotic simulation of articulated systems and contact dynamics in robotics and control research.
- **NVIDIA Isaac** is an E2E, GPU-accelerated simulation stack that comprises Isaac Sim, Isaac Gym [221], and Isaac Lab [222]. It offers photorealistic rendering and large-scale RL capabilities.
- **CARLA** [219] is an open-source simulator based on Unreal Engine for urban autonomous driving, providing realistic rendering, diverse sensors, and closed-loop evaluation protocols.
- **Habitat** [220] is a high-performance simulator for embodied AI, specializing in photorealistic 3D indoor navigation.

2) *Interactive Benchmarks:* Interactive benchmarks offer standardized task suites and protocols for reproducible closed-loop evaluation of world models.

- **DeepMind Control (DMC)** [224] is a standard MuJoCo-based suite for control tasks, offering a consistent basis for comparing agents that learn from state or pixel-based observations.
- **Atari** [223] is a suite of pixel-based, discrete-action games for evaluating agent performance. The Atari100k [239] specifically assesses sample efficiency by limiting interaction to 100k steps.
- **Meta-World** [225] is a benchmark for multi-task and meta-RL, featuring 50 diverse robotic manipulation tasks with a Sawyer arm in MuJoCo under standardized evaluation protocols.
- **RLBench** [226] offers 100 simulated tabletop manipulation tasks with sparse rewards and rich, multi-modal observations, designed to test complex skills and rapid adaptation.
- **LIBERO** [228] is a benchmark for lifelong robotic manipulation, providing 130 procedurally generated tasks and human demonstrations to evaluate sample-efficient and continual learning.
- **nuPlan** [227] is a planning benchmark for autonomous driving, using a lightweight closed-loop simulator and over 1 500 h of real-world driving logs to evaluate long-horizon performance.

3) *Offline Datasets:* Offline datasets are large-scale, pre-collected trajectories that eliminate interactive rollouts and provide a foundation for reproducible evaluation and data-efficient pretraining of world models.

- **RT-1** [233] is a real-world dataset for robot learning, collected over 17 months with 13 Everyday Robots mobile manipulators. It contains 130 000 demonstrations spanning more than 700 tasks, pairing language instructions and image observations with discretized 11-DoF actions for the arm and mobile base.
- **Open X-Embodiment (OXE)** [235] is a corpus aggregating 60 sources from 21 institutions, spanning 22 robot embodiments, 527 skills, and over one million trajectories in a unified format for cross-embodiment training. Models trained

为轨迹条件扩散推导紧凑表示方法, DynamicCity[99]将4D占用状态编码为带VAE的HexPlanes表示, 并采用条件DiT实现高保真可控动力学。通过COME[213]中将自我运动与场景演化解耦、DrivePhysica[214]中物理信息约束以及Liu等人中的跨视点地图对齐技术, 系统保真度与一致性得到显著提升。[215]以及基于光度畸变的PosePilot监督方法[216]。为实现可控条件化, DriveDreamer 2[97]将提示转化为智能体轨迹和高清地图以支持可定制视频生成, EOT-WM[217]将自我及周边轨迹编码为轨迹视频以实现轨迹一致性合成, ORV[65]则利用4D语义占用序列通过S2R迁移指导动作条件化视频生成。aether[77]在合成4D数据训练下实现了动态4D重建、动作条件化视频预测与基于视觉的规划的统一, 并达成对真实场景的零样本泛化能力。

**分解式渲染表征。**该范式通过将显式三维结构与视频生成先验相结合, 实现全局预测。

当前趋势是将视频生成与高斯点扩散技术相结合。DriveDreamer4D[105]利用车道变换等复杂驾驶轨迹来指导视频合成并优化4DGS模型, 从而从新颖视角提升重建保真度与时空一致性。ReconDreamer[106]引入在线修复模块及渐进式数据复用机制, 可修正高斯渲染视图中的伪影, 并实现大规模轨迹的可靠重建。MagicDrive3D[84]基于BEV地图、3D框和文本生成多视角街景, 并通过容错性GS将输出转换为完整的3D环境。相比之下, 隐式场方法用连续神经表征替代GS。UnO[17]利用未来点云学习NeRF风格的4D占用场, 实现无需标注的预测能力, 并在点云预测任务中超越监督基线模型展现出强大的迁移性能。

#### IV. 数据资源与指标

在具身人工智能领域, 世界模型需要应对涵盖物体操控、导航及自动驾驶等多样化任务, 这要求模型具备异构资源支持并经过严格评估。为此, 我们在第四章A节中阐述数据资源体系, 在第四章B节中提出评估指标体系, 重点聚焦主流平台与通用评估方法, 为跨领域评估构建统一基础框架。

##### A. 数据资源

为满足具身人工智能的多样化需求, 我们将数据资源划分为四大类: 仿真平台、交互式基准测试集、离线数据集和真实世界机器人平台, 具体分类详见下文各小节。表III对这些资源进行了全面概述。

1) **模拟平台:** 模拟平台为训练和评估世界模型提供可控且可扩展的虚拟环境。

- **MuJoCo[218]**是一款可定制的物理引擎, 因其在机器人学及控制研究中对铰接系统和接触动力学的高效机器人仿真能力而被广泛采用。
- **NVIDIA Isaac**是一个端到端、GPU加速的仿真堆栈, 包含IsaacSim、IsaacGym[221]和IsaacLab[222], 提供逼真渲染和大规模强化学习能力。
- **carla[219]**是一款基于Unreal Engine的开源城市自动驾驶模拟器, 具备逼真的渲染效果、多样化的传感器以及闭环评估协议。
- **Habitat[220]**是一款面向具身人工智能的高性能模拟器, 专注于逼真三维室内导航领域。

2) **交互式基准测试:** 交互式基准测试提供了标准化的任务套件和协议, 用于对世界模型进行可重复的闭环评估。

- **DeepMind Control (DMC)[224]**是一套基于MuJoC的标准控制任务工具套件, 为比较从状态或像素级观测中学习的智能体提供了统一的基础。
- **Atari[223]**是一套基于像素的离散动作游戏, 用于评估智能体性能。Atari100k[239]通过将交互限制在10万步内, 专门用于评估样本效率。
- **Meta-World[225]**是一个多任务与元强化学习(meta-RL)的基准测试平台, 该平台在MuJoCo环境中配备了Sawyer机械臂, 包含50项多样化的机器人操作任务, 并采用标准化评估协议进行测试。
- **RLBenchmark[226]**提供了100个模拟桌面操作任务, 这些任务具有稀疏奖励和丰富的多模态观测数据, 旨在测试复杂技能和快速适应能力。
- **LIBERO[228]**是终身机器人操作的基准测试集, 包含130个程序生成任务及人类演示案例, 用于评估样本效率与持续学习能力。
- **nuPlan[227]**是一款自动驾驶规划基准测试工具, 它采用轻量级闭环模拟器及超过1 500小时的真实驾驶数据记录, 用于评估自动驾驶系统的长期性能表现。

3) **离线数据集:** 离线数据集是大规模预收集的轨迹数据, 可消除交互式部署过程, 为可重复评估及数据高效预训练世界模型提供基础。

- **RT-1[233]**是一个用于机器人学习的真实世界数据集, 通过13台Everyday Robots移动机械臂在17个月内收集而成。该数据集包含130,000个演示样本, 涵盖700多项任务, 将语言指令与图像观测结果相结合, 并采用离散化的11自由度动作来描述机械臂和移动底座的运动状态。
- **Open X-Embodiment (OXE) [235]**是一个语料库, 整合了来自21家机构的60个数据源, 涵盖22种机器人实现方案、527项技能以及超过一百万条轨迹, 采用统一格式用于跨实现方案训练。所训练的模型



TABLE III  
AN OVERVIEW OF DATA RESOURCES FOR TRAINING AND EVALUATING EMBODIED WORLD MODELS.

| Category  | Name               | Year | Task                         | Input                         | Domain | Scale              | Protocol <sup>1</sup>  |
|-----------|--------------------|------|------------------------------|-------------------------------|--------|--------------------|------------------------|
| Platform  | MuJoCo [218]       | 2012 | Continuous control           | Proprio.                      | Sim    | -                  | -                      |
|           | CARLA [219]        | 2017 | Driving simulation           | RGB-D/Seg/LiDAR/Radar/GPS/IMU | Sim    | -                  | ✓                      |
|           | Habitat [220]      | 2019 | Embodied navigation          | RGB-D/Seg/GPS/Compass         | Sim    | -                  | ✓                      |
|           | Isaac Gym [221]    | 2021 | continuous control           | Proprio.                      | Sim    | -                  | -                      |
|           | Isaac Lab [222]    | 2023 | Robot learning suites        | RGB-D/Seg/LiDAR/Proprio.      | Sim    | -                  | -                      |
| Benchmark | Atari [223]        | 2013 | Discrete-action game         | RGB/State                     | Sim    | 55+ Games          | ✓                      |
|           | DMC [224]          | 2018 | Continuous control           | RGB/Proprio.                  | Sim    | 30+ Tasks          | ✓                      |
|           | Meta-World [225]   | 2019 | Multi-task manipulation      | RGB/Proprio.                  | Sim    | 50 tasks           | ✓                      |
|           | RLBench [226]      | 2020 | Robotic manipulation         | RGB-D/Seg/Proprio.            | Sim    | 100 tasks          | ✓                      |
|           | nuPlan [227]       | 2021 | Driving planning             | RGB/LiDAR/Map/Proprio.        | Real   | 1.5k hours         | ✓                      |
|           | LIBERO [228]       | 2023 | Lifelong manipulation        | RGB/Text/Proprio.             | Sim    | 130 tasks          | ✓                      |
| Dataset   | SSv2 [229]         | 2018 | Video-action understanding   | RGB/Text                      | Real   | 220k videos        | 169k/24k/27k           |
|           | nuScenes [230]     | 2020 | Driving perception           | RGB/LiDAR/Radar/GPS/IMU       | Real   | 1k scenes          | 700/150/150            |
|           | Waymo [231]        | 2020 | Driving perception           | RGB/LiDAR                     | Real   | 1.15k scenes       | 798/202/150            |
|           | HM3D [232]         | 2021 | Indoor navigation            | RGB-D                         | Real   | 1k scenes          | 800/100/100            |
|           | RT-1 [233]         | 2022 | Real-robot manipulation      | RGB/Text                      | Real   | 130k+ trajectories | -                      |
|           | Occ3D [234]        | 2023 | 3D occupancy                 | RGB/LiDAR                     | Real   | 1.9k scenes        | 600/150/150; 798/202/- |
|           | OXE [235]          | 2024 | Cross-embodiment pretraining | RGB-D/LiDAR/Text              | Real   | 1M+ trajectories   | -                      |
|           | OpenDV [90]        | 2024 | Driving video pretraining    | RGB/Text                      | Real   | 2k+ hours          | -                      |
|           | VideoMix22M [14]   | 2025 | Video pretraining            | RGB                           | Real   | 22M+ samples       | -                      |
| Robot     | Franka Emika [236] | 2022 | Manipulation                 | Proprio.                      | Real   | -                  | -                      |
|           | Unitree Go1 [237]  | 2021 | Quadruped locomotion         | RGB-D/LiDAR/Proprio.          | Real   | -                  | -                      |
|           | Unitree G1 [238]   | 2024 | Humanoid manipulation        | RGB-D/LiDAR/Proprio./Audio    | Real   | -                  | -                      |

<sup>1</sup> Protocol: For interactive benchmarks, a check mark (✓) indicates available evaluation protocols. For datasets, it indicates official data splits are provided.

on OXE demonstrate strong transfer beyond single-robot baselines, underscoring the effectiveness of cross-platform data sharing.

- **Habitat-Matterport 3D (HM3D)** [232] is a large-scale dataset of 1 000 indoor reconstructions with 112 500 m<sup>2</sup> navigable area, substantially expanding the scope and diversity of embodied AI simulation. Released for the Habitat platform, it provides the necessary metadata and resources for seamless use.
- **nuScenes** [230] is a large-scale multimodal driving dataset with a 360-degree sensor suite comprising six cameras, five radars, one LiDAR, and GPS/IMU. It contains 1 000 twenty-second scenes collected in Boston and Singapore with dense 3D annotations for 23 classes and HDMaps, providing a core benchmark for multimodal fusion and long-horizon prediction.
- **Waymo** [231] is a multimodal autonomous driving benchmark with 1 150 twenty-second scenes at 10 Hz from San Francisco, Phoenix, and Mountain View. It includes five LiDARs and five cameras, with about 12 million 3D and 2D annotations, making it a large-scale resource for modeling traffic dynamics.
- **Occ3D** [234] defines 3D occupancy prediction from surround-view images, providing voxel labels that distinguish free, occupied, and unobserved states. Occ3D-nuScenes contains about 40 000 frames at 0.4 m resolution, while Occ3D-Waymo offers about 200 000 frames at 0.05 m. This voxel-level supervision enables holistic scene understanding beyond bounding boxes.
- **Something-Something v2 (SSv2)** [229] is a video dataset for fine-grained action understanding. It contains 220 847 clips across 174 categories, collected from crowd workers following textual prompts (e.g., Putting something into

something) with splits of 168 913 train, 24 777 val, and 27 157 test videos.

- **OpenDV** [90] is the largest large-scale video-text dataset for autonomous driving, proposed by GenAD, which supports video prediction and world-model pretraining. It contains 2 059 hours and 65.1 million frames from YouTube and seven public datasets, covering over 40 countries and 244 cities. The dataset provides command and context annotations to enable language- and action-conditioned prediction and planning.
- **VideoMix22M** [14] is a large-scale dataset introduced with V-JEPA 2 for self-supervised pretraining. It scales from 2 million to 22 million samples, drawn from YT-Temporal-1B [240], HowTo100M [241], Kinetics [242], SSv2, and ImageNet [243]. The largest source, YT-Temporal-1B, is curated with retrieval-based filtering to suppress noise, while ImageNet images are converted into static video clips for consistency.

4) *Real-world Robot Platforms*: Real-world robot platforms provide physical embodiments for interaction, enabling closed-loop evaluation, high-fidelity data collection, and S2R validation under real-world constraints.

- **Franka Emika** [236] is a 7-DoF collaborative robot arm with joint torque sensors for precise force control. Through the control interface, it supports 1 kHz torque control for contact-rich tasks, while its ROS integration makes it a versatile platform.
- **Unitree Go1** [237] is a cost-effective and widely adopted quadrupedal robot equipped with a panoramic depth-sensing suite, onboard computing of 1.5 TFLOPS, and a maximum speed of 4.7 m/s, establishing it as a standard platform for locomotion and embodied-AI research.
- **Unitree G1** [238] is a compact humanoid robot for research,

表 III  
具身化世界模型训练与评估数据资源概述

| 分类  | 姓名                 | 年    | 任务             | 输入                                                     | 领域 | 规模, 规模效应   | 协议 <sup>1</sup>        |
|-----|--------------------|------|----------------|--------------------------------------------------------|----|------------|------------------------|
| 平台  | MuJoCo [218]       | 2012 | 持续控制           | 确实如此。                                                  | 同时 | -          | -                      |
|     | 卡拉[219]            | 2017 | 驾驶模拟           | RGB-D/Seg/LiDAR/雷达/GPS/惯性测量单元 (IMU) RGB-D/Seg/GPS/固有罗盘 | 同时 | -          | ✓                      |
|     | 栖息地[220]           | 2019 | 具身导航           | RGB-D/Seg/GPS/固有罗盘                                     | 同时 | -          | ✓                      |
|     | 艾萨克体育馆[221]        | 2021 | 连续控制           | RGB-D/分割/激光雷达/本体感知                                     | 同时 | -          | -                      |
|     | 艾萨克实验室[222]        | 2023 | 机器学习套件         | RGB-D/分割/激光雷达/本体感知                                     | 同时 | -          | -                      |
| 基准  | 雅达利[223]           | 2013 | 离散动作博弈         | RGB/状态                                                 | 同时 | 55+款游戏     | ✓                      |
|     | DMC [224]          | 2018 | 持续控制           | RGB/Proprio.                                           | 同时 | 30+项任务     | ✓                      |
|     | 元世界[225]           | 2019 | 多任务操作          | RGB/Proprio.                                           | 同时 | 50项任务      | ✓                      |
|     | RLBench [226]      | 2020 | 机器人操作          | RGB-D/分割/本体信息。RGB/激光雷达/地图/本体信息。RGB/文本/本体信息。            | 同时 | 100项任务     | ✓                      |
|     | nuPlan [227]       | 2021 | 驾驶规划           | RGB-D/分割/本体信息。RGB/激光雷达/地图/本体信息。RGB/文本/本体信息。            | 真的 | 1.5千小时     | ✓                      |
|     | LIBERO [228]       | 2023 | 终身操作           | RGB-D/分割/本体信息。RGB/激光雷达/地图/本体信息。RGB/文本/本体信息。            | 同时 | 130项任务     | ✓                      |
| 数据集 | SSv2 [229]         | 2018 | 视频动作理解与驾驶感知    | RGB/文本                                                 | 真的 | 220,000个视频 | 169k/24k/27k           |
|     | nuScenes [230]     | 2020 | 驾驶感知 室内导航 实体机器 | RGB/激光雷达/雷达/GPS/惯性测量单元                                 | 真的 | 1000个场景    | 700/150/150            |
|     | Waymo [231]        | 2020 | 驾驶感知 室内导航 实体机器 | RGB/激光雷达                                               | 真的 | 1,150个场景   | 798/202/150            |
|     | HM3D [232]         | 2021 | 驾驶感知 室内导航 实体机器 | RGB-D                                                  | 真的 | 1000个场景    | 800/100/100            |
|     | RT-1 [233]         | 2022 | 人操作 三维空间占用率    | RGB/文本                                                 | 真的 | 13万条以上轨迹   | -                      |
|     | Occ3D [234]        | 2023 | 跨实体预训练 驾驶视频预训练 | RGB/激光雷达 RGB-D/激光雷达/文本                                 | 真的 | 1.9千个场景    | 600/150/150; 798/202/- |
|     | OXE [235]          | 2024 | 跨实体预训练 驾驶视频预训练 | RGB/激光雷达 RGB-D/激光雷达/文本                                 | 真的 | 100万条以上轨迹  | -                      |
|     | OpenDV [90]        | 2024 | 跨实体预训练 驾驶视频预训练 | RGB/激光雷达 RGB-D/激光雷达/文本                                 | 真的 | 2000小时以上   | -                      |
| 机器人 | VideoMix22M [14]   | 2025 | 训练 视频预训练       | 本 RGB/文本 RGB                                           | 真的 | 22M+样本     | -                      |
|     | Franka Emika [236] | 2022 | 操作             | 确实如此。                                                  | 真的 | -          | -                      |
|     | Unitree Go1 [237]  | 2021 | 四足动物运动         | RGB-D/激光雷达/本体感知                                        | 真的 | -          | -                      |
|     | Unitree G1 [238]   | 2024 | 人形操控           | RGB-D/激光雷达/本体感知/音频                                     | 真的 | -          | -                      |

<sup>1</sup> 协议: 对于交互式基准测试, 勾选标记 ( ) 表示可用的评估协议; 对于数据集, 则表示提供了官方数据划分方案。

OXE 证明了超越单机器人基准测试的显著性能提升, 充分彰显了跨平台数据共享的优势。

- **Habitat-Matterport 3D (HM3D)[232]**是一个包含1 000 个室内重建数据集的大规模数据集, 其可导航区域总面积达112,500平方米<sup>2</sup>, 显著拓展了具身人工智能模拟的应用范围与多样性。该数据集专为Habitat平台发布, 提供了无缝使用的必要元数据与资源。
- **nuScenes[230]**是一个大规模多模态驾驶数据集, 其配备的360度传感器组包含六台摄像头、五台雷达、一台激光雷达 (LiDAR) 以及GPS/IMU。该数据集收录了在波士顿和新加坡采集的1 000 个22秒时长的场景, 包含针对23个类别的详细3D标注及高清地图 (HDMaps), 为多模态融合与远距离预测提供了核心基准。
- **Waymo[231]**是一个多模态自动驾驶基准数据集, 包含来自旧金山、凤凰城和山景城的1 150 个20秒场景, 采样频率为10Hz。该数据集配备5个激光雷达和5个摄像头, 包含约1200万条3D和2D标注数据, 是建模交通动态的大型资源库。
- **Occ3D[234]**通过全景图像实现三维空间占用状态预测, 提供可区分空闲、被占用及未观测状态的体素标签。Occ 3D-nuScenes包含约40,000帧分辨率0.4米的图像, 而Occ3 DWaymo则提供约200,000帧分辨率0.05米的图像。这种基于体素级别的监督机制能够实现超越边界框的全景场景整体理解。
- **Something-Something v2 (SSv2)[229]**是一个用于细粒度动作理解的视频数据集。该数据集包含来自众包用户根据文本提示 (例如 “将某物放入某物中”) 生成的220,847个视频片段, 涵盖174个类别, 其中训练集包含168,913个视

频, 验证集包含24,777个视频, 测试集包含27,157个视频。

- **OpenDV[90]**是由GenAD提出的最大规模自动驾驶视频文本数据集, 支持视频预测和世界模型预训练。该数据集包含来自YouTube和七个公开数据集的2 059 小时6510 万帧视频, 覆盖40多个国家和244个城市。数据集提供指令和上下文注释, 以实现语言和动作条件下的预测与规划。
- **VideoMix22M[14]**是随 V- JEPa 2 一同发布的大规模数据集, 用于自监督预训练。其样本量从 200 万到 2200 万不等, 数据来源于 YT-Temporal-1B[240]、HowTo100 M[241]、Kinetics[242]、SSv2 和 ImageNet[243]。其中最大的数据源 YT-Temporal-1B 通过基于检索的过滤技术进行了清洗以消除噪声; 而 ImageNet 的图像则被转换为静态视频片段以确保数据一致性。

4) **现实世界机器人平台:** 现实世界机器人平台为交互提供物理载体, 可在真实环境约束条件下实现闭环评估、高保真数据采集及系统与机器人 (S2R) 验证。

- **Franka Emika[236]**是一款配备关节扭矩传感器的7自由度协作机器人手臂, 可实现精准力控制。通过控制接口, 其支持1 kHz扭矩控制以适应高接触任务需求, 同时ROS集成功能使其成为多功能平台。
- **Unitree Go1[237]**是一款经济高效且被广泛采用的四足机器人, 配备全景深度传感套件、1.5 TFLOPS 的车载计算能力以及4.7 m/s的最大速度, 使其成为运动学和具身人工智能研究的标准平台。
- **Unitree G1[238]**是一款用于科研的紧凑型人形机器人,

offering up to 43-DoF and knee torques of 120 N·m, with integrated 3D LiDAR and depth cameras. With multimodal sensing, onboard compute, ROS support, and swappable batteries, this low-cost platform provides a practical real-robot testbed for training and evaluating embodied world models.

## B. Metrics

Metrics evaluate the capability of world models to capture dynamics, generalize to unseen scenarios, and scale with additional resources. We organize them into three abstraction levels: §IV-B1 Pixel Prediction Quality, §IV-B2 State-level Understanding, and §IV-B3 Task Performance, representing a progression from low-level signal fidelity to high-level goal attainment.

1) *Pixel Generation Quality*: At the most fundamental level, world models are evaluated by their ability to reconstruct sensory inputs and generate realistic sequences. Metrics assess image fidelity, temporal consistency, and perceptual similarity, providing quantitative measures of the extent to which models capture raw environmental dynamics.

**Fréchet Inception Distance (FID)** [244]. FID is a metric for assessing the realism and diversity of generated images. It compares real and generated image distributions in the feature space of an ImageNet-pretrained Inception-v3 [245], modeling embeddings as Gaussians with means  $\mu_x, \mu_y$  and covariances  $\Sigma_x, \Sigma_y$ . Defined as

$$\text{FID}(x, y) = \|\mu_x - \mu_y\|_2^2 + \text{Tr} \left( \Sigma_x + \Sigma_y - 2(\Sigma_x \Sigma_y)^{1/2} \right), \quad (6)$$

a lower FID denotes a closer alignment between real and generated distributions. By comparing the first and second moments, it penalizes fidelity loss (mean shift) and mode collapse (covariance mismatch), offering a holistic measure of generative performance.

**Fréchet Video Distance (FVD)** [246]. FVD extends FID to videos, evaluating both per-frame quality and temporal consistency. It replaces the image-based Inception network with an I3D [247] pretrained on Kinetics-400 [248]. Using the same Fréchet distance as Eq. (6) on motion-aware features, FVD yields a holistic video quality score. A lower value indicates a closer alignment of distributions in appearance and dynamics while penalizing temporal artifacts like unnatural motion or flickering.

**Structural Similarity Index Measure (SSIM)** [249]. SSIM is a perceptual metric for image quality that compares luminance, contrast, and structure between a generated image and its reference. For two patches  $x$  and  $y$  with means  $\mu_x, \mu_y$ , variances  $\Sigma_x^2, \Sigma_y^2$ , and covariance  $\Sigma_{xy}$ , SSIM is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\Sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\Sigma_x^2 + \Sigma_y^2 + C_2)}. \quad (7)$$

The final score is obtained by averaging SSIM over sliding windows, and values closer to 1 indicate higher similarity.

**Peak Signal-to-Noise Ratio (PSNR)** [250]. PSNR measures pixel-wise distortion between a reconstruction and its reference. Let the mean-squared error (MSE) over  $N$  pixels be

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2, \quad (8)$$

and let MAX denote the maximum possible pixel value (e.g., 255 for RGB or 1 for normalized images). Then

$$\text{PSNR}(x, y) = 10 \cdot \log_{10} \left( \frac{\text{MAX}^2}{\text{MSE}} \right). \quad (9)$$

Higher PSNR values indicate lower distortion and greater fidelity.

**Learned Perceptual Image Patch Similarity (LPIPS)** [251]. LPIPS is a metric that correlates with human judgments by comparing features extracted from pretrained networks. Let  $\hat{f}_x^l$  and  $\hat{f}_y^l$  denote the unit-normalized activations at layer  $l$  for inputs  $x$  and  $y$ , and  $w_l$  the channel-wise weights. LPIPS is defined as

$$\text{LPIPS}(x, y) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \left\| w_l \odot (\hat{f}_{h,w,x}^l - \hat{f}_{h,w,y}^l) \right\|_2^2. \quad (10)$$

Lower LPIPS values imply greater similarity, offering enhanced fidelity compared to pixel-based metrics and robustness against distortions.

**VBench** [252]. VBench is a comprehensive benchmark for video generation that assesses performance across 16 dimensions grouped into two categories: Video Quality (e.g., subject consistency, motion smoothness) and Video-Condition Consistency (e.g., object class, human action). It provides carefully curated prompt suites and large-scale human preference annotations to ensure strong perceptual alignment, thereby enabling fine-grained evaluation of model capabilities and limitations.

2) *State-level Understanding*: Beyond pixel fidelity, state-level understanding assesses whether models capture objects, layouts, and semantics, and can predict their evolution. Metrics span semantic, BEV, and 3D segmentation, detection, occupancy, geometry, and trajectory accuracy, emphasizing structural understanding beyond appearance.

**mean Intersection over Union (mIoU)**. mIoU evaluates semantic segmentation by averaging the Intersection over Union (IoU) across classes. For class  $c$ ,

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \quad (11)$$

where TP, FP, and FN denote the true positives, false positives, and false negatives. IoU quantifies overlap with the ground truth while penalizing segmentation errors. The dataset-level score is

$$\text{mIoU} = \frac{1}{|C|} \sum_{c \in C} \text{IoU}_c. \quad (12)$$

A higher mIoU reflects more precise semantic scene understanding.

**mean Average Precision (mAP)**. mAP evaluates detection and instance segmentation by averaging per-class Average



该平台提供高达43自由度和120 N·m的膝关节扭矩，并集成3D激光雷达和深度摄像头。通过多模态传感、车载计算、ROS支持以及可更换电池，这一低成本平台为训练和评估具身世界模型提供了实用的实机测试平台。

## B. 指标

指标用于评估世界模型捕捉动态变化、对未见场景进行泛化以及随资源增加而扩展的能力。我们将这些指标划分为三个抽象层级：§IV-B1像素预测质量、§IV-B2状态级理解和§IV-B3任务性能，代表从低级信号保真度到高级目标达成的递进过程。

1) **像素生成质量**：从最基础层面来看，世界模型的评估标准在于其重建感官输入和生成真实场景序列的能力。评估指标包括图像保真度、时间一致性及感知相似性，这些量化指标可反映模型对原始环境动态的捕捉程度。

**Frechet Inception Distance (FID)**[244]。FID是一种用于评估生成图像真实性和多样性的度量指标。它通过比较ImageNet预训练Inception-v3模型[245]特征空间中的真实图像与生成图像分布，将嵌入向量建模为均值 $\mu_x, \mu_y$ 和协方差 $\Sigma_x, \Sigma_y$ 的高斯分布。其定义为

$$FID(x, y) = \|\mu_x - \mu_y\|_2^2 + \text{Tr}(\Sigma_x + \Sigma_y - 2(\Sigma_x \Sigma_y)^{1/2}), \quad (6)$$

较低的FID值表明真实分布与生成分布之间的对齐程度更高。通过比较一阶矩与二阶矩，该指标能够有效抑制保真度损失（均值漂移）和峰度塌缩（协方差失配），从而为生成性能提供全面评估依据。

**弗雷歇视频距离 (FVD)** [246]。FVD将FID扩展至视频领域，同时评估每帧质量与时间一致性。它用基于Kinetics-400[248]预训练的I3D[247]替换了基于图像的Inception网络。通过在运动感知特征上采用与等式(6)相同的弗雷歇距离，FVD得出综合视频质量评分。评分越低表明视频在外观与动态特征上的分布越接近，同时会对不自然运动或闪烁等时间伪影进行惩罚。

**结构相似性指数度量 (SSIM)** [249]。SSIM是一种用于评估图像质量的感知指标，通过比较生成图像与其参考图像之间的亮度、对比度和结构特征。对于两个均值分别为 $\mu_x, \mu_y$ 、方差 $\Sigma_x, \Sigma_y$ 及协方差 $\Sigma_{xy}$ 的图像块 $x$ 与 $y$ ，SSIM定义为

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\Sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\Sigma_x^2 + \Sigma_y^2 + C_2)}. \quad (7)$$

最终得分是通过滑动窗口对SSIM取平均值获得的，数值越接近1表示相似度越高。

**峰值信噪比 (PSNR)** [250]。PSNR用于衡量重建图像与参考图像之间的像素级失真。设 $N$ 个像素上的均方误差 (MSE) 为

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2, \quad (8)$$

令MAX表示可能的最大像素值（例如，RGB图像为255，归一化图像为1）。

$$PSNR(x, y) = 10 \cdot \log_{10} \left( \frac{MAX^2}{MSE} \right). \quad (9)$$

较高的PSNR值表明畸变程度较低且保真度更高。

**习得性感知图像块相似性 (LPIPS)** [251]。

LPIPS是一种与人类判断相关联的度量指标，比较从预训练网络中提取的特征。设 $f_l^x$ 和 $f_l^y$ 表示输入 $x$ 和 $y$ 在第 $l$ 层的单位归一化激活值， $w_l$ 表示通道权重。LPIPS定义为

$$LPIPS(x, y) = \sum_l \frac{1}{H_l W_l} \sum_{h, w} \left\| w_l \odot (f_{h, w, x}^l - f_{h, w, y}^l) \right\|_2^2. \quad (10)$$

较低的LPIPS值意味着更高的相似度，相较于基于像素的度量指标，其保真度更高且对图像失真具有更强的鲁棒性。

**VBench**[252]。VBench是一款全面的视频生成基准测试工具，通过16个维度评估性能表现，这些维度分为两大类：视频质量（例如主体一致性、运动平滑度）和视频条件一致性（例如物体类别、人物动作）。该工具提供精心设计的提示词集和大规模人工偏好标注数据，确保高度一致的感知效果，从而能够对模型能力与局限性进行精细化评估。

2) **状态级理解**：该指标不仅关注像素级保真度，更评估模型对物体、布局及语义的捕捉能力，并能预测其演变趋势。评估指标涵盖语义理解、BEV（边界扩展视图）分割、三维分割与检测、空间占用率、几何结构分析及轨迹预测精度，着重体现超越表面特征的结构理解能力。

**平均交并比 (mIoU)**。mIoU通过计算各类别间交并比 (IoU) 的平均值来评估语义分割效果。对于类别 $c$ ,

$$IoU = \frac{TP}{TP + FP + FN}, \quad (11)$$

其中TP、FP和FN分别表示真阳性、假阳性和假阴性。IoU指标通过量化与真实标签的重叠程度来评估分割误差。数据集级别的评分结果为

$$mIoU = \frac{1}{|C|} \sum_{c \in C} IoU_c. \quad (12)$$

较高的mIoU值反映了更精确的语义场景理解能力。

**平均精度 (mAP)**。mAP通过计算各类别的平均值来评估检测与实例分割性能。

Precision (AP). For a class  $c$  at IoU threshold  $\tau$ , predictions are ranked by confidence and matched one-to-one with ground truths when  $\text{IoU} \geq \tau$  with unmatched predictions counted as FP and unmatched ground truths as FN. Precision and recall are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (13)$$

Let  $P_{c,\tau}(r)$  denote the precision-recall envelope obtained via monotonic interpolation. The AP for class  $c$  at threshold  $\tau$  is

$$\text{AP}_{c,\tau} = \int_0^1 P_{c,\tau}(r) dr. \quad (14)$$

mAP averages AP across classes and thresholds  $T$ :

$$\text{mAP} = \frac{1}{|C|} \sum_{c \in C} \left( \frac{1}{|T|} \sum_{\tau \in T} \text{AP}_{c,\tau} \right). \quad (15)$$

A higher mAP indicates better instance recognition, more accurate localization, and more calibrated confidence estimates.

**Displacement Error.** Displacement error metrics assess state-level understanding by measuring spatial accuracy for keypoints, object centers, and trajectory waypoints. The L2 trajectory error computes the Euclidean distance between predicted and ground-truth waypoints. Common variants include Average Displacement Error (ADE), which calculates the average displacement, and Final Displacement Error (FDE), which measures the displacement at the final step. Lower values indicate more accurate localization.

**Chamfer Distance (CD) [253].** CD quantifies geometric similarity between a prediction  $S_1$  and ground truth  $S_2$  by summing squared nearest-neighbor distances across the two sets:

$$\text{CD}(S_1, S_2) = \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2^2 + \sum_{y \in S_2} \min_{x \in S_1} \|x - y\|_2^2. \quad (16)$$

Unlike pixel-level metrics, CD captures surfaces, occupancy, BEV, and 3D structures, and its differentiability enables use as both a training loss and an evaluation metric that complements IoU.

3) *Task Performance:* Ultimately, the value of a world model lies in supporting effective decision-making, with task-level metrics assessing goal achievement under safety and efficiency constraints in embodied settings.

**Success Rate (SR).** SR quantifies performance as the fraction of evaluation episodes that satisfy a predefined success condition. In navigation and manipulation, the condition is typically binary, such as reaching a target or placing an object correctly. In autonomous driving, the requirement is stricter, demanding route completion without collisions or major violations. The final SR is reported as the average of binary outcomes across all test episodes.

**Sample Efficiency (SE).** SE quantifies the samples needed to reach target performance. It is evaluated by fixed-budget benchmarks (e.g., Atari-100k), data-performance curves, or in robotics by the demonstrations required to achieve a given success rate.

TABLE IV  
PERFORMANCE COMPARISON OF VIDEO GENERATION ON THE NUSCENES.

| Method              | Pub.       | Resolution | FID↓       | FVD↓        |
|---------------------|------------|------------|------------|-------------|
| MagicDrive3D [84]   | arXiv'24   | 224 × 400  | 20.7       | 164.7       |
| Delphi [86]         | arXiv'24   | 512 × 512  | 15.1       | 113.5       |
| Drive-WM [88]       | CVPR'24    | 192 × 384  | 15.8       | 122.7       |
| GenAD [90]          | CVPR'24    | 256 × 448  | 15.4       | 184.0       |
| DriveDreamer [91]   | ECCV'24    | 128 × 192  | 52.6       | 452.0       |
| Vista [96]          | NeurIPS'24 | 576 × 1024 | 6.9        | 89.4        |
| DrivePhysica [214]  | arXiv'24   | 256 × 448  | <b>4.0</b> | 38.1        |
| DrivingWorld [133]  | arXiv'24   | 512 × 1024 | 7.4        | 90.9        |
| DriveDreamer-2 [97] | AAAI'25    | 256 × 448  | 11.2       | 55.7        |
| UniFuture [206]     | arXiv'25   | 320 × 576  | 11.8       | 99.9        |
| MiLA [189]          | arXiv'25   | 360 × 640  | 4.1        | <b>14.9</b> |
| GeoDrive [170]      | arXiv'25   | 480 × 720  | 4.1        | 61.6        |
| LongDWM [188]       | arXiv'25   | 480 × 720  | 12.3       | 102.9       |
| MaskGWM [104]       | CVPR'25    | 288 × 512  | 8.9        | 65.4        |
| GEM [102]           | CVPR'25    | 576 × 1024 | 10.5       | 158.5       |
| Epona [148]         | ICCV'25    | 512 × 1024 | 7.5        | 82.8        |
| STAGE [198]         | IROS'25    | 512 × 768  | 11.0       | 242.8       |
| DriVerse [109]      | ACMMM'25   | 480 × 832  | 18.2       | 95.2        |

**Reward.** In RL, the reward is a signal  $r_t$  at timestep  $t$ . The goal is to maximize the discounted return  $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$ . Results are reported as cumulative reward or average return, often with normalization for cross-task comparison.

**Collision.** Safety is evaluated with collision-based metrics. The primary measure, collision rate, is the proportion of evaluation episodes with at least one collision and is common in indoor navigation. In autonomous driving, exposure-normalized variants are used, such as collisions per kilometer or collisions per hour.

## V. PERFORMANCE COMPARISON

Given the proliferation of world-model variants and heterogeneous metrics, we organize comparisons by task objectives and rely on standardized benchmarks, reporting concise tables that highlight each method's strengths and limitations.

### A. Pixel Generation

**Generation on nuScenes.** Driving video generation is treated as a world-modeling task that synthesizes plausible scene dynamics in fixed-length clips. Typical protocols produce short sequences and evaluate quality with *FID* for appearance fidelity and *FVD* for temporal consistency. For a fair comparison on the nuScenes validation split, recent approaches have achieved remarkable progress, as shown in Tab. IV. DrivePhysica delivers the best visual fidelity, while MiLA achieves the strongest temporal coherence, together establishing new state-of-the-art performance.

### B. Scene Understanding

**4D Occupancy Forecasting on Occ3D-nuScenes.** 4D occupancy forecasting is treated as a representative world modeling task. Given 2 s of past 3D occupancy, models predict the subsequent 3 s of scene dynamics. Evaluation follows the

精确度（AP）。对于IoU阈值 $\tau$ 下的类别 $c$ ，当 $\text{IoU} \geq \tau$ 时，预测结果按置信度排序并与真实值一一匹配，未匹配的预测结果计为假阳性（FP），未匹配的真实值计为假阴性（FN）。精确度与召回率

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (13)$$

令 $P_{c,\tau}(r)$ 表示通过单调插值获得的精确率-召回率包络线。阈值 $\tau$ 下类别 $c$ 的平均精度为

$$\text{AP}_{c,\tau} = \int_0^1 P_{c,\tau}(r) dr. \quad (14)$$

mAP通过跨类别和阈值 $T$ 对AP进行平均计算：

$$\text{mAP} = \frac{1}{|C|} \sum_{c \in C} \left( \frac{1}{|T|} \sum_{\tau \in T} \text{AP}_{c,\tau} \right). \quad (15)$$

较高的mAP值表明实例识别能力更优、定位精度更高且置信度估计更为校准准确。

**位移误差。**位移误差指标通过测量关键点、物体中心及轨迹航点的空间精度来评估系统对状态层面的理解能力。L2轨迹误差通过计算预测航点与真实航点之间的欧氏距离得出。常见变体包括平均位移误差（ADE，计算平均位移值）和最终位移误差（FDE，测量最后一步位移量）。数值越低表明定位精度越高。

**倒角距离（CD）[253]。**CD通过计算两个集合中最近邻距离的平方和，量化预测结果 $S_1$ 与真实值 $S_2$ 之间的几何相似性：

$$\text{CD}(S_1, S_2) = \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2^2 + \sum_{y \in S_2} \min_{x \in S_1} \|x - y\|_2^2. \quad (16)$$

与像素级指标不同，CD能够捕捉表面特征、占用率、边界扩展值（BEV）及三维结构，其可微性使其既可作为训练损失函数，也可作为评估指标与IoU指标形成互补。

3) **任务表现：**世界模型的核心价值在于支持有效决策制定，通过任务层面的评估指标，在具身化场景中分析在安全与效率约束条件下的目标达成情况。

**成功率（SR）。**成功率通过量化满足预设成功条件的评估回合比例来衡量系统性能。在导航与物体操控场景中，成功条件通常为二元判断（如抵达目标位置或正确放置物体）。而在自动驾驶领域，要求更为严苛，需确保路线完整且无碰撞或重大违规行为。最终成功率以所有测试回合中二元结果的平均值形式呈现。

**样本效率（SE）。**样本效率用于量化达到目标性能所需的样本量。其评估方法包括固定预算基准测试（例如Atari-100k）、数据-性能曲线，或在机器人领域中通过实现特定成功率所需的演示次数来衡量。

表 IV  
视频生成性能对比分析  
NUSCENES。

| 方法                  | 公共的：公开的     | 分辨率        | FID↓       | FVD↓        |
|---------------------|-------------|------------|------------|-------------|
| MagicDrive3D [84]   | arXiv '24   | 224 × 400  | 20.7       | 164.7       |
| Delphi [86]         | arXiv '24   | 512 × 512  | 15.1       | 113.5       |
| Drive-WM [88]       | CVPR '24    | 192 × 384  | 15.8       | 122.7       |
| GenAD [90]          | CVPR '24    | 256 × 448  | 15.4       | 184.0       |
| DriveDreamer [91]   | ECCV '24    | 128 × 192  | 52.6       | 452.0       |
| Vista [96]          | NeurIPS '24 | 576 × 1024 | 6.9        | 89.4        |
| DrivePhysica [214]  | arXiv '24   | 256 × 448  | <b>4.0</b> | 38.1        |
| DrivingWorld [133]  | arXiv '24   | 512 × 1024 | 7.4        | 90.9        |
| DriveDreamer-2 [97] | AAAI '25    | 256 × 448  | 11.2       | 55.7        |
| UniFuture [206]     | arXiv '25   | 320 × 576  | 11.8       | 99.9        |
| MiLA [189]          | arXiv '25   | 360 × 640  | 4.1        | <b>14.9</b> |
| GeoDrive [170]      | arXiv '25   | 480 × 720  | 4.1        | 61.6        |
| LongDWM [188]       | arXiv '25   | 480 × 720  | 12.3       | 102.9       |
| MaskGWM [104]       | CVPR '25    | 288 × 512  | 8.9        | 65.4        |
| GEM [102]           | CVPR '25    | 576 × 1024 | 10.5       | 158.5       |
| Epona [148]         | ICCV '25    | 512 × 1024 | 7.5        | 82.8        |
| 阶段[198]             | IROS '25    | 512 × 768  | 11.0       | 242.8       |
| DriVerse [109]      | ACMMM '25   | 480 × 832  | 18.2       | 95.2        |

**奖励。**在强化学习中，奖励是时间步 $t$ 处的信号 $r_t$ 。目标是最大化折扣回报 $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$ 。结果以累积奖励或平均回报形式呈现，通常会进行标准化处理以便跨任务比较。

**碰撞。**安全性评估采用基于碰撞的指标。主要指标碰撞率是指发生至少一次碰撞的评估事件比例，在室内导航系统中较为常见。在自动驾驶领域，通常使用暴露量标准化的指标，例如每公里碰撞次数或每小时碰撞次数。

## V. 性能比较

鉴于全球模型变体的激增及度量指标的异质性，我们按任务目标对比较结果进行分类，并采用标准化基准进行评估，通过简洁表格呈现各方法的优势与局限性。

### A. 像素生成

**nuScenes上的生成任务。**视频生成被视为一项世界建模任务，旨在固定长度的视频片段中合成合理的场景动态。典型方法生成短序列，并采用FID评估外观保真度，以及FVD评估时间一致性。为在nuScenes验证集上实现公平比较，近期方法已取得显著进展（详见表IV）。DrivePhysica在视觉保真度方面表现最佳，而MiLA则在时间一致性方面表现最强，两者共同确立了新的最先进性能标准。

### B. 场景理解

**基于Occ3D-nuScenes数据集的4D空间占用预测。**4D空间占用预测被视作具有代表性的世界建模任务。给定过去2秒的三维空间占用数据，模型可预测后续3秒的场景动态变化。评估方法遵循以下标准：



Occ3D-nuScenes protocol and reports *mIoU* and per horizon *IoU*. As summarized in Tab. V, we compare methods by input modality, auxiliary supervision, and ego trajectory usage to reveal design choices for spatiotemporal forecasting. Methods using occupancy inputs outperform camera-only variants, and adding auxiliary supervision with a GT ego trajectory further mitigates performance decay at 2–3 s. Among all methods, COME (with GT ego) achieves the best average *mIoU* and per-horizon *IoU*.

### C. Control Tasks

**Evaluation on DMC.** Most studies probe the capacity of world models to learn control-relevant dynamics, typically adopting a pixel-based setting with  $64 \times 64 \times 3$  observations. The primary metric is *Episode Return*, defined as the cumulative reward over 1 000 steps, with a theoretical maximum of 1 000 given  $r_t \in [0, 1]$ . For comparability, Tab. VI reports the step budget and summarizes performance by task score and task count. The results indicate improved data efficiency, with recent models reaching strong performance in far fewer training steps. However, inconsistent evaluation protocols and task subsets impede a fair assessment of generalization, and building a broadly transferable model across tasks, modalities, and datasets remains an open challenge.

**Evaluation on RL Bench.** RL Bench evaluates manipulation with a 7-DoF simulated Franka arm and is widely used to assess whether world models capture task-relevant dynamics and support conditioned action generation. The primary metric is *Success Rate*, defined as the fraction of episodes that reach the goal within the step limit. As summarized in Tab. VII, implementations differ in episode budgets, resolution, and modalities, which complicates like-for-like comparison. Despite this heterogeneity, several trends are evident. Recent methods increasingly leverage multimodal inputs and adopt stronger backbones such as 3DGS and DiT. VidMan achieves a high average success rate on the broadest task, revealing IDM as a promising architectural direction.

**Planning on nuScenes.** Open-loop planning is treated as a world modeling task on the nuScenes validation split, where models predict ego motion from a limited history. Methods observe 2 s of past trajectories and forecast the next 3 s as 2D BEV waypoints. Evaluation reports *L2* error and *collision rate* at multiple horizons, and Tab. VIII summarizes results by input modality, auxiliary supervision, and metric settings. Under this shared protocol, a clear tradeoff emerges. UniAD+DriveWorld achieves the lowest *L2* with extensive auxiliary supervision, whereas SSR attains the best collision rate with competitive *L2* without extra supervision. Camera-based methods now surpass models that use privileged occupancy, reflecting the growing maturity of E2E planning.

## VI. CHALLENGES AND TRENDS

This section reviews the open challenges and emerging directions for world models in embodied AI. We discuss them across three dimensions: §VI-A Data & Evaluation, §VI-B Computational Efficiency, and §VI-C Modeling Strategies.

### A. Data & Evaluation

**Challenges.** From a data perspective, the central challenge lies in the scarcity and heterogeneity of existing corpora. Although embodied AI spans diverse domains such as navigation, manipulation, and autonomous driving, a unified large-scale dataset remains lacking. This fragmentation constrains the capacity of world models and substantially hinders their ability to generalize.

Evaluation practices face similar limitations. Metrics such as FID and FVD emphasize pixel fidelity while ignoring physical consistency, dynamics, and causality. Recent benchmarks, such as EWM-Bench [255], introduce new measures but remain task-specific and lack cross-domain standards.

**Future Directions.** Recent initiatives such as OpenDV-2K [90] and VideoMix22M [14] highlight the growing focus on large-scale pretraining and broader modality coverage, yet resources remain fragmented and domain specific. Future work should prioritize constructing unified multimodal, cross-domain datasets to enable transferable pretraining, while advancing evaluation frameworks that move beyond perceptual realism to assess physical consistency, causal reasoning, and long-horizon dynamics.

### B. Computational Efficiency

**Challenges.** Embodied AI tasks encounter significant challenges in computational efficiency, particularly in real-time applications. Although models such as Transformers and Diffusion networks exhibit strong performance, their high inference costs conflict with the real-time control demands of robotic systems. Consequently, traditional methods like RNNs and global latent vectors remain widely employed, as they offer greater computational efficiency, despite limitations in capturing long-term dependencies.

**Future Directions.** To address this challenge, future research should focus on optimizing model architectures using techniques like quantization, pruning, and sparse computation to reduce inference latency without compromising performance. Additionally, exploring novel temporal methods, such as SSMs, could enhance long-range reasoning while maintaining real-time efficiency, offering a promising solution for robotic systems.

### C. Modeling Strategy

**Challenges.** Despite rapid progress, world models still struggle with long-horizon temporal dynamics and efficient spatial representations. The main difficulty lies in balancing recurrent simulation and global prediction: autoregressive designs are compact and sample-efficient but accumulate errors over time, whereas global prediction improves multi-step coherence at the cost of heavy computation and weaker closed-loop interactivity. On the spatial side, efficiency remains a bottleneck. Latent vectors, token sequences, and spatial grids each present trade-offs between efficiency and expressiveness, while decomposed rendering approaches (e.g., NeRF and 3DGS) offer high fidelity yet scale poorly in dynamic scenes. Taken

Occ3D-nuScenes协议会报告 $mIoU$ 及每个时间范围内的 $IoU$ 值。如表V所示，我们通过输入模态、辅助监督方式及自我轨迹使用情况对各方法进行比较，以揭示时空预测的设计选择。采用占用率输入的方法优于仅使用摄像头的变体；而在2–3秒时间范围内，加入基于真实自我轨迹的辅助监督可进一步缓解性能下降。在所有方法中，COME（结合真实自我轨迹）实现了最佳的平均 $mIoU$ 及每个时间范围内的 $IoU$ 指标。

### C. 控制任务

**关于DMC的评估。**大多数研究探讨了世界模型学习控制相关动态的能力，通常采用基于像素的设置，包含 $64 \times 64 \times 3$ 个观测值。主要评估指标是回合回报，定义为1 000步内的累积奖励；在 $r_t \in [0, 1]$ 条件下，其理论最大值为1 000。为便于比较，表VI列出了步数预算，并按任务得分和任务数量汇总了性能表现。结果表明数据效率有所提升，近期模型在更少的训练步数下即可取得优异性能。然而，不一致的评估协议和任务子集限制了对泛化的公平评估，而构建能在不同任务、模态和数据集上广泛迁移的模型仍是亟待解决的挑战。

**RLBenchmark评估结果。**RLBenchmark通过7自由度模拟Franka机械臂进行操作评估，被广泛用于检验世界模型是否能捕捉任务相关动态并支持条件动作生成。主要评估指标是成功率，定义为在步数限制内达成目标的回合比例。如表VII所示，不同实现方案在回合预算、分辨率和输入模态方面存在差异，这使得同类比较变得复杂。尽管存在这种异质性，仍可观察到若干趋势：近期方法日益采用多模态输入，并采用3DGS、DiT等更强的骨干网络；Vid-Man在最广泛任务中实现了高平均成功率，表明IDM架构方向具有广阔前景。

**nuScenes上的规划。**在nuScenes验证集上，开环规划被视为世界建模任务，模型通过有限历史数据预测自我运动。方法学观测过去2秒的轨迹，并将未来3秒预测为二维BEV航点。评估报告在多个时间范围内报告 $L2$ 误差与碰撞率，表VIII按输入模态、辅助监督方式及指标设置汇总结果。在此共享协议下，明显存在权衡关系：UniAD+DriveWorld通过大量辅助监督实现最低 $L2$ 误差，而SSR在无需额外监督的情况下以具有竞争力的 $L2$ 指标获得最佳碰撞率。基于相机的方法现已超越使用特权占用率的模型，这反映了端到端规划技术日益成熟的趋势。

## VI. 挑战与趋势

本节综述了具身人工智能领域世界模型面临的开放性挑战及新兴发展方向。我们从三个维度进行探讨：§VI-A数据与评估、§VI-B计算效率以及§VI-C建模策略。

### A. 数据与评估

**挑战。**从数据层面来看，核心挑战在于现有语料库的稀缺性与异质性。尽管具身人工智能已涵盖导航、操作控制及自动驾驶等多元领域，但统一的大规模数据集仍显不足。这种碎片化现状不仅限制了世界模型的构建能力，更严重制约了其泛化学习能力。

评估方法同样存在类似局限性。诸如FID和FVD等指标侧重像素保真度，却忽视物理一致性、动态变化及因果关系。近期基准测试（如EWM-Bench[255]）虽引入了新评估指标，但仍局限于特定任务且缺乏跨领域标准。

**未来发展方向。**近期诸如OpenDV-2K[90]和VideoMix22M[14]等项目凸显了对大规模预训练及更广泛模态覆盖的日益重视，但现有资源仍存在碎片化与领域特定性问题。未来研究应优先构建统一的多模态跨域数据集以实现可迁移预训练，同时推进超越感知真实性的评估框架，以评估物理一致性、因果推理能力及长时程动态表现。

### B. 计算效率

**挑战。**嵌入式人工智能任务在计算效率方面面临重大挑战，尤其是在实时应用场景中。尽管Transformer模型和扩散网络等算法展现出强劲性能，但其高昂的推理成本与机器人系统的实时控制需求存在矛盾。因此，尽管存在捕捉长期依赖关系的局限性，传统方法如循环神经网络（RNN）和全局潜在向量仍被广泛采用，因其具有更高的计算效率优势。

**未来研究方向。**为应对这一挑战，未来研究应着重通过量化技术、剪枝算法及稀疏计算等方法优化模型架构，在保证性能的前提下降低推理延迟。此外，探索时序自编码器（SSMs）等新型时序方法，可在保持实时处理效率的同时提升长距离推理能力，为机器人系统提供极具前景的解决方案。

### C. 建模策略

**挑战。**尽管技术发展迅速，全球模型在处理长时域动态变化和高效空间表征方面仍面临挑战。核心难点在于如何平衡循环模拟与全局预测：自回归架构虽结构紧凑且样本效率高，但会随时间累积误差；而全局预测虽能提升多步预测一致性，却需要消耗大量计算资源且闭环交互性较弱。在空间维度上，效率始终是瓶颈所在。潜在向量、标记序列和空间网格在效率与表达能力之间存在权衡关系，而分解式渲染方法（例如NeRF和3DGS）虽能实现高保真效果，但在动态场景中扩展性较差。

TABLE V  
PERFORMANCE COMPARISON OF 4D OCCUPANCY FORECASTING ON THE OCC3D-NUSCENES BENCHMARK<sup>1</sup>.

| Method                    | Input  | Aux. Sup       | Ego traj. | mIoU (%) ↑ |       |       |       |       | IoU (%) ↑ |       |       |       |       |
|---------------------------|--------|----------------|-----------|------------|-------|-------|-------|-------|-----------|-------|-------|-------|-------|
|                           |        |                |           | Recon.     | 1s    | 2s    | 3s    | Avg.  | Recon.    | 1s    | 2s    | 3s    | Avg.  |
| Copy & Paste <sup>2</sup> | Occ    | None           | Pred.     | 66.38      | 14.91 | 10.54 | 8.52  | 11.33 | 62.29     | 24.47 | 19.77 | 17.31 | 20.52 |
| OccWorld-O [93]           | Occ    | None           | Pred.     | 66.38      | 25.78 | 15.14 | 10.51 | 17.14 | 62.29     | 34.63 | 25.07 | 20.18 | 26.63 |
| OccLLaMA-O [18]           | Occ    | None           | Pred.     | 75.20      | 25.05 | 19.49 | 15.26 | 19.93 | 63.76     | 34.56 | 28.53 | 24.41 | 29.17 |
| RenderWorld-O [156]       | Occ    | None           | Pred.     | -          | 28.69 | 18.89 | 14.83 | 20.80 | -         | 37.74 | 28.41 | 24.08 | 30.08 |
| DTT-O [98]                | Occ    | None           | Pred.     | 85.50      | 37.69 | 29.77 | 25.10 | 30.85 | 92.07     | 76.60 | 74.44 | 72.71 | 74.58 |
| DFIT-OccWorld-O [174]     | Occ    | None           | Pred.     | -          | 31.68 | 21.29 | 15.18 | 22.71 | -         | 40.28 | 31.24 | 25.29 | 32.27 |
| COME-O [213]              | Occ    | None           | Pred.     | -          | 30.57 | 19.91 | 13.38 | 21.29 | -         | 36.96 | 28.26 | 21.86 | 29.03 |
| DOVE-O [94]               | Occ    | None           | GT        | 83.08      | 35.11 | 25.89 | 20.29 | 27.10 | 77.25     | 43.99 | 35.36 | 29.74 | 36.36 |
| COME-O [213]              | Occ    | None           | GT        | -          | 42.75 | 32.97 | 26.98 | 34.23 | -         | 50.57 | 43.47 | 38.36 | 44.13 |
| OccWorld-T [93]           | Camera | Semantic LiDAR | Pred.     | 7.21       | 4.68  | 3.36  | 2.63  | 3.56  | 10.66     | 9.32  | 8.23  | 7.47  | 8.34  |
| OccWorld-S [93]           | Camera | None           | Pred.     | 0.27       | 0.28  | 0.26  | 0.24  | 0.26  | 4.32      | 5.05  | 5.01  | 4.95  | 5.00  |
| RenderWorld-S [156]       | Camera | None           | Pred.     | -          | 2.83  | 2.55  | 2.37  | 2.58  | -         | 14.61 | 13.61 | 12.98 | 13.73 |
| COME-S [213]              | Camera | None           | Pred.     | -          | 25.57 | 18.35 | 13.41 | 19.11 | -         | 45.36 | 37.06 | 30.46 | 37.63 |
| OccWorld-D [93]           | Camera | Occ            | Pred.     | 18.63      | 11.55 | 8.10  | 6.22  | 8.62  | 22.88     | 18.90 | 16.26 | 14.43 | 16.53 |
| OccWorld-F [93]           | Camera | Occ            | Pred.     | 20.09      | 8.03  | 6.91  | 3.54  | 6.16  | 35.61     | 23.62 | 18.13 | 15.22 | 18.99 |
| OccLLaMA-F [18]           | Camera | Occ            | Pred.     | 37.38      | 10.34 | 8.66  | 6.98  | 8.66  | 38.92     | 25.81 | 23.19 | 19.97 | 22.99 |
| DFIT-OccWorld-F [174]     | Camera | Occ            | Pred.     | -          | 13.38 | 10.16 | 7.96  | 10.50 | -         | 19.18 | 16.85 | 15.02 | 17.02 |
| DTT-F [98]                | Camera | Occ            | Pred.     | 43.52      | 24.87 | 18.30 | 15.63 | 19.60 | 54.31     | 38.98 | 37.45 | 31.89 | 36.11 |
| DOVE-F [94]               | Camera | Occ            | GT        | 75.00      | 24.12 | 17.41 | 13.24 | 18.25 | 74.31     | 35.18 | 27.90 | 23.44 | 28.84 |
| COME-F [213]              | Camera | Occ            | GT        | -          | 26.56 | 21.73 | 18.49 | 22.26 | -         | 48.08 | 43.84 | 40.28 | 44.07 |

<sup>1</sup> Note: Method variants are denoted by their input source: O for ground-truth; camera-based variants include D (TPVFormer), F (FBOCC), T (TPVFormer with semantic LiDAR), and S (self-supervised TPVFormer).

<sup>2</sup> Copy & Paste: A naive baseline that repeats the final input frame for all future predictions.

TABLE VI  
PERFORMANCE COMPARISON ON THE DMC BENCHMARK<sup>1</sup>.

| Method            | Step | Episode Return <sup>†</sup> |             |             |             | Avg. / Total |
|-------------------|------|-----------------------------|-------------|-------------|-------------|--------------|
|                   |      | Reacher Easy                | Cheetah Run | Finger Spin | Walker Walk |              |
| PlaNet [38]       | 5M   | 469                         | 496         | 495         | 945         | 333/20       |
| Dreamer [110]     | 5M   | 935                         | 895         | 499         | 962         | 823/20       |
| Dreaming [110]    | 500k | 905                         | 566         | 762         | 469         | 610/12       |
| TransDreamer [28] | 2M   | -                           | <u>865</u>  | -           | <u>933</u>  | <u>893/4</u> |
| DreamerPro [111]  | 1M   | 873                         | 897         | 811         | -           | 857/6        |
| MWM [41]          | 1M   | -                           | <u>670</u>  | -           | -           | <u>690/7</u> |
| HRSSM [25]        | 500k | <u>910</u>                  | -           | <u>960</u>  | -           | <u>938/3</u> |
| DisWM [112]       | 1M   | <u>960</u>                  | <u>820</u>  | -           | <u>920</u>  | <u>879/5</u> |

<sup>1</sup> Note: Performance comparison on the DMC. Underlined entries indicate scores approximated from their respective reward curves. Average scores (Avg.) are provided as a coarse indicator, given the varying difficulty across tasks.

TABLE VII  
PERFORMANCE COMPARISON FOR MANIPULATION TASKS ON RLBENCH.

| Criteria                  | VidMan [55] | ManiGaussian [53] | Methods             |            |                |
|---------------------------|-------------|-------------------|---------------------|------------|----------------|
|                           |             |                   | ManiGaussian++ [80] | DreMa [60] | Tesseract [78] |
| Episode                   | 125         | 25                | 25                  | 250        | 100            |
| Pixel                     | 224         | 128               | 256                 | 128        | 512            |
| Depth                     | ✓           | ✓                 | ✓                   | ✓          | ✓              |
| Language                  | ✓           | ✓                 | ✓                   | ✓          | ✓              |
| Proprioception            | ✓           | ✓                 | ✓                   | ✓          | ✓              |
| Characteristic            | IDM         | GS                | Bimanual            | GS         | DT             |
| Success Rate (%)          |             |                   |                     |            |                |
| Stack Blocks              | 48          | 12                | -                   | 12         | -              |
| Close Jar                 | 88          | 28                | -                   | 51         | 44             |
| Open Drawer               | 94          | 76                | -                   | -          | 80             |
| Sweep to Dustpan          | 93          | 64                | 92                  | -          | 56             |
| Slide Block               | 98          | 24                | -                   | 62         | -              |
| Avg. <sup>1</sup> / Total | 67/18       | 45/10             | 35/10               | 25/9       | 63/10          |

<sup>1</sup> Avg.: Average scores are reported only as a coarse indicator, given varying task difficulty.

together, temporal and spatial modeling are still constrained by structural trade-offs that limit scalability and adaptability.

**Future Directions.** Several promising avenues have emerged to address current bottlenecks. SSMs (e.g., Mamba), aligned with autoregressive modeling, offer linear-time scalability and strong potential for long-horizon reasoning. In contrast, masked approaches (e.g., JEPa), closer to global prediction, improve representation learning and efficiency, though their integration

into closed-loop control remains challenging. Furthermore, a promising approach lies in integrating the strengths of both autoregressive and global prediction methods. Explicit memory or hierarchical planning can enhance long-horizon prediction stability, while task decomposition inspired by CoT can improve temporal consistency through intermediate goal setting. Future frameworks should prioritize optimizing long-range reasoning, computational efficiency, and generative fidelity, while seamlessly integrating temporal and spatial modeling into unified architectures that strike an effective balance between efficiency, fidelity, and interactivity.

## VII. CONCLUSION

This survey organizes world models in embodied AI using a novel three-part framework: functionality, temporal modeling, and spatial representation. Based on this, we review existing research, datasets, and metrics to establish a standard for comparison. However, significant challenges remain, including a lack of unified datasets and evaluation methods that overlook physical causality. A core modeling challenge involves reconciling the trade-off between efficient autoregressive approaches and robust global prediction paradigms. Future work should address these gaps by creating unified, physically-grounded benchmarks and exploring efficient architectures. Developing hybrid methods that balance fidelity, efficiency, and interactivity is key, as world models form the foundation for the next generation of embodied AI by unifying perception, prediction, and decision-making.

## REFERENCES

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表V  
4D 占用率预测在OCC3D-NUSCENES基准集上的性能比较<sup>1</sup>.

| 方法                    | 输入    | Aux. Sup | 自我轨迹 | mIoU (%) ↑ |       |       |       |       | IoU (%) ↑ |       |       |       |       |
|-----------------------|-------|----------|------|------------|-------|-------|-------|-------|-----------|-------|-------|-------|-------|
|                       |       |          |      | 推荐         | 1s    | 2s    | 3s    | 平均值   | 推荐        | 1s    | 2s    | 3s    | 平均值   |
| 复制并粘贴 <sup>2</sup>    | 偶尔：随机 | 没有一个     | 总统。  | 66.38      | 14.91 | 10.54 | 8.52  | 11.33 | 62.29     | 24.47 | 19.77 | 17.31 | 20.52 |
| OccWorld-O [93]       | 偶尔    | 没有一个     | 总统。  | 66.38      | 25.78 | 15.14 | 10.51 | 17.14 | 62.29     | 34.63 | 25.07 | 20.18 | 26.63 |
| OccLLaMA-O [18]       | 偶尔    | 没有一个     | 总统。  | 75.20      | 25.05 | 19.49 | 15.26 | 19.93 | 63.76     | 34.56 | 28.53 | 24.41 | 29.17 |
| RenderWorld-O [156]   | 偶尔：随机 | 没有一个     | 总统。  | -          | 28.69 | 18.89 | 14.83 | 20.80 | -         | 37.74 | 28.41 | 24.08 | 30.08 |
| DTT-O [98]            | 偶尔：随机 | 没有一个     | 总统。  | 85.50      | 37.69 | 29.77 | 25.10 | 30.85 | 92.07     | 76.60 | 74.44 | 72.71 | 74.58 |
| DFIT-OccWorld-O [174] | 偶尔    | 没有一个     | 总统。  | -          | 31.68 | 21.29 | 15.18 | 22.71 | -         | 40.28 | 31.24 | 25.29 | 32.27 |
| COME-O [213]          | 偶尔：随机 | 没有一个     | 总统。  | -          | 30.57 | 19.91 | 13.38 | 21.29 | -         | 36.96 | 28.26 | 21.86 | 29.03 |
| DOME-O [94]           | 偶尔：随机 | 没有一个     | GT   | 83.08      | 35.11 | 25.89 | 20.29 | 27.10 | 77.25     | 43.99 | 35.36 | 29.74 | 36.36 |
| COME-O [213]          | 偶尔    | 没有一个     | GT   | -          | 42.75 | 32.97 | 26.98 | 34.23 | -         | 50.57 | 43.47 | 38.36 | 44.13 |
| OccWorld-T [93]       | 照相机   | 语义激光雷达   | 总统。  | 7.21       | 4.68  | 3.36  | 2.63  | 3.56  | 10.66     | 9.32  | 8.23  | 7.47  | 8.34  |
| OccWorld-S [93]       | 照相机   | 没有一个     | 总统。  | 0.27       | 0.28  | 0.26  | 0.24  | 0.26  | 4.32      | 5.05  | 5.01  | 4.95  | 5.00  |
| RenderWorld-S [156]   | 照相机   | 没有一个     | 总统。  | -          | 2.83  | 2.55  | 2.37  | 2.58  | -         | 14.61 | 13.61 | 12.98 | 13.73 |
| COME-S [213]          | 照相机   | 没有一个     | 总统。  | -          | 25.57 | 18.35 | 13.41 | 19.11 | -         | 45.36 | 37.06 | 30.46 | 37.63 |
| OccWorld-D [93]       | 照相机   | 偶尔       | 总统。  | 18.63      | 11.55 | 8.10  | 6.22  | 8.62  | 22.88     | 18.90 | 16.26 | 14.43 | 16.53 |
| OccWorld-F [93]       | 照相机   | 偶尔       | 总统。  | 20.09      | 8.03  | 6.91  | 3.54  | 6.16  | 35.61     | 23.62 | 18.13 | 15.22 | 18.99 |
| OccLLaMA-F [18]       | 照相机   | 偶尔       | 总统。  | 37.38      | 10.34 | 8.66  | 6.98  | 8.66  | 38.92     | 25.81 | 23.19 | 19.97 | 22.99 |
| DFIT-OccWorld-F [174] | 照相机   | 偶尔       | 总统。  | -          | 13.38 | 10.16 | 7.96  | 10.50 | -         | 19.18 | 16.85 | 15.02 | 17.02 |
| DTT-F [98]            | 照相机   | 偶尔       | 总统。  | 43.52      | 24.87 | 18.30 | 15.63 | 19.60 | 54.31     | 38.98 | 37.45 | 31.89 | 36.11 |
| DOME-F [94]           | 照相机   | 偶尔       | GT   | 75.00      | 24.12 | 17.41 | 13.24 | 18.25 | 74.31     | 35.18 | 27.90 | 23.44 | 28.84 |
| COME-F [213]          | 照相机   | 偶尔       | GT   | -          | 26.56 | 21.73 | 18.49 | 22.26 | -         | 48.08 | 43.84 | 40.28 | 44.07 |

<sup>1</sup> 注：方法变体通过输入源标识：O表示真实数据；基于相机的变体包括D（TPVFormer）、F（FBOCC）、T（带语义激光雷达的TPVFormer）和S（自监督TPVFormer）。**复制粘贴**：一种简单基线方法，将最终输入帧重复用于所有后续预测。  
<sup>2</sup>

表VI  
DMC 基准测试中的性能对比。1.

| 方法                | 步骤   | 跟踪返回 / 里物简单跑手指旋转步行器 |     |     |     | 平均值 / 总计 |
|-------------------|------|---------------------|-----|-----|-----|----------|
|                   |      | 469                 | 496 | 495 | 945 |          |
| PlaNet [38]       | 5M   | 469                 | 496 | 495 | 945 | 333/20   |
| 梦想家[10]           | 5M   | 935                 | 895 | 499 | 962 | 823/20   |
| 做梦[110]           | 500k | 905                 | 566 | 762 | 469 | 610/12   |
| TransDreamer [28] | 2M   | -                   | 865 | -   | 933 | 893/4    |
| DreamerPro [111]  | 1M   | 873                 | 897 | 811 | -   | 857/6    |
| MWM [41]          | 1M   | -                   | 670 | -   | -   | 690/7    |
| HRSSM [25]        | 500k | 910                 | -   | 960 | -   | 938/3    |
| DisWM [112]       | 1M   | 960                 | 820 | -   | 920 | 879/5    |

<sup>1</sup> 注：DMC上的性能比较。带下划线的条目表示根据各自奖励曲线估算的分数。由于任务难度存在差异，平均分数（Avg.）仅作为粗略指标提供。

表VII  
RLBenchmark平台上操作任务的性能比较。

| 标准   | 方法         | ManiGaussian [53] | ManiGaussian++ [80] | DreMa [60] | TesserAct [78] |
|------|------------|-------------------|---------------------|------------|----------------|
|      |            |                   |                     |            |                |
| 视觉感知 | 回合         | 125               | 25                  | 25         | 250            |
|      | 像素         | 224               | 128                 | 256        | 128            |
|      | 深          | ✓                 | ✓                   | ✓          | ✓              |
|      | 语言         | ✓                 | ✓                   | ✓          | ✓              |
|      | 本体感觉       | ✓                 | ✓                   | ✓          | ✓              |
| 交互性  | 特征性的       | IDM               | GS                  | 双手的        | GS             |
|      | 堆栈块        | 48                | 12                  | -          | 12             |
|      | 关闭 jar 文件夹 | 88                | 28                  | -          | 51             |
|      | 打开抽屉       | 94                | 76                  | -          | 80             |
|      | 扫入簸箕       | 93                | 64                  | 92         | 56             |
| 5    | 幻灯片块       | 98                | 24                  | -          | 62             |
|      | 平均值/总数     | 67/18             | 45/10               | 35/10      | 25/9           |

<sup>1</sup> 平均值：由于任务难度各异，平均分数仅作为粗略指标进行报告。

总体而言，时空建模仍受结构权衡的限制，这些权衡制约了其可扩展性和适应性。

未来发展方向。目前已有若干具有前景的研究路径可解决当前瓶颈问题。与自回归建模相契合的SSMs（例如Mamba）具备线性时间扩展性及强大的长时程推理潜力。相比之下，更接近全局预测的掩码方法（例如JEPA）虽能提升表征学习效果与计算效率，但其在闭环控制中的集成

仍具挑战性。此外，融合自回归与全局预测方法优势的策略颇具前景：显式记忆或分层规划可增强长时程预测稳定性，而受CoT启发的任务分解则通过中间目标设定提升时间一致性。未来框架应优先优化长程推理能力、计算效率与生成保真度，同时将时空建模无缝整合至统一架构中，在效率、保真度与交互性之间实现有效平衡。

VII.结论

本研究采用创新性的三要素框架——功能模块、时序建模与空间表征，对具身人工智能领域的世界模型进行系统梳理。基于此框架，我们通过梳理现有研究成果、数据集及评估指标，建立了统一的比较标准。然而仍存在重大挑战，包括缺乏统一数据集及忽视物理因果关系的评估方法。核心建模难题在于如何平衡高效自回归方法与稳健全局预测范式之间的权衡关系。未来研究应通过构建统一的物理基础基准测试框架、探索高效架构来填补这些空白。开发兼具保真度、效率与交互性的混合方法至关重要，因为世界模型通过整合感知、预测与决策机制，将成为下一代具身人工智能发展的基石。

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TABLE VIII  
PERFORMANCE COMPARISON FOR OPEN-LOOP PLANNING ON THE NUSCENES VALIDATION SPLIT<sup>1</sup>.

| Method                | Input  |                                      | L2 (m) ↓ |      |      |      | Collision Rate (%) ↓ |      |      |      |
|-----------------------|--------|--------------------------------------|----------|------|------|------|----------------------|------|------|------|
|                       |        |                                      | 1s       | 2s   | 3s   | Avg. | 1s                   | 2s   | 3s   | Avg. |
| UniAD [254]           | Camera | Map & Box & Motion & Tracklets & Occ | 0.48     | 0.96 | 1.65 | 1.03 | 0.05                 | 0.17 | 0.71 | 0.31 |
| UniAD+DriveWorld [87] | Camera | Map & Box & Motion & Tracklets & Occ | 0.34     | 0.67 | 1.07 | 0.69 | 0.04                 | 0.12 | 0.41 | 0.19 |
| GenAD [92]            | Camera | Map & Box & Motion                   | 0.36     | 0.83 | 1.55 | 0.91 | 0.06                 | 0.23 | 1.00 | 0.43 |
| FSDrive [101]         | Camera | Map & Box & QA                       | 0.40     | 0.89 | 1.60 | 0.96 | 0.07                 | 0.12 | 1.02 | 0.40 |
| OccWorld-T [93]       | Camera | Semantic LiDAR                       | 0.54     | 1.36 | 2.66 | 1.52 | 0.12                 | 0.40 | 1.59 | 0.70 |
| Doe-1 [134]           | Camera | QA                                   | 0.50     | 1.18 | 2.11 | 1.26 | 0.04                 | 0.37 | 1.19 | 0.53 |
| SSR [160]             | Camera | None                                 | 0.24     | 0.65 | 1.36 | 0.75 | 0.00                 | 0.10 | 0.36 | 0.15 |
| OccWorld-S [93]       | Camera | None                                 | 0.67     | 1.69 | 3.13 | 1.83 | 0.19                 | 1.28 | 4.59 | 2.02 |
| Epona [148]           | Camera | None                                 | 0.61     | 1.17 | 1.98 | 1.25 | 0.01                 | 0.22 | 0.85 | 0.36 |
| RenderWorld [156]     | Camera | None                                 | 0.48     | 1.30 | 2.67 | 1.48 | 0.14                 | 0.55 | 2.23 | 0.97 |
| Drive-OccWorld [157]  | Camera | None                                 | 0.32     | 0.75 | 1.49 | 0.85 | 0.05                 | 0.17 | 0.64 | 0.29 |
| OccWorld-D [93]       | Camera | Occ                                  | 0.52     | 1.27 | 2.41 | 1.40 | 0.12                 | 0.40 | 2.08 | 0.87 |
| OccWorld-F [93]       | Camera | Occ                                  | 0.45     | 1.33 | 2.25 | 1.34 | 0.08                 | 0.42 | 1.71 | 0.73 |
| OccLLaMA-F [18]       | Camera | Occ                                  | 0.38     | 1.07 | 2.15 | 1.20 | 0.06                 | 0.39 | 1.65 | 0.70 |
| DTT-F [98]            | Camera | Occ                                  | 0.35     | 1.01 | 1.89 | 1.08 | 0.08                 | 0.33 | 0.91 | 0.44 |
| DFIT-OccWorld-V [174] | Camera | Occ                                  | 0.42     | 1.14 | 2.19 | 1.25 | 0.09                 | 0.19 | 1.37 | 0.55 |
| NeMo [161]            | Camera | Occ                                  | 0.39     | 0.74 | 1.39 | 0.84 | 0.00                 | 0.09 | 0.82 | 0.30 |
| OccWorld-O [93]       | Occ    | None                                 | 0.43     | 1.08 | 1.99 | 1.17 | 0.07                 | 0.38 | 1.35 | 0.60 |
| OccLLaMA-O [18]       | Occ    | None                                 | 0.37     | 1.02 | 2.03 | 1.14 | 0.04                 | 0.24 | 1.20 | 0.49 |
| RenderWorld-O [156]   | Occ    | None                                 | 0.35     | 0.91 | 1.84 | 1.03 | 0.05                 | 0.40 | 1.39 | 0.61 |
| DTT-O [98]            | Occ    | None                                 | 0.32     | 0.91 | 1.76 | 1.00 | 0.08                 | 0.32 | 0.51 | 0.30 |
| DFIT-OccWorld-O [174] | Occ    | None                                 | 0.38     | 0.96 | 1.73 | 1.02 | 0.07                 | 0.39 | 0.90 | 0.45 |

<sup>1</sup> Note: Results are reported following the UniAD [254] protocol. Method variants are denoted by their input source: O (ground-truth), V (camera-predicted), D (TPVFormer), F (FBOCC), etc.

<sup>2</sup> Aux. Sup.: Abbreviation for auxiliary supervision, which refers to additional training signals beyond ground-truth trajectories.

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表 VIII  
基于 NUSCENES 验证分割集1的开环规划性能比较

| 方法                    | 输入  | Aux. Sup. <sup>2</sup> | L2 (m) ↓ |      |      |      | 碰撞率 (%) ↓ |      |      |      |
|-----------------------|-----|------------------------|----------|------|------|------|-----------|------|------|------|
|                       |     |                        | 1s       | 2s   | 3s   | 平均值  | 1s        | 2s   | 3s   | 平均值  |
| UniAD [254]           | 照相机 | 地图与方框与运动与轨迹片段与发生       | 0.48     | 0.96 | 1.65 | 1.03 | 0.05      | 0.17 | 0.71 | 0.31 |
| UniAD+DriveWorld [87] | 照相机 | 地图与方框与运动与轨迹片段与发生       | 0.34     | 0.67 | 1.07 | 0.69 | 0.04      | 0.12 | 0.41 | 0.19 |
| GenAD [92] FS-        | 照相机 | 地图、方框与动态效果             | 0.36     | 0.83 | 1.55 | 0.91 | 0.06      | 0.23 | 1.00 | 0.43 |
| Drive [101]           | 照相机 | 地图、盒子与质量保证             | 0.40     | 0.89 | 1.60 | 0.96 | 0.07      | 0.12 | 1.02 | 0.40 |
| OccWorld-T [93]       | 照相机 | 语义激光雷达质量保证             | 0.54     | 1.36 | 2.66 | 1.52 | 0.12      | 0.40 | 1.59 | 0.70 |
| Doe-1 [134]           | 照相机 |                        | 0.50     | 1.18 | 2.11 | 1.26 | 0.04      | 0.37 | 1.19 | 0.53 |
| SSR [160] OccWorld-   | 照相机 | 没有一个                   | 0.24     | 0.65 | 1.36 | 0.75 | 0.00      | 0.10 | 0.36 | 0.15 |
| S [93] Epona [148]    | 照相机 | 没有一个                   | 0.67     | 1.69 | 3.13 | 1.83 | 0.19      | 1.28 | 4.59 | 2.02 |
| RenderWorld [156]     | 照相机 | 没有一个                   | 0.61     | 1.17 | 1.98 | 1.25 | 0.01      | 0.22 | 0.85 | 0.36 |
| 驱动-发生世界[157]发生世       | 照相机 | 没有一个                   | 0.48     | 1.30 | 2.67 | 1.48 | 0.14      | 0.55 | 2.23 | 0.97 |
| 界-D[93]               | 照相机 | 偶见                     | 0.32     | 0.75 | 1.49 | 0.85 | 0.05      | 0.17 | 0.64 | 0.29 |
| OccWorld-F [93]       | 照相机 | 偶见                     | 0.52     | 1.27 | 2.41 | 1.40 | 0.12      | 0.40 | 2.08 | 0.87 |
| OccLLaMA-F [18]       | 照相机 | 偶见                     | 0.45     | 1.33 | 2.25 | 1.34 | 0.08      | 0.42 | 1.71 | 0.73 |
| DTT-F [98]            | 照相机 | 偶见                     | 0.38     | 1.07 | 2.15 | 1.20 | 0.06      | 0.39 | 1.65 | 0.70 |
| DFIT-OccWorld-V [174] | 照相机 | 偶见                     | 0.35     | 1.01 | 1.89 | 1.08 | 0.08      | 0.33 | 0.91 | 0.44 |
| NeMo [161]            | 照相机 | 偶见                     | 0.42     | 1.14 | 2.19 | 1.25 | 0.09      | 0.19 | 1.37 | 0.55 |
| OccWorld-O [93]       | 偶见  | 没有一个                   | 0.39     | 0.74 | 1.39 | 0.84 | 0.00      | 0.09 | 0.82 | 0.30 |
| OccLLaMA-O [18]       | 偶见  | 没有一个                   | 0.43     | 1.08 | 1.99 | 1.17 | 0.07      | 0.38 | 1.35 | 0.60 |
| RenderWorld-O [156]   | 偶见  | 没有一个                   | 0.37     | 1.02 | 2.03 | 1.14 | 0.04      | 0.24 | 1.20 | 0.49 |
| DTT-O [98]            | 偶见  | 没有一个                   | 0.35     | 0.91 | 1.84 | 1.03 | 0.05      | 0.40 | 1.39 | 0.61 |
| DFIT-OccWorld-O [174] | 偶见  | 没有一个                   | 0.32     | 0.91 | 1.76 | 1.00 | 0.08      | 0.32 | 0.51 | 0.30 |
|                       |     |                        | 0.38     | 0.96 | 1.73 | 1.02 | 0.07      | 0.39 | 0.90 | 0.45 |

<sup>1</sup> 注：结果报告遵循UniAD[254]方案。方法变体按输入源标注：O（真实值）、V（相机预测）、D（TPVFormer）、F（FBOCC）、等。

<sup>2</sup> Aux. Sup.: 辅助监督的缩写，指超出真实轨迹范围的附加训练信号。

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